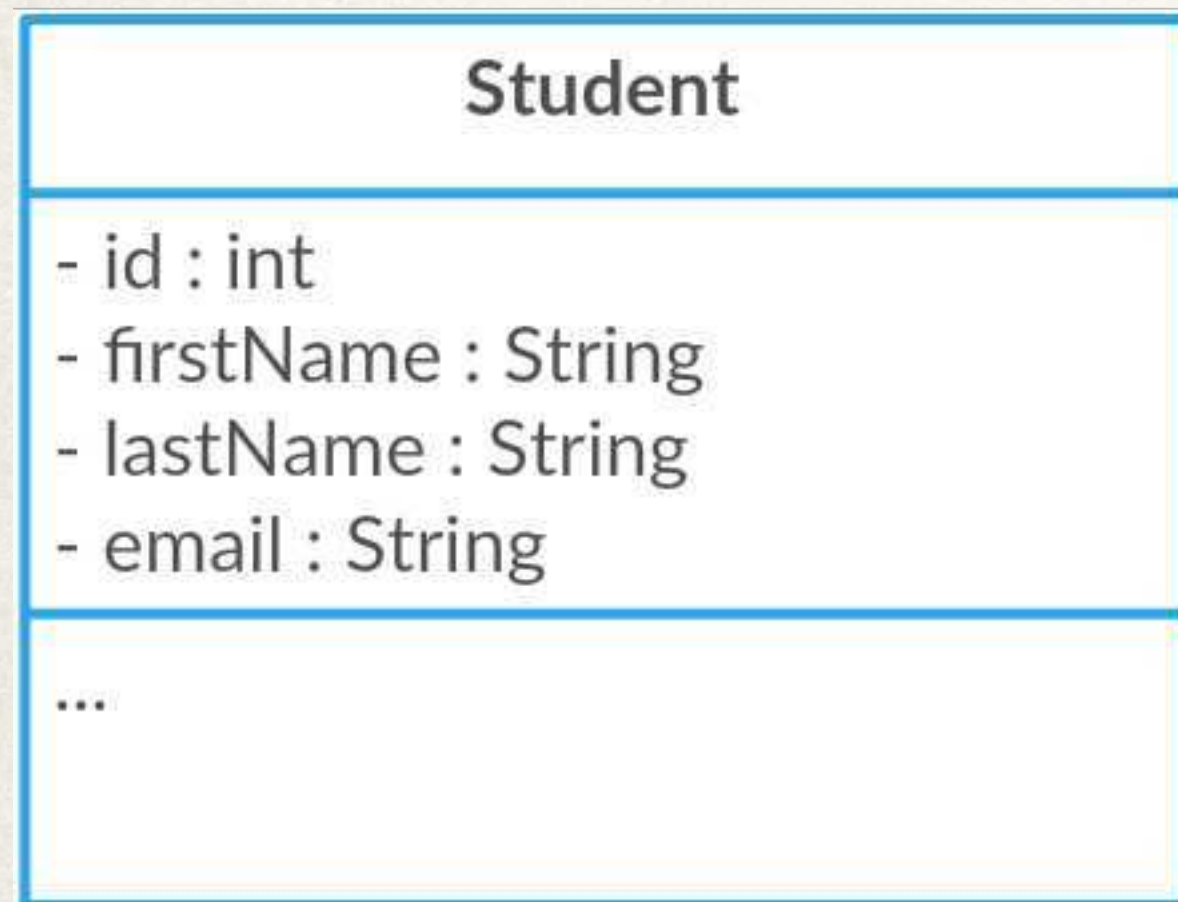


JPA / Hibernate Advanced Mappings



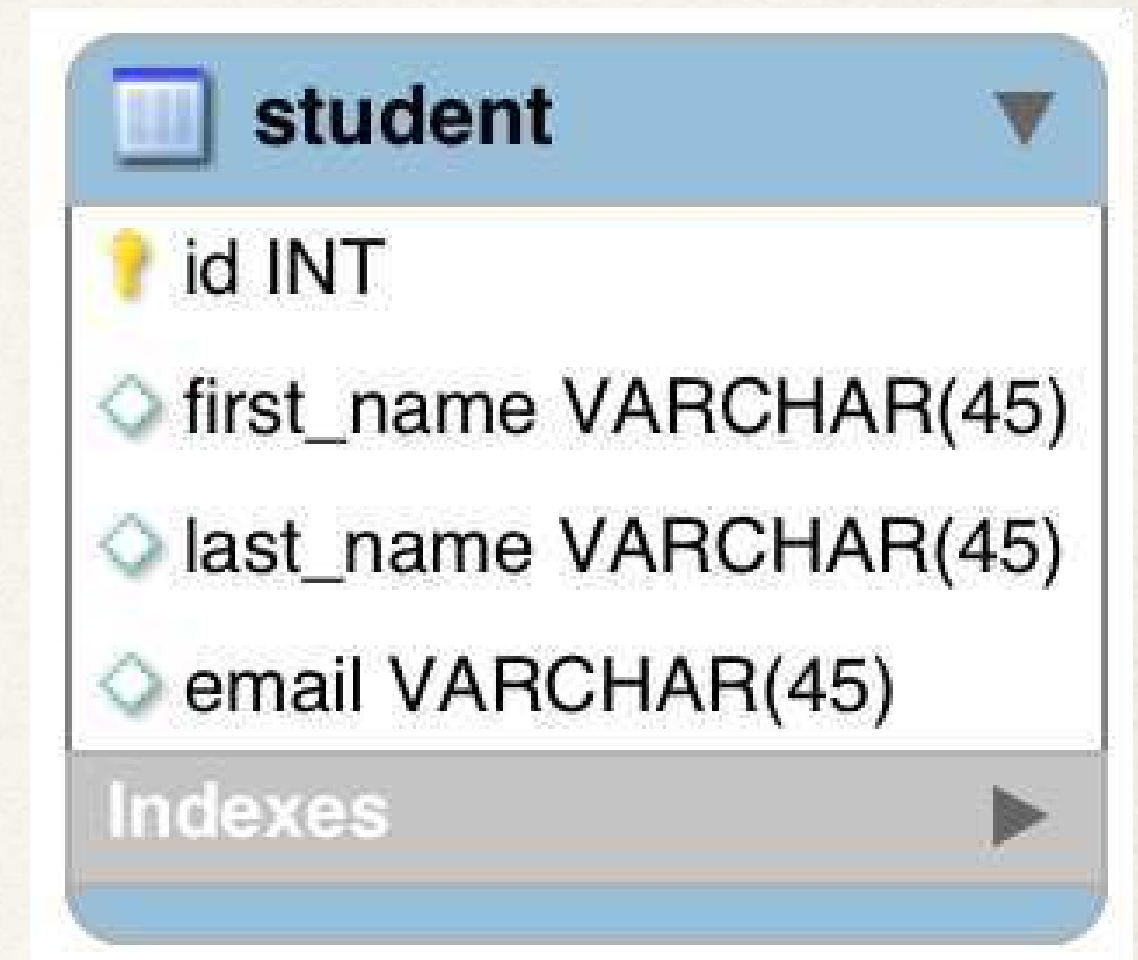
Basic Mapping

Java Class



Hibernate

Database Table



Advanced Mappings

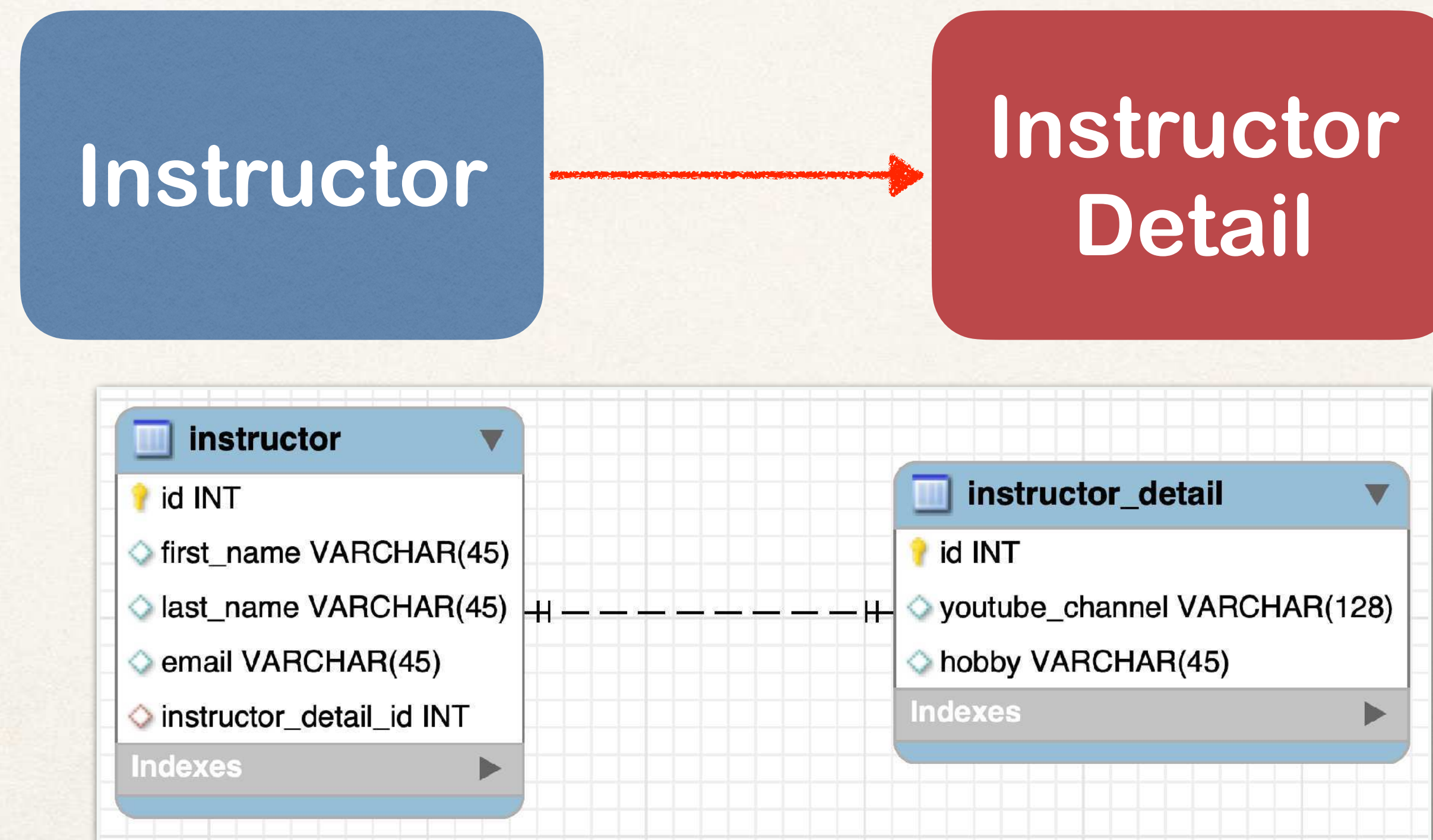
- In the database, you most likely will have
 - Multiple Tables
 - Relationships between Tables
- Need to model this with Hibernate

Advanced Mappings

- One-to-One
- One-to-Many, Many-to-One
- Many-to-Many

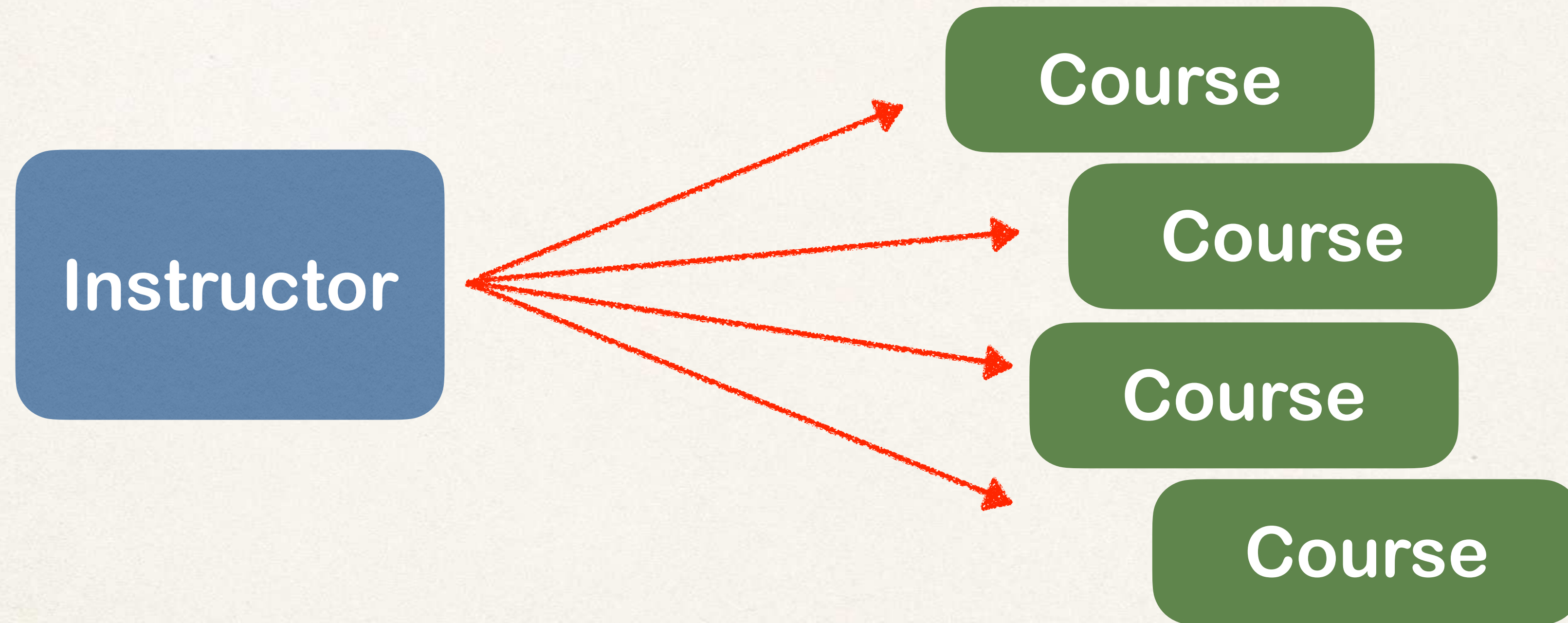
One-to-One Mapping

- An instructor can have an “instructor detail” entity
- Similar to an “instructor profile”



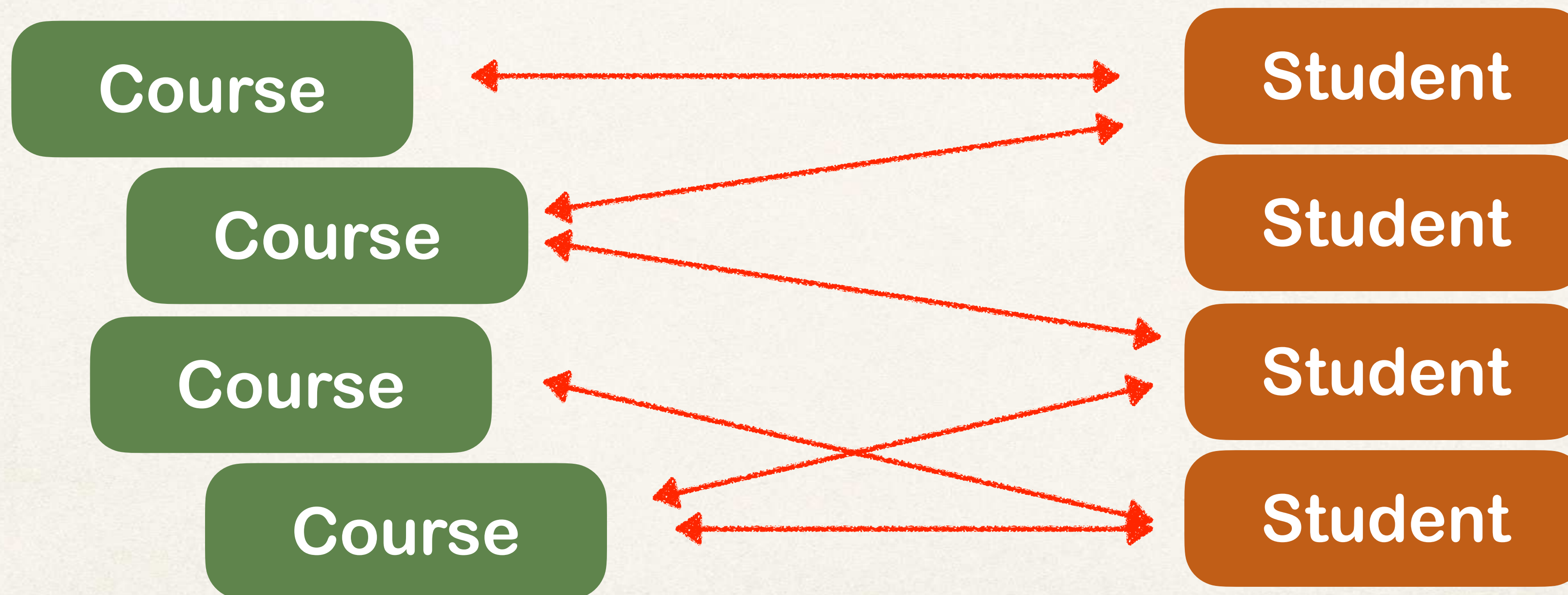
One-to-Many Mapping

- An instructor can have many courses



Many-to-Many Mapping

- A course can have many students
- A student can have many courses



Important Database Concepts

- Primary key and foreign key
- Cascade

Primary Key and Foreign Key

- Primary key: identify a unique row in a table
- Foreign key:
 - Link tables together
 - a field in one table that refers to primary key in another table

Foreign Key Example

Foreign key column

Table: instructor

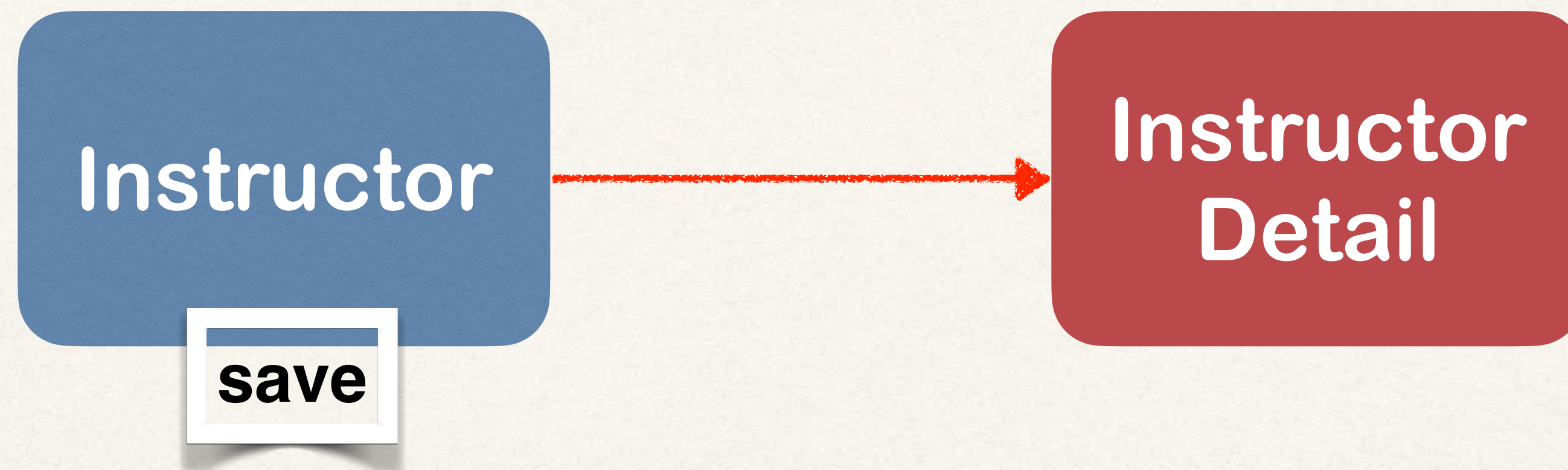
id	first_name	last_name	instructor_detail_id
1	Chad	Darby	100
2	Madhu	Patel	200

Table: instructor_detail

id	youtube_channel	hobby
100	www.luv2code.com/youtube	Luv 2 Code!!!
200	www.youtube.com	Guitar

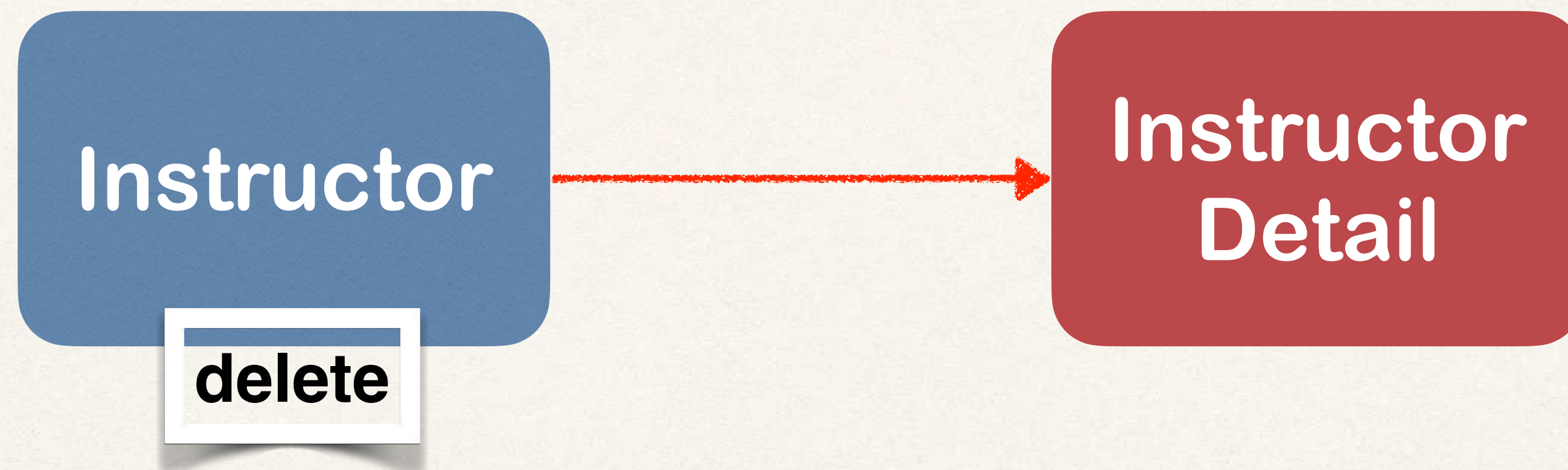
Cascade

- You can **cascade** operations
- Apply the same operation to related entities



Cascade

- If we delete an **instructor**, we should also delete their **instructor_detail**
- This is known as “CASCADE DELETE”



Cascade Delete

Foreign key
column

Table: instructor

id	first_name	last_name	instructor_detail_id
1	Chad	Darby	100
2	Madhu	Patel	200

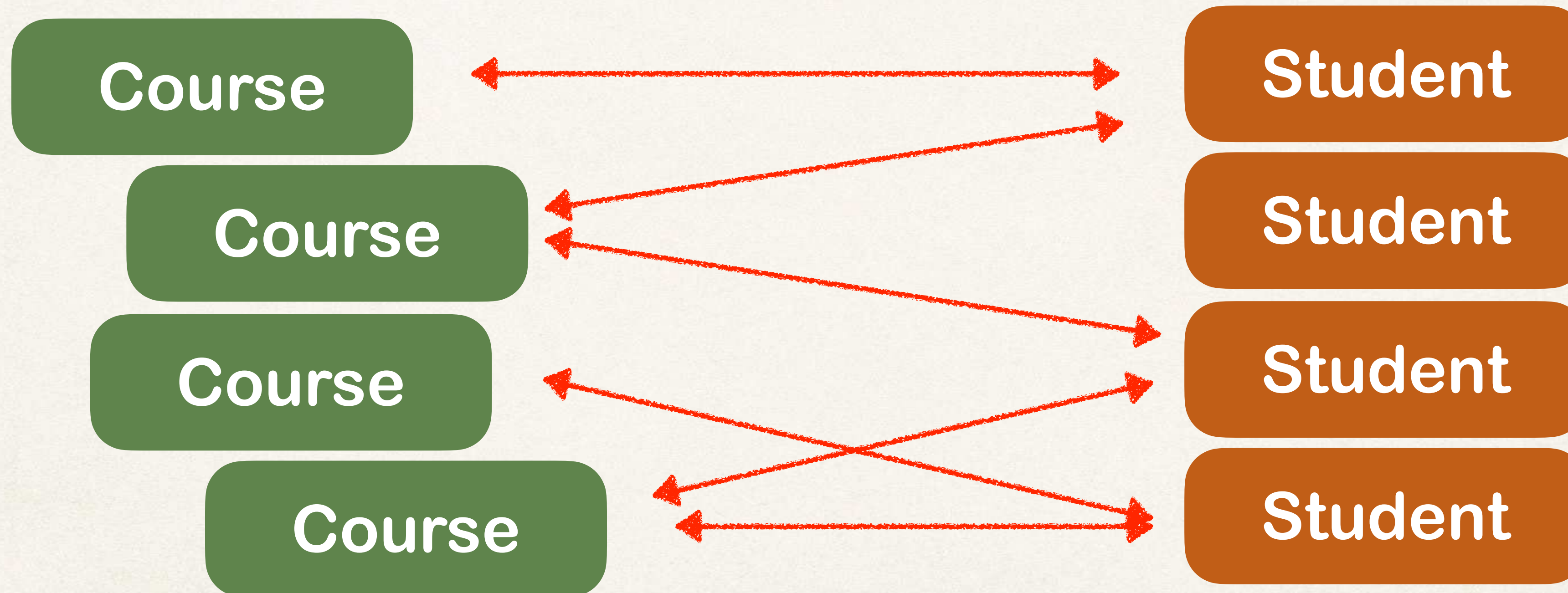
Table: instructor_detail

id	youtube_channel	hobby
100	www.luv2code.com/youtube	Luv 2 Code!!!
200	www.youtube.com	Guitar

Cascade Delete

- Cascade delete depends on the use case
- Should we do cascade delete here???

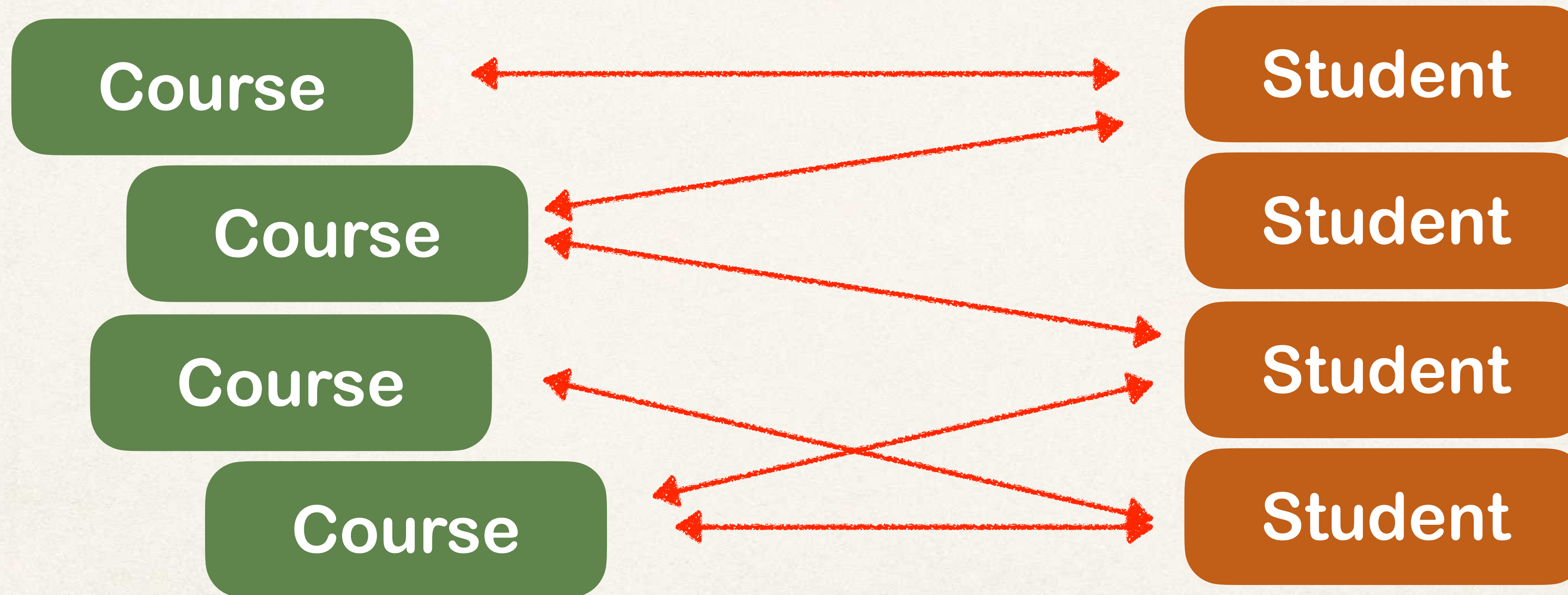
No way!!!



Cascade Delete

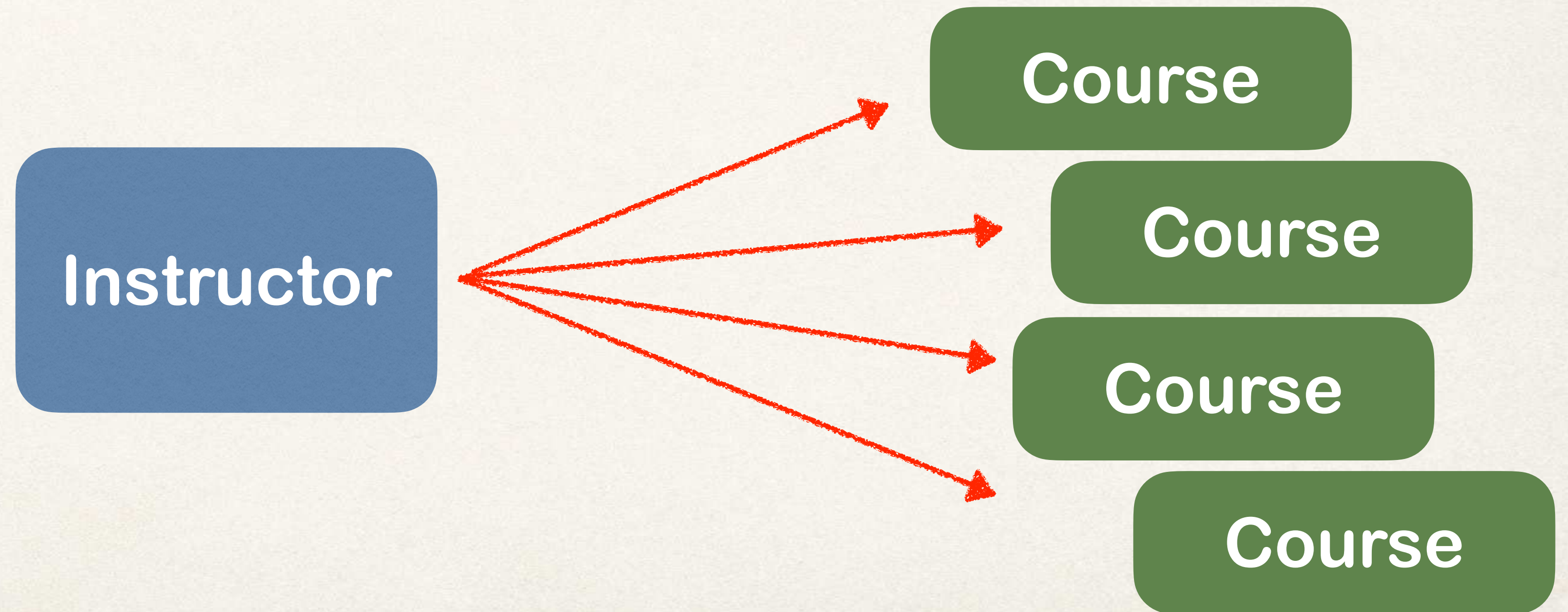
- Cascade delete depends on the use case
- Should we do cascade delete here???

Developer can
configure cascading



Fetch Types: Eager vs Lazy Loading

- When we fetch / retrieve data, should we retrieve EVERYTHING?
- **Eager** will retrieve everything
- **Lazy** will retrieve on request



Uni-Directional



Bi-Directional



Multiple Valid Designs

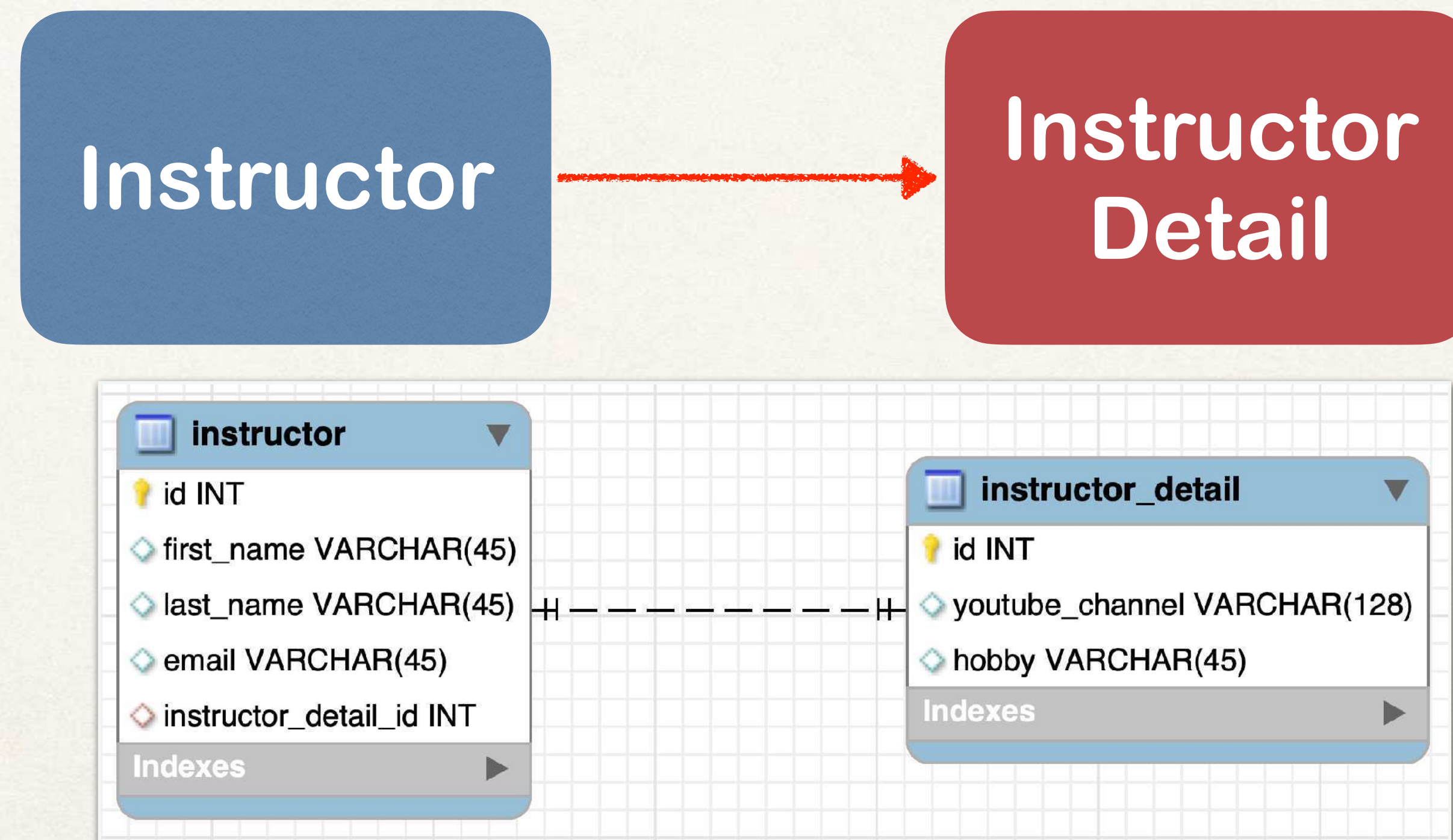
- For JPA / Hibernate relationships, there isn't one "right" mapping
- JPA / Hibernate supports several ways to model
 - @OneToOne, @OneToMany / @ManyToOne and @ManyToMany
- You may find other solutions online with different approaches
- In this course,
 - Treat the examples as a general guide
 - Adapt when your application requirements and domain needs differ

JPA / Hibernate One-to-One

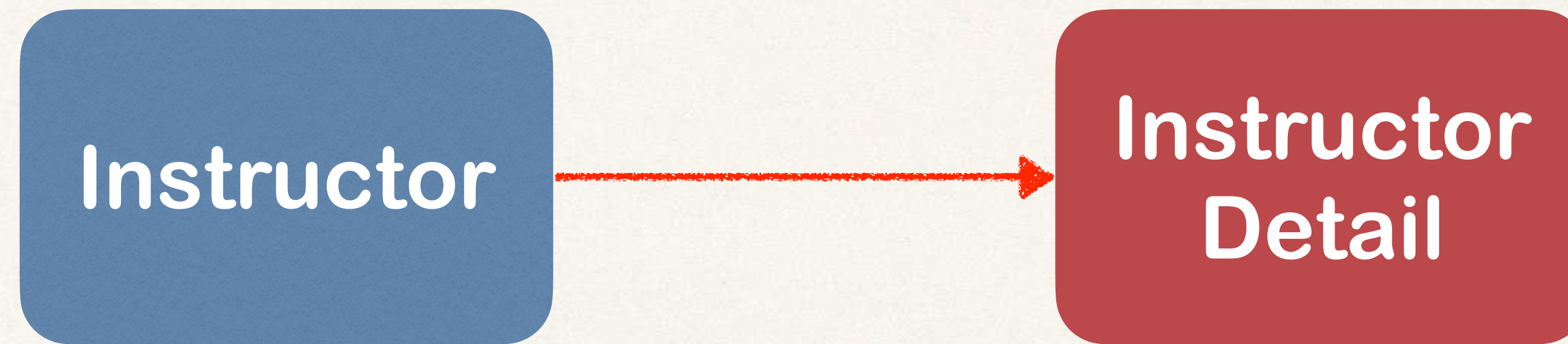


One-to-One Mapping

- An instructor can have an “instructor detail” entity
- Similar to an “instructor profile”



Uni-Directional



Development Process: One-to-One

1. Prep Work - Define database tables
2. Create InstructorDetail class
3. Create Instructor class
4. Create Main App

Step-By-Step

table: instructor_detail

File: create-db.sql

```
CREATE TABLE `instructor_detail` (  
  `id`          int(11)    NOT NULL AUTO_INCREMENT,  
  `youtube_channel` varchar(128) DEFAULT NULL,  
  `hobby`       varchar(45)  DEFAULT NULL,  
  
);  
  
...
```

instructor_detail	
💡	id INT(11)
🔹	youtube_channel VARCHAR(128)
🔹	hobby VARCHAR(45)
Indexes	

table: instructor_detail

File: create-db.sql

```
CREATE TABLE `instructor_detail` (  
  
  `id`          int(11)    NOT NULL AUTO_INCREMENT,  
  `youtube_channel` varchar(128) DEFAULT NULL,  
  `hobby`       varchar(45)  DEFAULT NULL,  
  
  PRIMARY KEY (`id`)  
);  
  
...
```

instructor_detail	
💡	id INT(11)
🔹	youtube_channel VARCHAR(128)
🔹	hobby VARCHAR(45)
Indexes	

table: instructor

File: create-db.sql

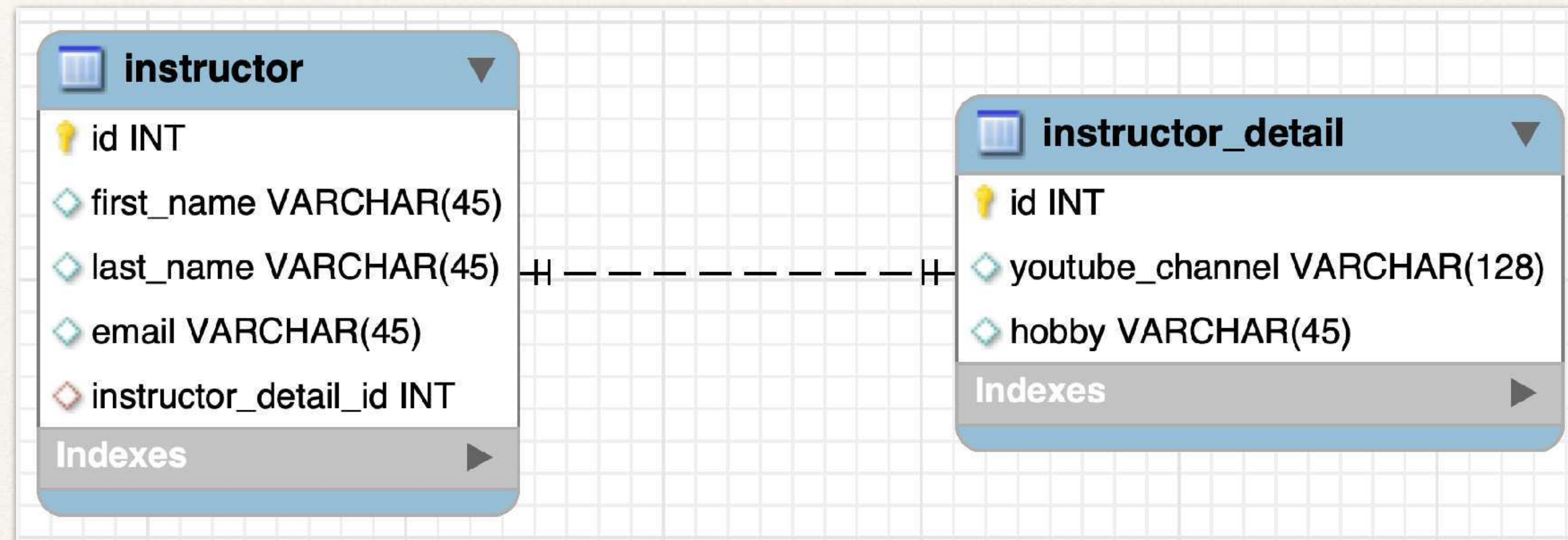
...

```
CREATE TABLE `instructor` (  
  
  `id`      int(11) NOT NULL AUTO_INCREMENT,  
  `first_name` varchar(45) DEFAULT NULL,  
  `last_name` varchar(45) DEFAULT NULL,  
  `email`    varchar(45) DEFAULT NULL,  
  `instructor_detail_id` int(11) DEFAULT NULL,  
  
  PRIMARY KEY (`id`)  
  ...  
);
```

instructor	
💡	id INT(11)
◇	first_name VARCHAR(45)
◇	last_name VARCHAR(45)
◇	email VARCHAR(45)
◇	instructor_detail_id INT(11)
Indexes	

Foreign Key

- Link tables together
- A field in one table that refers to primary key in another table



Foreign Key Example

Foreign key column

Table: instructor

id	first_name	last_name	instructor_detail_id
1	Chad	Darby	100
2	Madhu	Patel	200

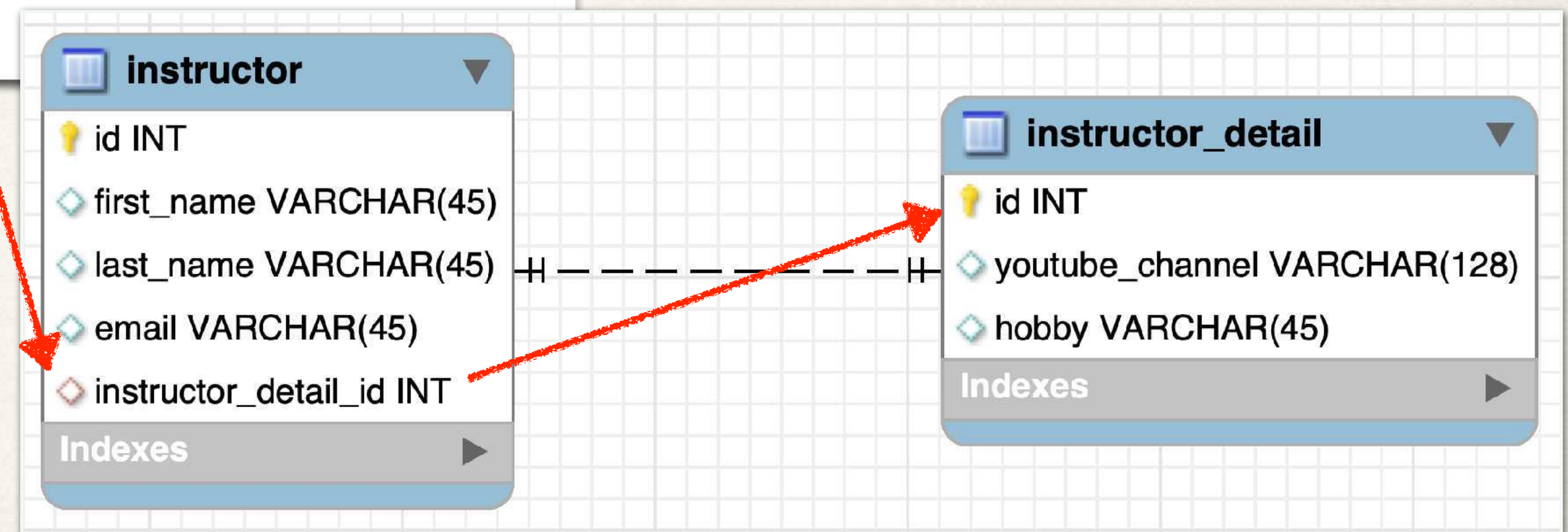
Table: instructor_detail

id	youtube_channel	hobby
100	www.luv2code.com/youtube	Luv 2 Code!!!
200	www.youtube.com	Guitar

Defining Foreign Key

File: create-db.sql

```
...  
CREATE TABLE `instructor` (  
  ...  
  
  CONSTRAINT `FK_DETAIL` FOREIGN KEY (`instructor_detail_id`)  
  REFERENCES `instructor_detail` (`id`)  
  
);
```



More on Foreign Key

- Main purpose is to preserve relationship between tables
 - Referential Integrity
- Prevents operations that would destroy relationship
- Ensures only valid data is inserted into the foreign key column
 - Can only contain valid reference to primary key in other table

Development Process: One-to-One

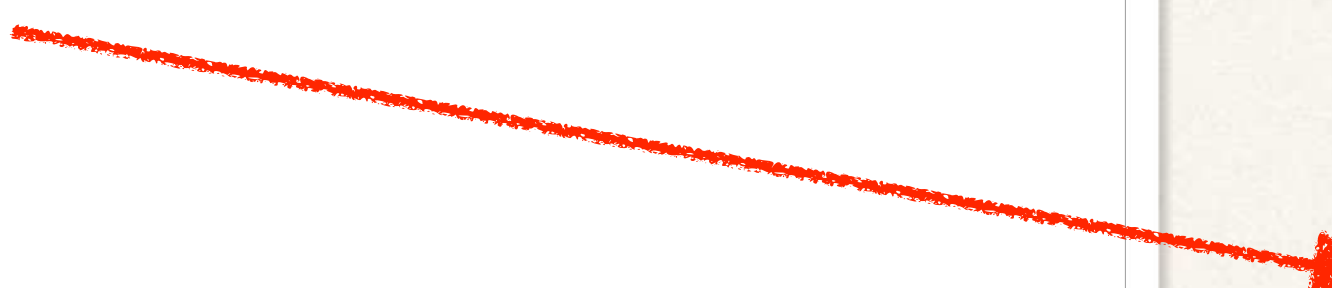
Step-By-Step

1. Prep Work - Define database tables
2. Create InstructorDetail class
3. Create Instructor class
4. Create Main App

Step 2: Create InstructorDetail class

```
@Entity
@Table(name="instructor_detail")
public class InstructorDetail {

}
```



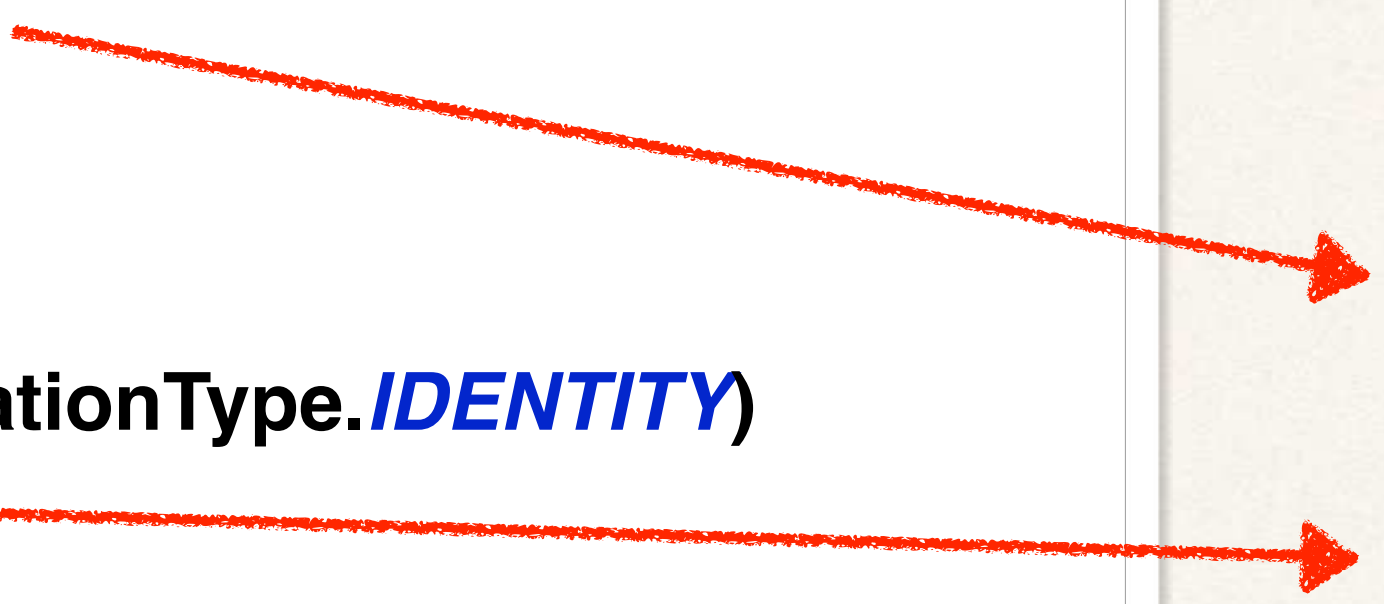
instructor_detail	
id	INT(11)
youtube_channel	VARCHAR(128)
hobby	VARCHAR(45)
Indexes	

Step 2: Create InstructorDetail class

```
@Entity
@Table(name="instructor_detail")
public class InstructorDetail {

    @Id
    @GeneratedValue(strategy=GenerationType.IDENTITY)
    @Column(name="id")
    private int id;

}
```



instructor_detail	
id	INT(11)
youtube_channel	VARCHAR(128)
hobby	VARCHAR(45)
Indexes	

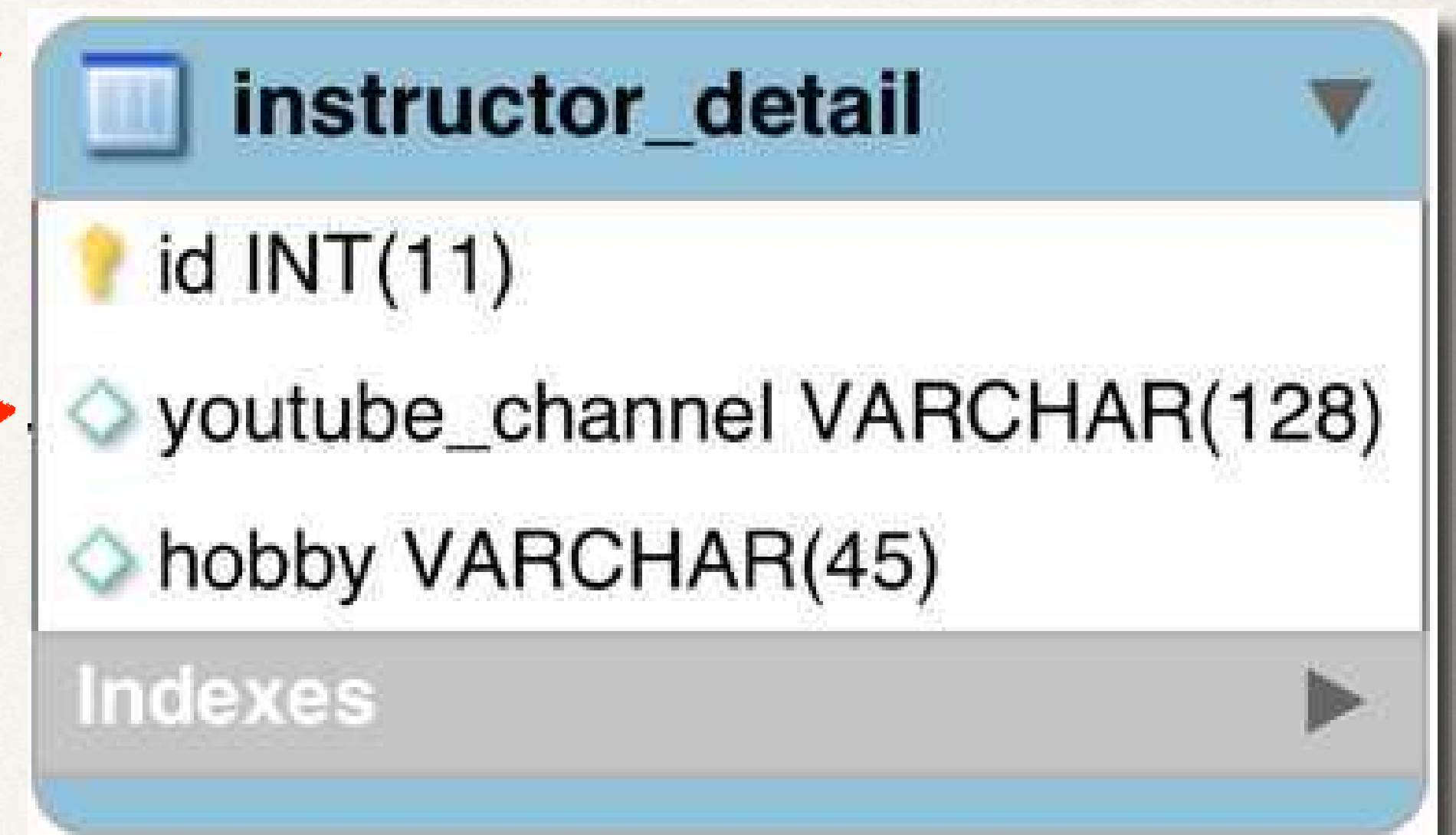
Step 2: Create InstructorDetail class

```
@Entity
@Table(name="instructor_detail")
public class InstructorDetail {

    @Id
    @GeneratedValue(strategy=GenerationType.IDENTITY)
    @Column(name="id")
    private int id;

    @Column(name="youtube_channel")
    private String youtubeChannel;

}
```



Step 2: Create InstructorDetail class

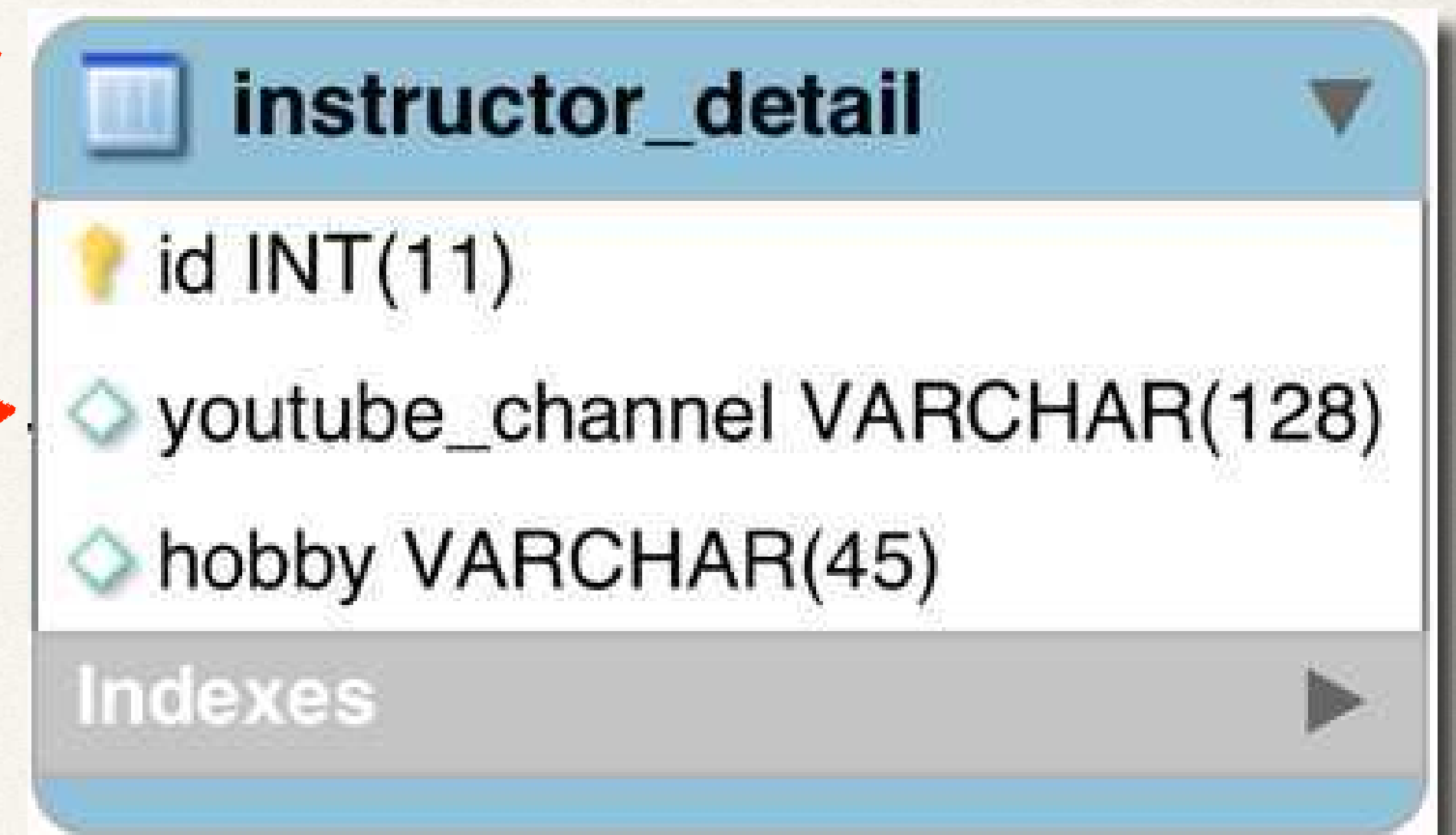
```
@Entity
@Table(name="instructor_detail")
public class InstructorDetail {

    @Id
    @GeneratedValue(strategy=GenerationType.IDENTITY)
    @Column(name="id")
    private int id;

    @Column(name="youtube_channel")
    private String youtubeChannel;

    @Column(name="hobby")
    private String hobby;

}
```



Step 2: Create InstructorDetail class

```
@Entity
@Table(name="instructor_detail")
public class InstructorDetail {

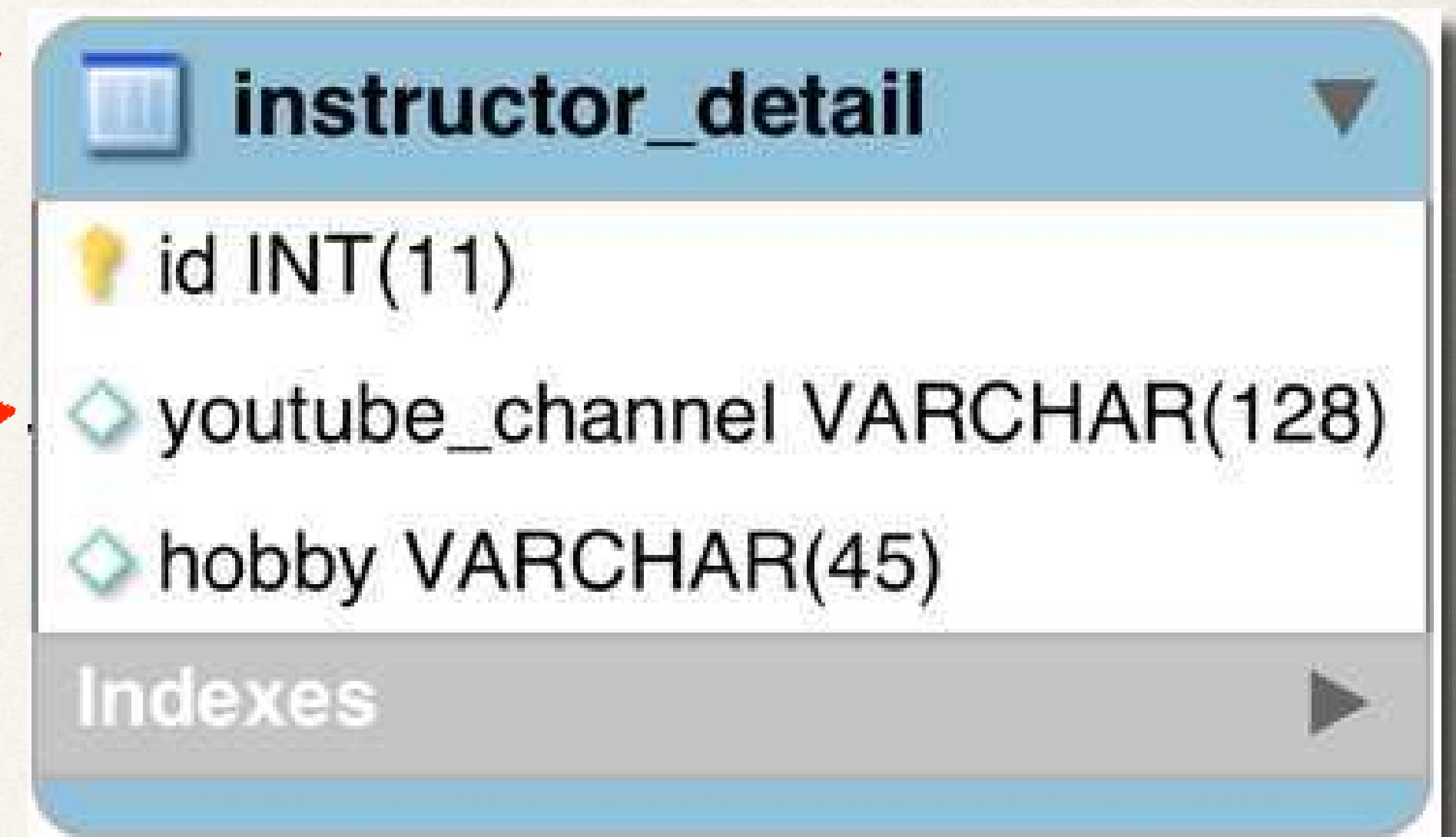
    @Id
    @GeneratedValue(strategy=GenerationType.IDENTITY)
    @Column(name="id")
    private int id;

    @Column(name="youtube_channel")
    private String youtubeChannel;

    @Column(name="hobby")
    private String hobby;

    // constructors

    // getters / setters
}
```



Step 3: Create Instructor class

```
@Entity
@Table(name="instructor")
public class Instructor {

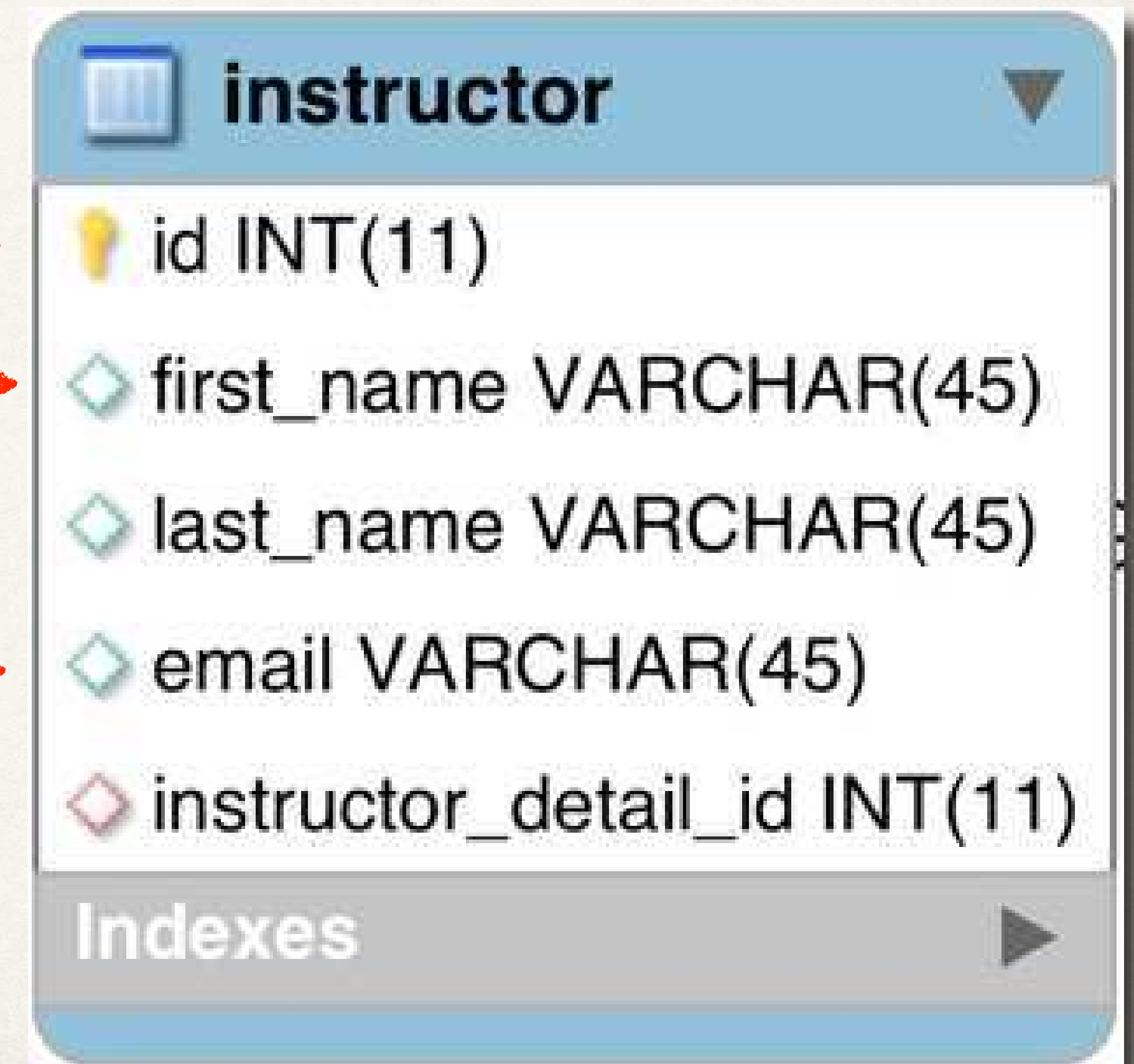
    @Id
    @GeneratedValue(strategy=GenerationType.IDENTITY)
    @Column(name="id")
    private int id;

    @Column(name="first_name")
    private String firstName;

    @Column(name="last_name")
    private String lastName;

    @Column(name="email")
    private String email;

    ...
    // constructors, getters / setters
}
```



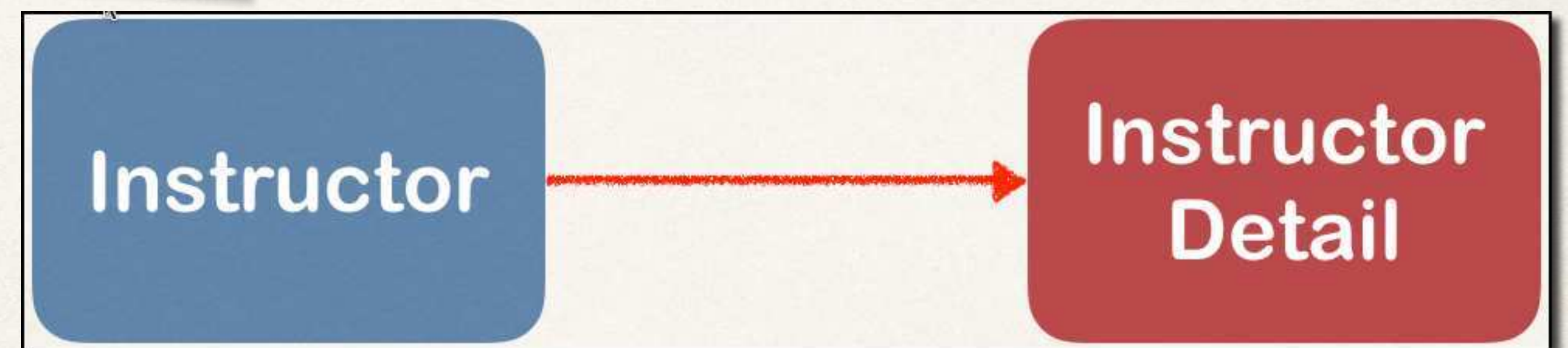
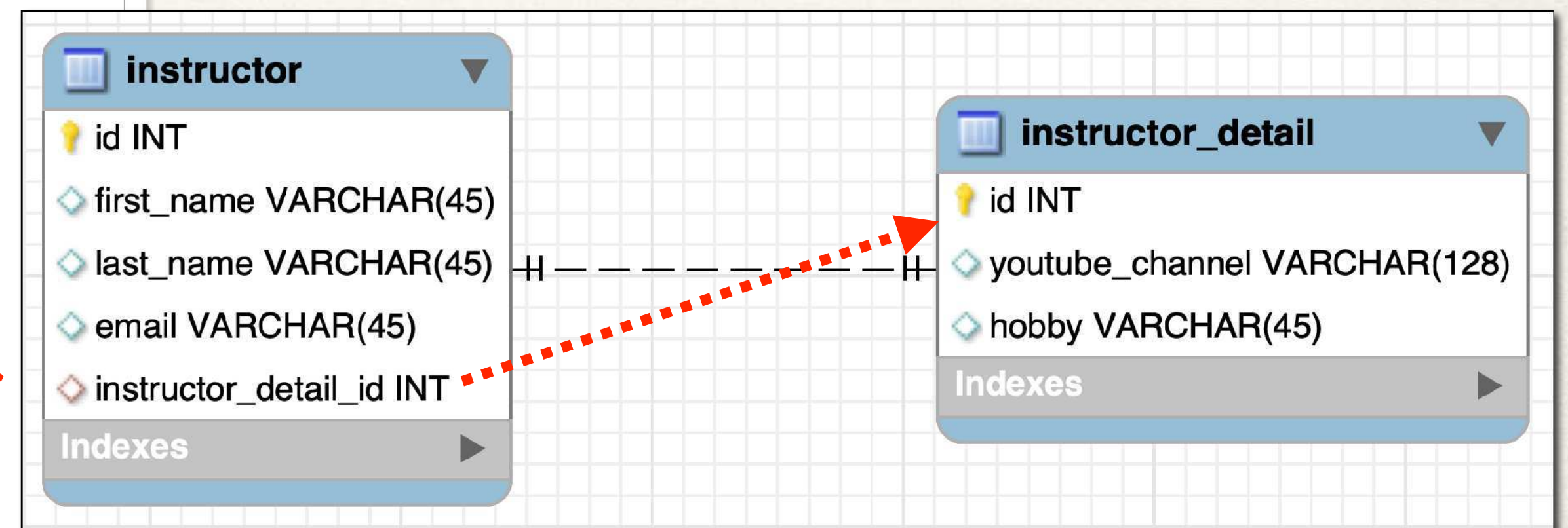
Step 3: Create Instructor class - @OneToOne

```
@Entity
@Table(name="instructor")
public class Instructor {
```

```
    ...

    @OneToOne
    @JoinColumn(name="instructor_detail_id")
    private InstructorDetail instructorDetail;
```

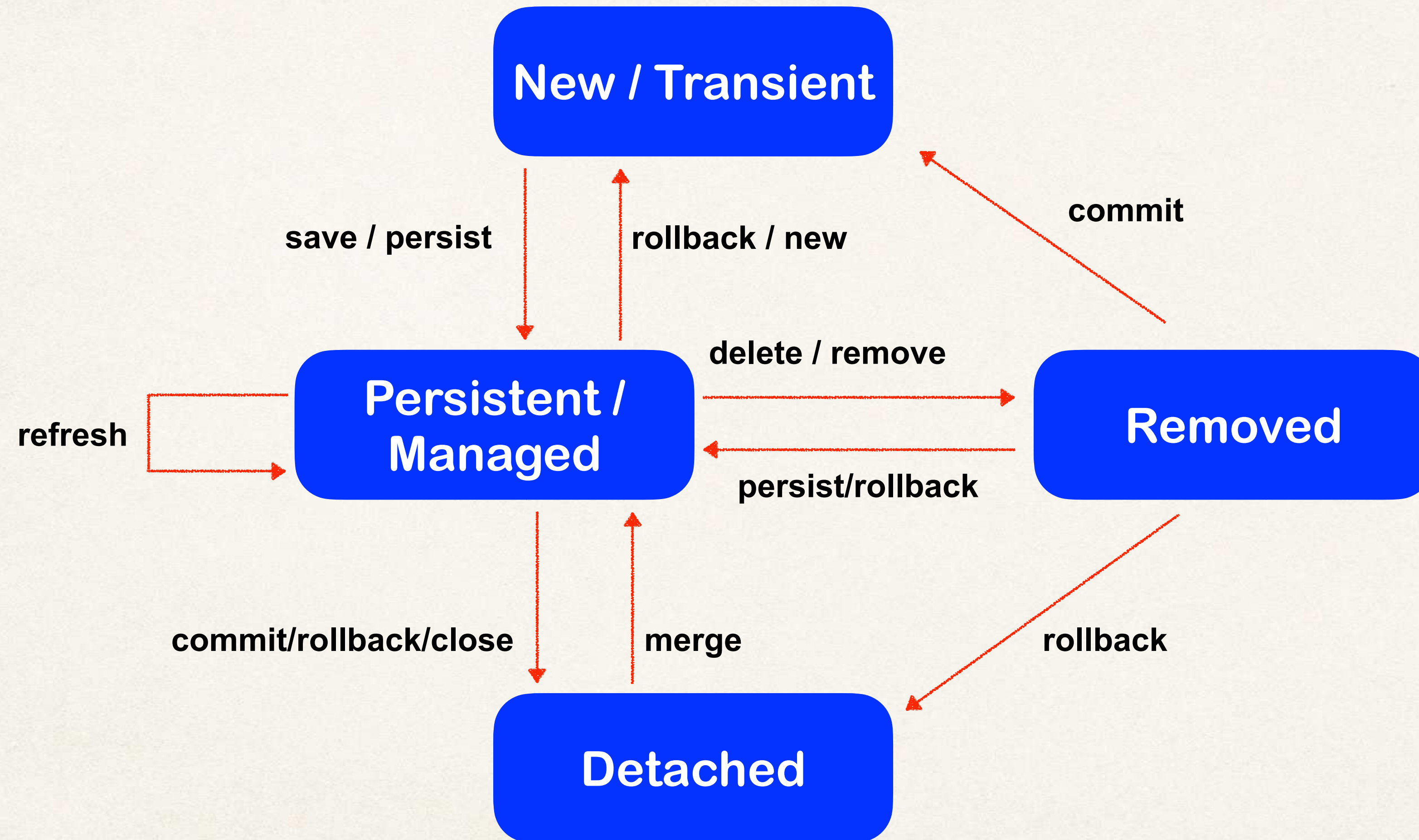
```
    ...
    // constructors, getters / setters
}
```



Entity Lifecycle

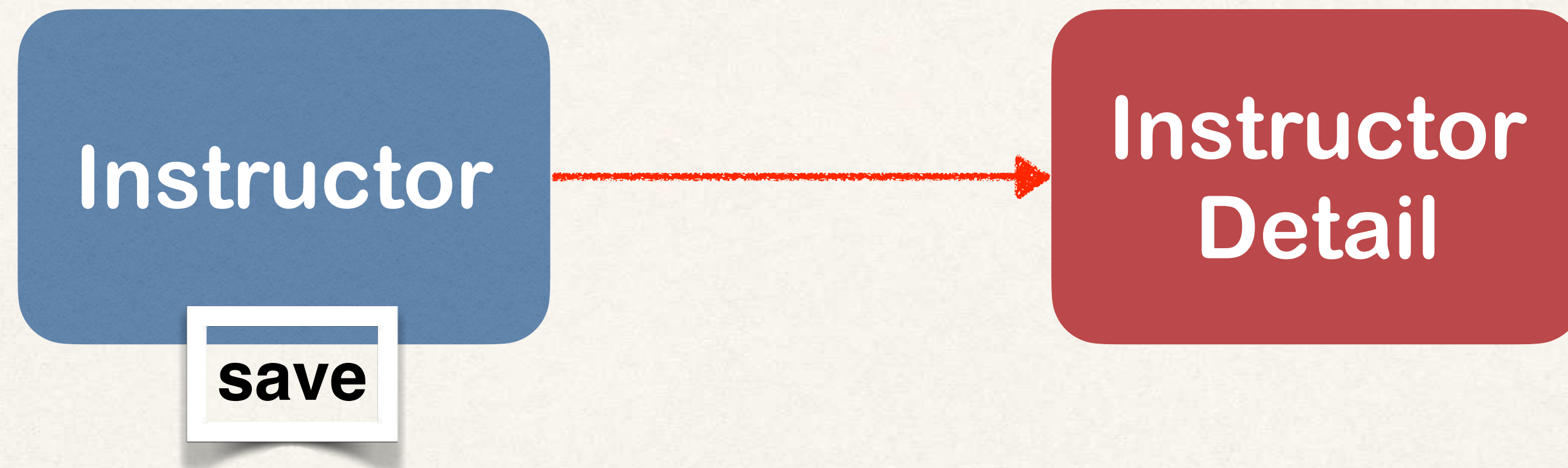
Operations	Description
Detach	If entity is detached, it is not associated with a Hibernate session
Merge	If instance is detached from session, then merge will reattach to session
Persist	Transitions new instances to managed state. Next flush / commit will save in db.
Remove	Transitions managed entity to be removed. Next flush / commit will delete from db.
Refresh	Reload / synch object with data from db. Prevents stale data

Entity Lifecycle - session method calls



Cascade

- Recall: You can **cascade** operations
- Apply the same operation to related entities



Cascade Delete

Foreign key
column

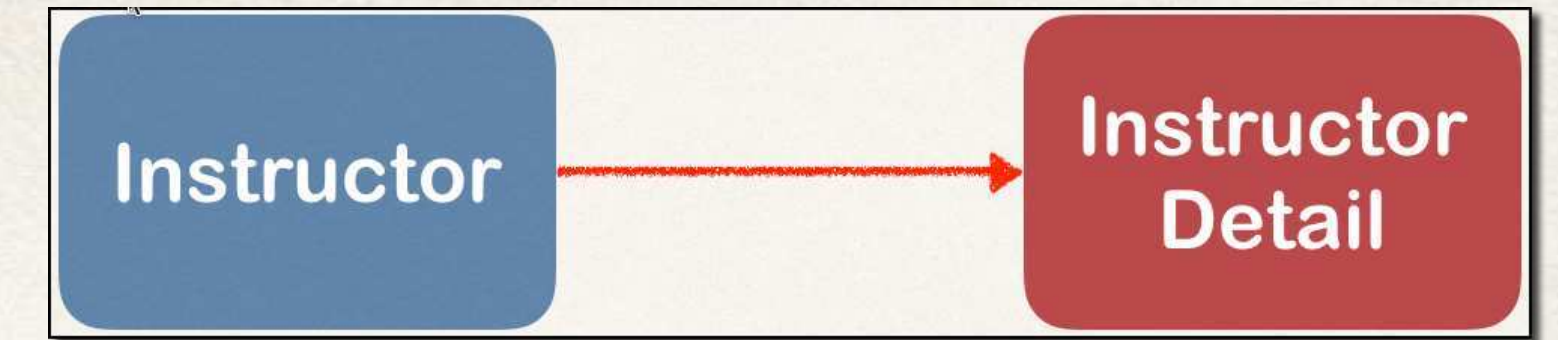
Table: instructor

id	first_name	last_name	instructor_detail_id
1	Chad	Darby	100
2	Madhu	Patel	200

Table: instructor_detail

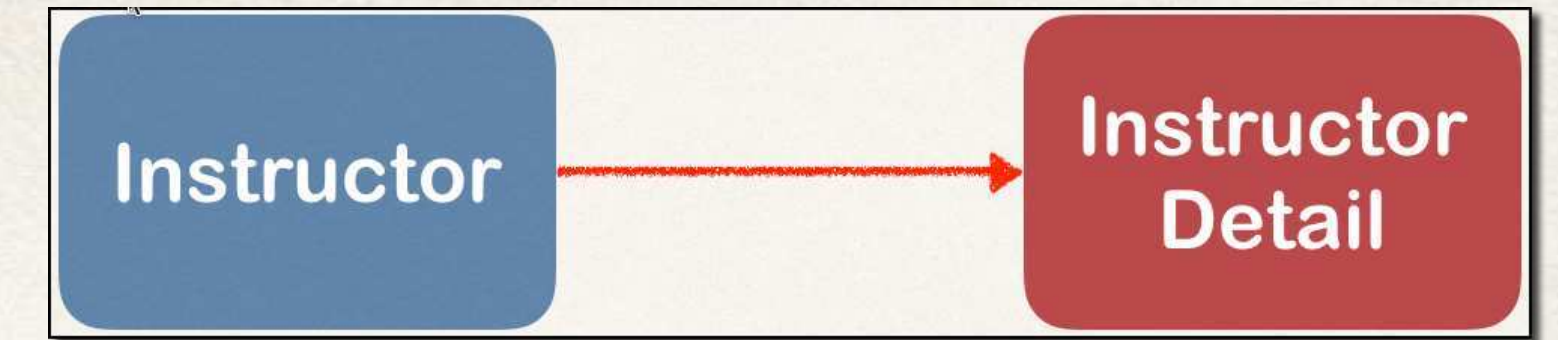
id	youtube_channel	hobby
100	www.luv2code.com/youtube	Luv 2 Code!!!
200	www.youtube.com	Guitar

@OneToOne - Cascade Types



Cascade Type	Description
PERSIST	If entity is persisted / saved, related entity will also be persisted
REMOVE	If entity is removed / deleted, related entity will also be deleted
REFRESH	If entity is refreshed, related entity will also be refreshed
DETACH	If entity is detached (not associated w/ session), then related entity will also be detached
MERGE	If entity is merged, then related entity will also be merged
ALL	All of above cascade types

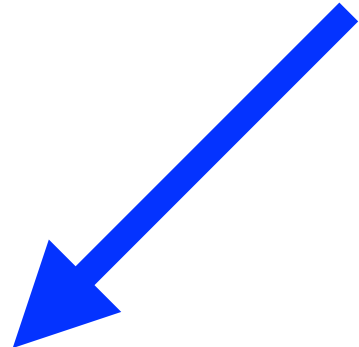
Configure Cascade Type



```
@Entity
@Table(name="instructor")
public class Instructor {
    ...

    @OneToOne(cascade=CascadeType.ALL)
    @JoinColumn(name="instructor_detail_id")
    private InstructorDetail instructorDetail;

    ...
    // constructors, getters / setters
}
```



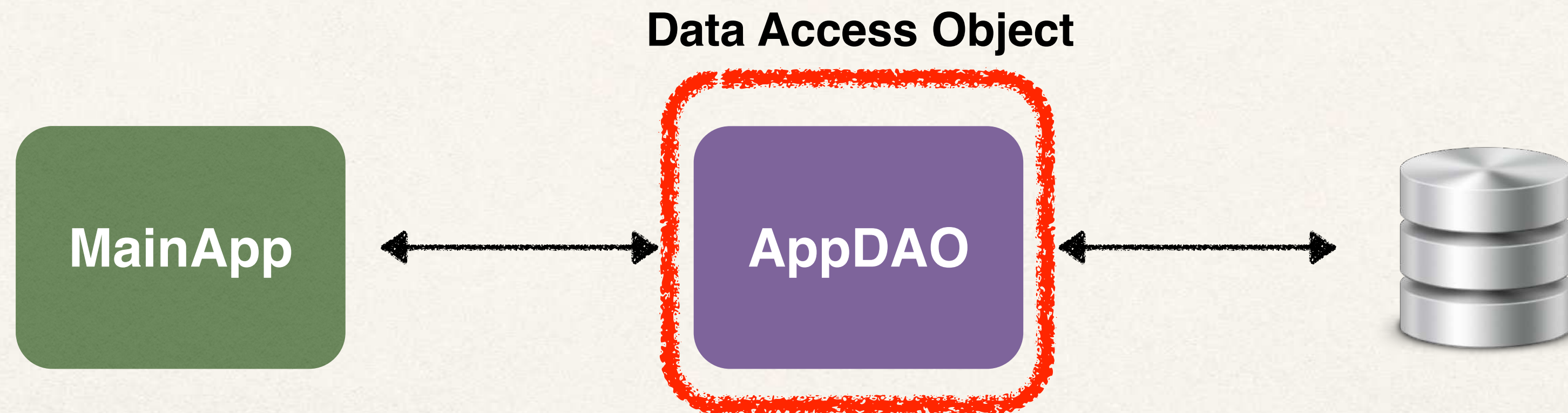
By default, no operations are cascaded.

Configure Multiple Cascade Types

```
@OneToOne(cascade={CascadeType.DETACH,  
                    CascadeType.MERGE,  
                    CascadeType.PERSIST,  
                    CascadeType.REFRESH,  
                    CascadeType.REMOVE})
```


Step 4 - Creating Spring Boot - Command Line App

- We will create a Spring Boot - Command Line App
- This will allow us to focus on JPA / Hibernate
- Leverage our DAO pattern as in previous videos



Define DAO interface

```
import com.luv2code.cruddemo.entity.Instructor;  
  
public interface AppDAO {  
    → void save(Instructor theInstructor);  
}
```

Instructor

Instructor
Detail

Define DAO implementation

```
import com.luv2code.cruddemo.entity.Instructor;
import jakarta.persistence.EntityManager;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Repository;
import org.springframework.transaction.annotation.Transactional;

@Repository
public class AppDAOImpl implements AppDAO {

    // define field for entity manager
    private EntityManager entityManager;

    // inject entity manager using constructor injection
    @Autowired
    public AppDAOImpl(EntityManager entityManager) {
        this.entityManager = entityManager;
    }

    @Override
    @Transactional
    public void save(Instructor theInstructor) {
        entityManager.persist(theInstructor);
    }

}
```

Inject the Entity Manager

Save the
Java object

Instructor

Instructor
Detail

Define DAO implementation

```
import com.luv2code.cruddemo.entity.Instructor;
import jakarta.persistence.EntityManager;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Repository;
import org.springframework.transaction.annotation.Transactional;

@Repository
public class AppDAOImpl implements AppDAO {

    // define field for entity manager
    private EntityManager entityManager;

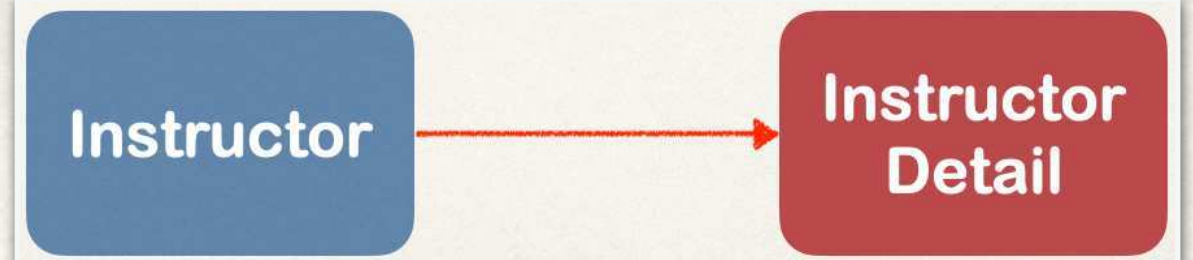
    // inject entity manager using constructor injection
    @Autowired
    public AppDAOImpl(EntityManager entityManager) {
        this.entityManager = entityManager;
    }

    @Override
    @Transactional
    public void save(Instructor theInstructor) {
        entityManager.persist(theInstructor);
    }
}
```

This will ALSO save the details object
Because of CascadeType.ALL



Update main app



Inject the AppDAO

```
@SpringBootApplication
public class MainApplication {

    public static void main(String[] args) {
        SpringApplication.run(MainApplication.class, args);
    }

    @Bean
    public CommandLineRunner commandLineRunner(AppDAO appDAO) {
        return runner -> {
            createInstructor(appDAO);
        }
    }

    ...
}
```

```
private void createInstructor(AppDAO appDAO) {

    // create the instructor
    Instructor tempInstructor =
        new Instructor("Chad", "Darby", "darby@luv2code.com");

    // create the instructor detail
    InstructorDetail tempInstructorDetail =
        new InstructorDetail(
            "http://www.luv2code.com/youtube",
            "Luv 2 code!!!");

    // associate the objects
    tempInstructor.setInstructorDetail(tempInstructorDetail);

    // save the instructor
    System.out.println("Saving instructor: " + tempInstructor);
    appDAO.save(tempInstructor);

    System.out.println("Done!");
}
```

Remember:

This will ALSO save the details object

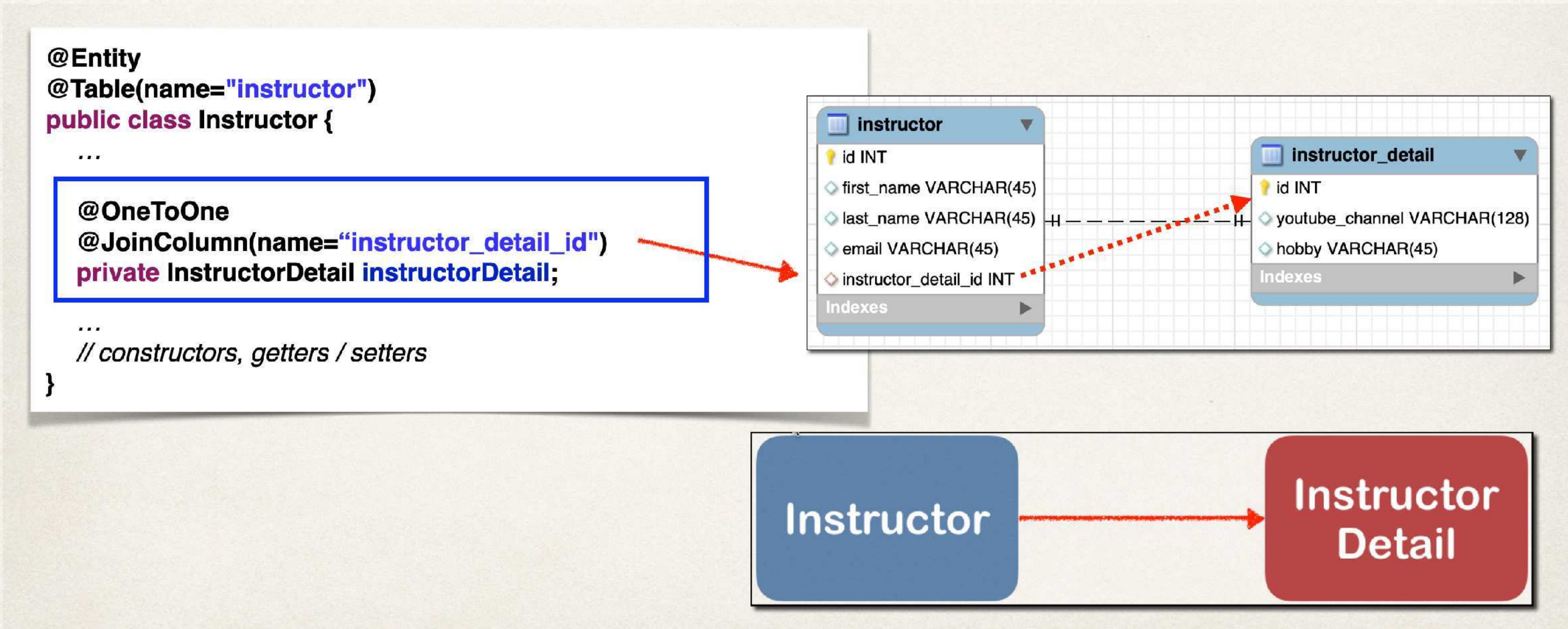
Because of CascadeType.ALL

*In AppDAO, delegated to
entityManager.persist(...)*

Recap: @OneToOne

Defines physical mapping of the relationship

Mapping	Foreign Key location in database table	Owning Side
@OneToOne - Unidirectional Instructor -> InstructorDetails	instructor table	Instructor @OneToOne + @JoinColumn



Use this course's examples as a general guide.

Adapt them to your application's requirements and domain needs.

JPA / Hibernate

One-to-One: Find an entity



Define DAO implementation

```
@Repository
public class AppDAOImpl implements AppDAO {

    ...

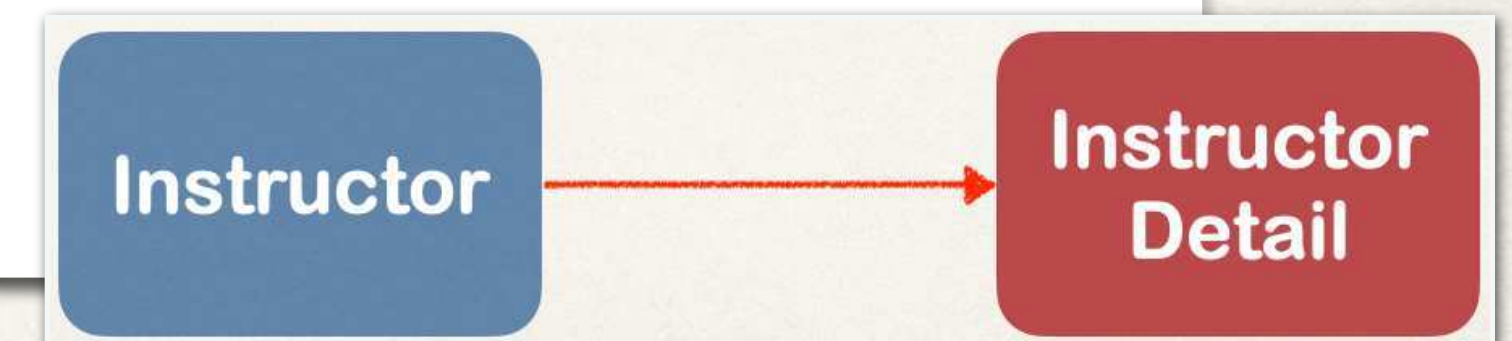
    @Override
    public Instructor findInstructorById(int theId) {
        return entityManager.find(Instructor.class, theId);
    }

}
```

We'll add supporting code in the video:
interface, main app

This will ALSO retrieve the instructor details object

Because of default behavior of @OneToOne
fetch type is eager ... more on fetch types later



JPA / Hibernate

One-to-One: Delete an entity



Define DAO implementation

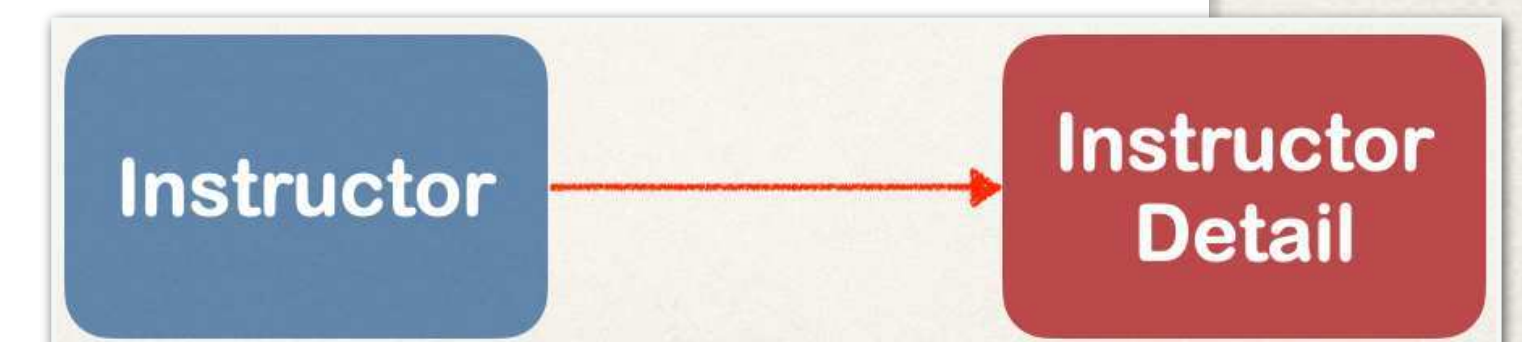
```
@Repository
public class AppDAOImpl implements AppDAO {
    ...

    @Override
    @Transactional
    public void deleteInstructorById(int theId) {

        // retrieve the instructor
        Instructor tempInstructor = entityManager.find(Instructor.class, theId);

        // delete the instructor
        entityManager.remove(tempInstructor);
    }
}
```

We'll add supporting code in the video:
interface, main app



This will ALSO delete the instructor details object
Because of CascadeType.ALL

JPA / Hibernate

One-to-One: Bi-Directional



One-to-One Mapping

- We currently have a uni-directional mapping



New Use Case

- If we load an InstructorDetail
 - Then we'd like to get the associated Instructor
- Can't do this with current uni-directional relationship :-)



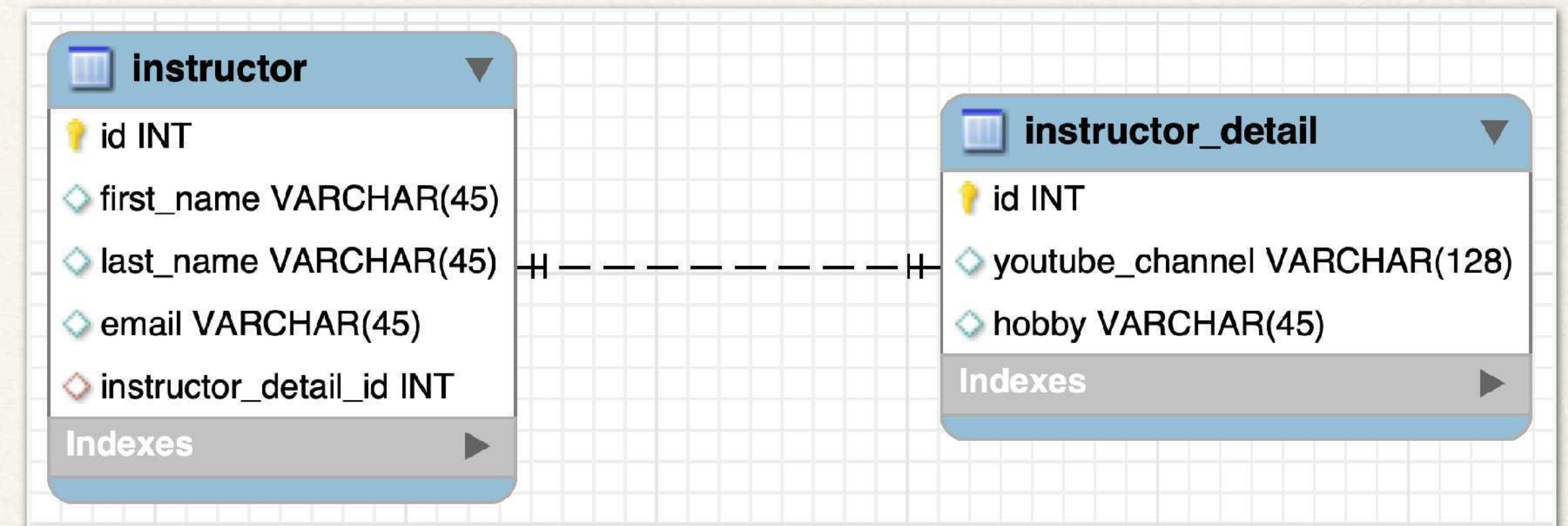
Bi-Directional

- Bi-Directional relationship is the solution
- We can start with InstructorDetail and make it back to the Instructor



Bi-Directional - The Good News

- To use Bi-Directional, we can keep the existing database schema
 - No changes required to database
- Simply update the Java code



Development Process: One-to-One (Bi-Directional)

Step-By-Step

1. Make updates to InstructorDetail class:
 1. Add new field to reference Instructor
 2. Add getter / setter methods for Instructor
 3. Add @OneToOne annotation
2. Create Main App

Step 1.1: Add new field to reference Instructor

```
@Entity
@Table(name="instructor_detail")
public class InstructorDetail {
    ...

    private Instructor instructor;

    ...
}
```


Step 1.2: Add getter/setter methods Instructor

```
@Entity
@Table(name="instructor_detail")
public class InstructorDetail {
```

```
...
```

```
private Instructor instructor;
```

```
public Instructor getInstructor() {
    return instructor;
}
```

```
public void setInstructor(Instructor instructor) {
    this.instructor = instructor;
}
```

```
...
```

```
}
```


Step 1.3: Add @OneToOne annotation

```
@Entity
@Table(name="instructor_detail")
public class InstructorDetail {
    ...
```

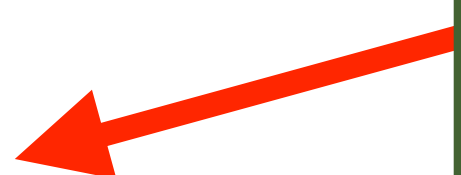
```
@OneToOne(mappedBy="instructorDetail")
private Instructor instructor;
```

```
public Instructor getInstructor() {
    return instructor;
}
```

```
public void setInstructor(Instructor instructor) {
    this.instructor = instructor;
}
...
```

```
}
```

Refers to “instructorDetail” property
in “Instructor” class



More on mappedBy

```
public class InstructorDetail {  
    ...  
  
    @OneToOne(mappedBy="instructorDetail")  
    private Instructor instructor;  
}
```

- **mappedBy** tells Hibernate
 - Look at the instructorDetail property in the Instructor class
 - Use information from the Instructor class @JoinColumn
 - To help find associated instructor



```
public class Instructor {  
    ...  
  
    @OneToOne(cascade=CascadeType.ALL)  
    @JoinColumn(name="instructor_detail_id")  
    private InstructorDetail instructorDetail;  
}
```


Add support for Cascading

```
@Entity
@Table(name="instructor_detail")
public class InstructorDetail {
    ...

    @OneToOne(mappedBy="instructorDetail", cascade=CascadeType.ALL)
    private Instructor instructor;

    public Instructor getInstructor() {
        return instructor;
    }

    public void setInstructor(Instructor instructor) {
        this.instructor = instructor;
    }
    ...
}
```

Cascade all operations
to the associated Instructor



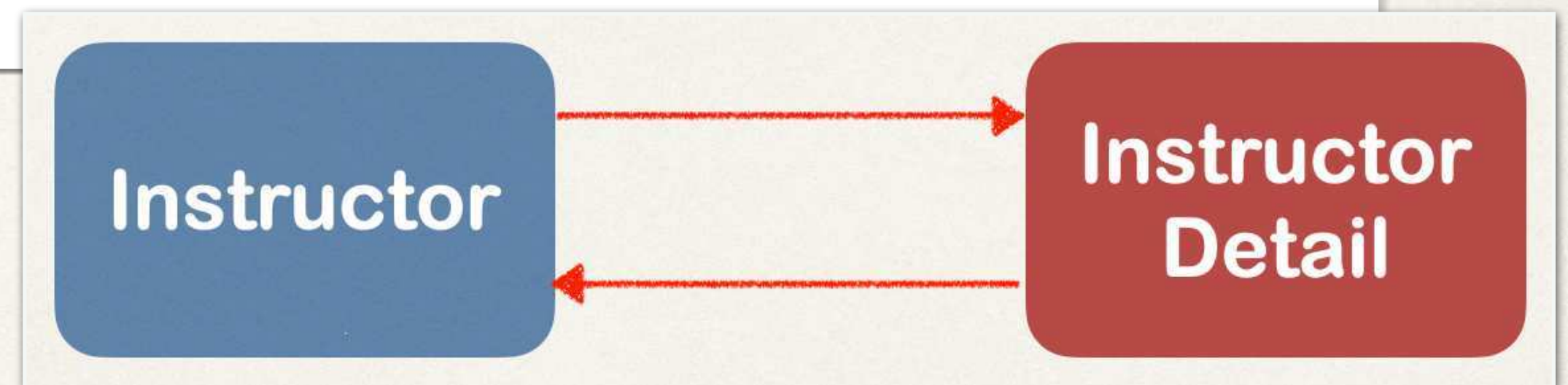
Define DAO interface

```
import com.luv2code.cruddemo.entity.Instructor;
```

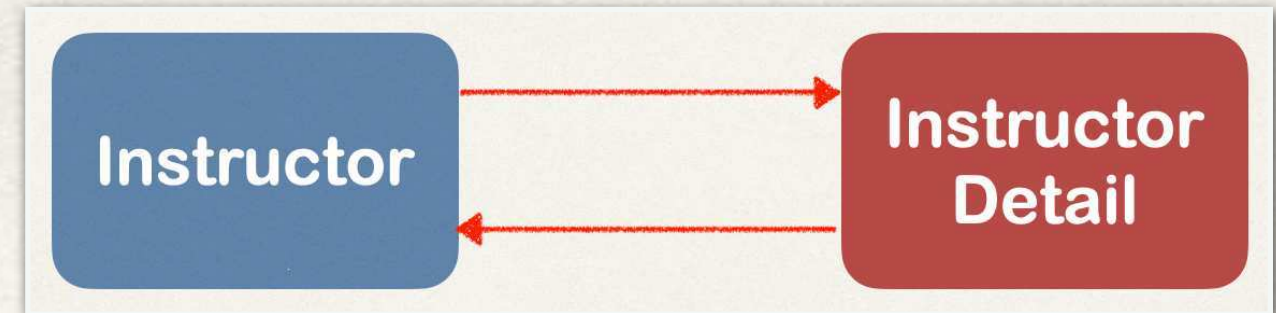
```
public interface AppDAO {
```

```
→ InstructorDetail findInstructorDetailById(int theId);
```

```
}
```



Define DAO implementation



```
import com.luv2code.cruddemo.entity.InstructorDetail;
...

@Repository
public class AppDAOImpl implements AppDAO {

    // define field for entity manager
    private EntityManager entityManager;

    // inject entity manager using constructor
    ...

    @Override
    public InstructorDetail findInstructorDetailById(int theId) {

        return entityManager.find(InstructorDetail.class, theId);

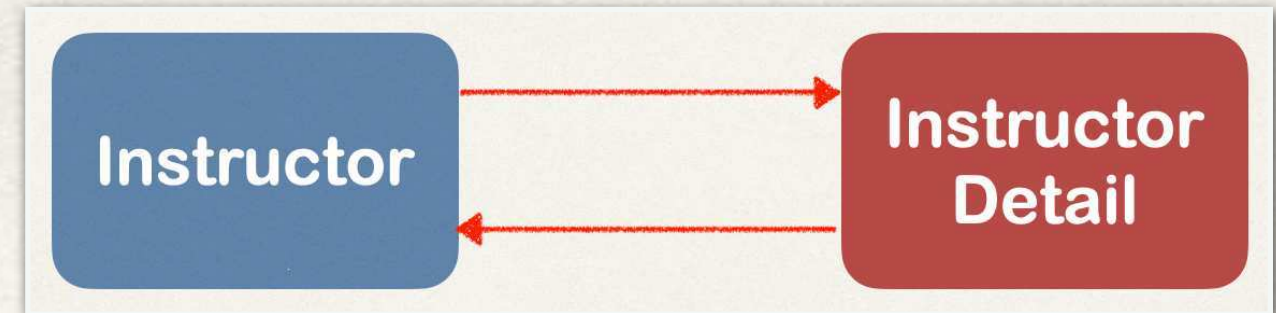
    }

}
```

This will ALSO retrieve the instructor object
Because of default behavior of @OneToOne

Retrieve the
InstructorDetail

Update main app



Inject the AppDAO

```
@SpringBootApplication
public class MainApplication {

    public static void main(String[] args) {
        SpringApplication.run(MainApplication.class, args);
    }

    @Bean
    public CommandLineRunner commandLineRunner(AppDAO appDAO) {
        return runner -> {
            findInstructorDetail(appDAO);
        }
    }

    ...
}
```

```
private void findInstructorDetail(AppDAO appDAO) {

    int theId = 1;
    System.out.println("Finding instructor detail id: " + theId);

    InstructorDetail tempInstructorDetail = appDAO.findInstructorDetailById(theId);

    System.out.println("tempInstructorDetail: " + tempInstructorDetail);
    System.out.println("the associated instructor: " + tempInstructorDetail.getInstructor());

}
}
```


Mappings: @OneToOne

Defines physical mapping of the relationship

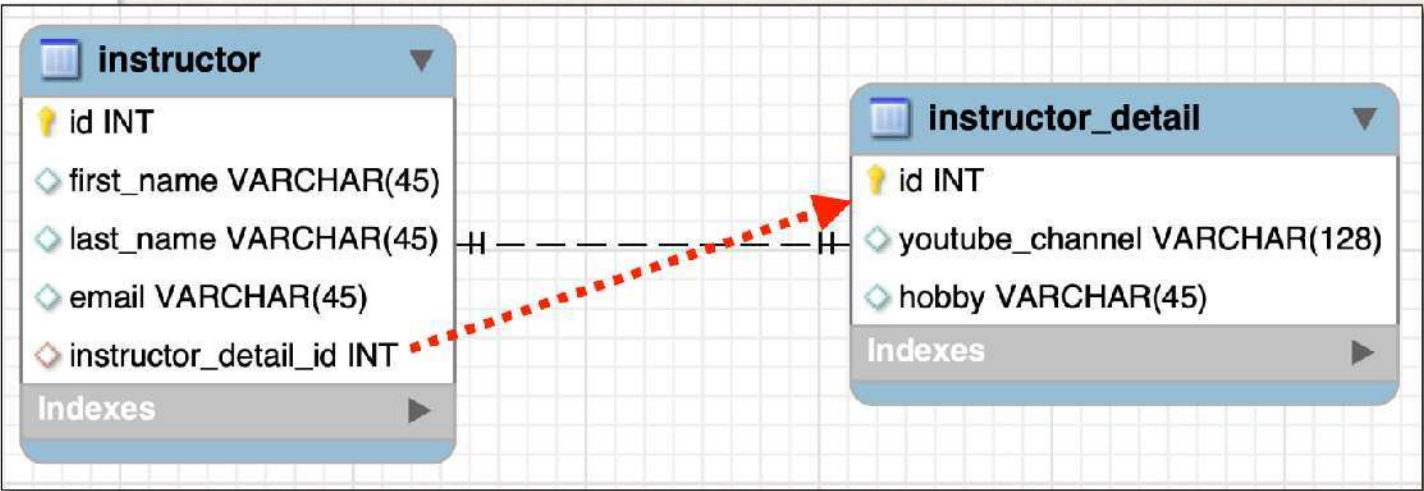
Mirrors the owning side

Mapping	Foreign Key location in database table	Owning Side	Inverse side
@OneToOne - Bidirectional Instructor <-> InstructorDetails	instructor table	Instructor @OneToOne + @JoinColumn	InstructorDetail @OneToOne(mappedBy="instructorDetail")

```
@Entity
@Table(name="instructor")
public class Instructor {
    ...

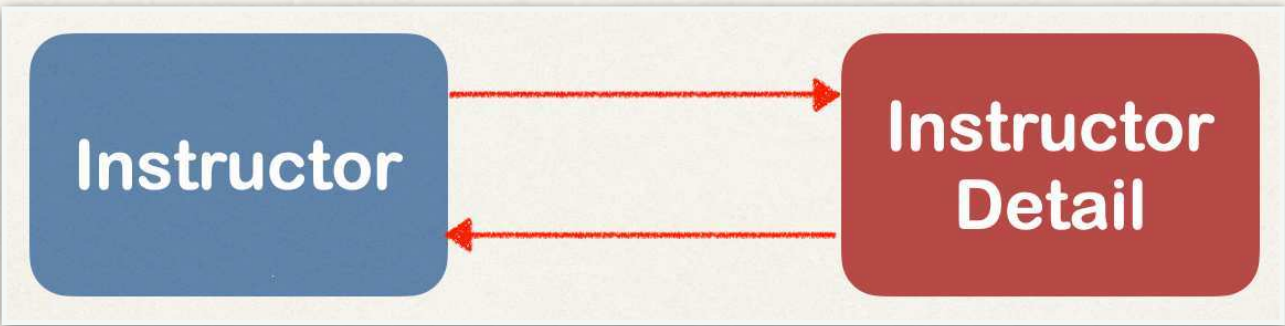
    @OneToOne
    @JoinColumn(name="instructor_detail_id")
    private InstructorDetail instructorDetail;

    ...
    // constructors, getters / setters
}
```



```
@Entity
@Table(name="instructor_detail")
public class InstructorDetail {
    ...

    @OneToOne(mappedBy="instructorDetail")
    private Instructor instructor;
```



Use this course's examples as a general guide.

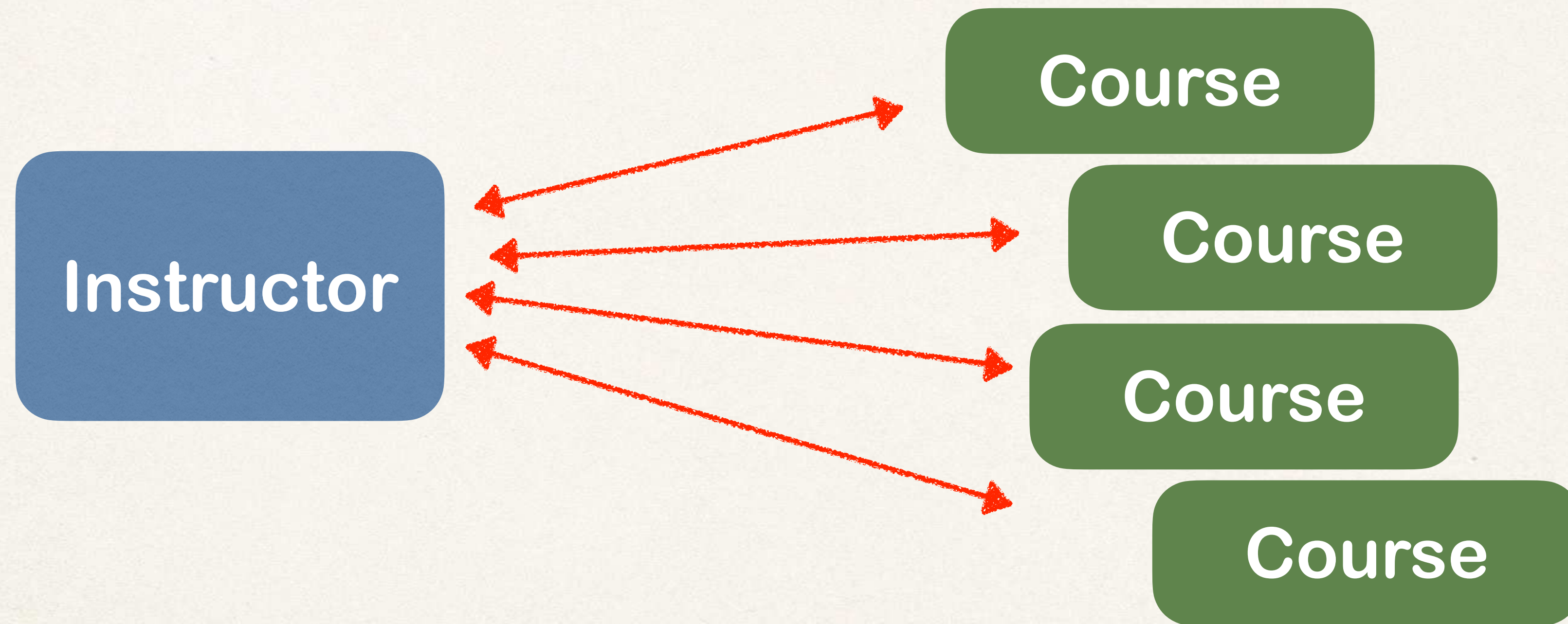
Adapt them to your application's requirements and domain needs.

JPA / Hibernate One-to-Many



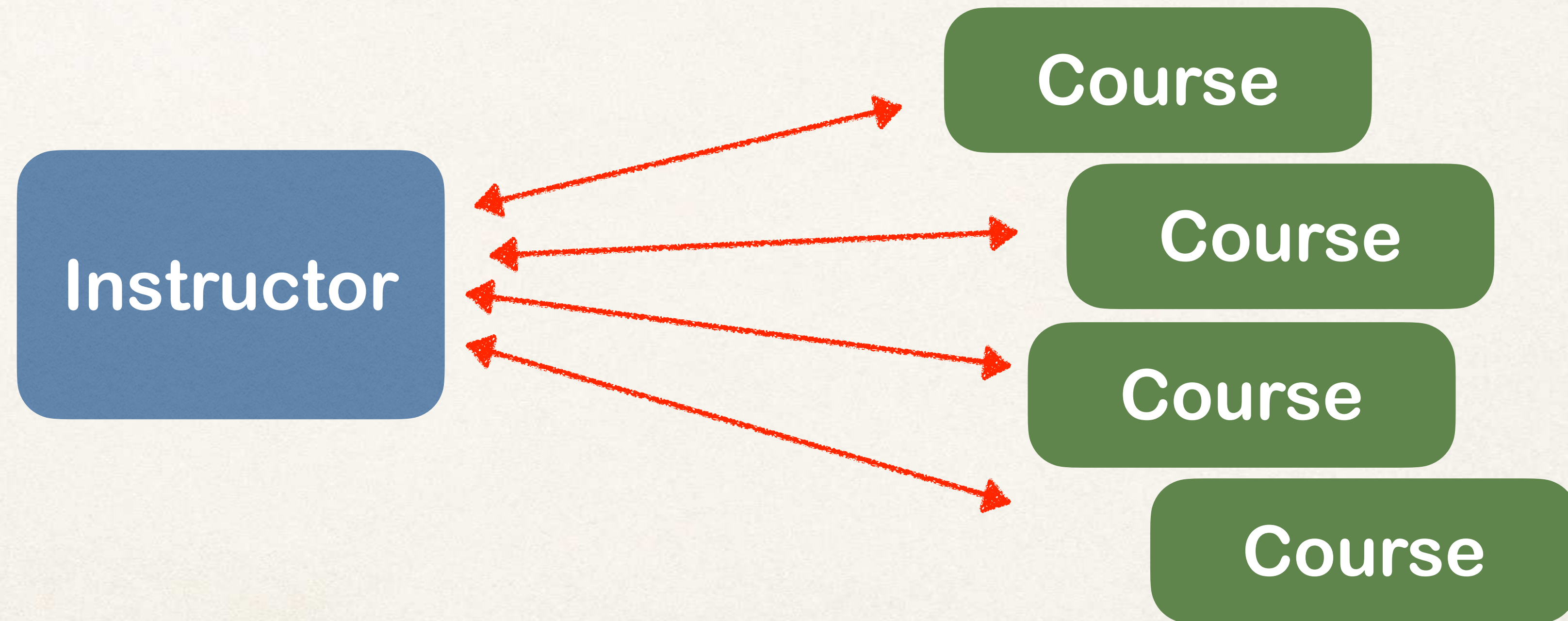
One-to-Many Mapping

- An instructor can have many courses
- Bi-directional



Many-to-One Mapping

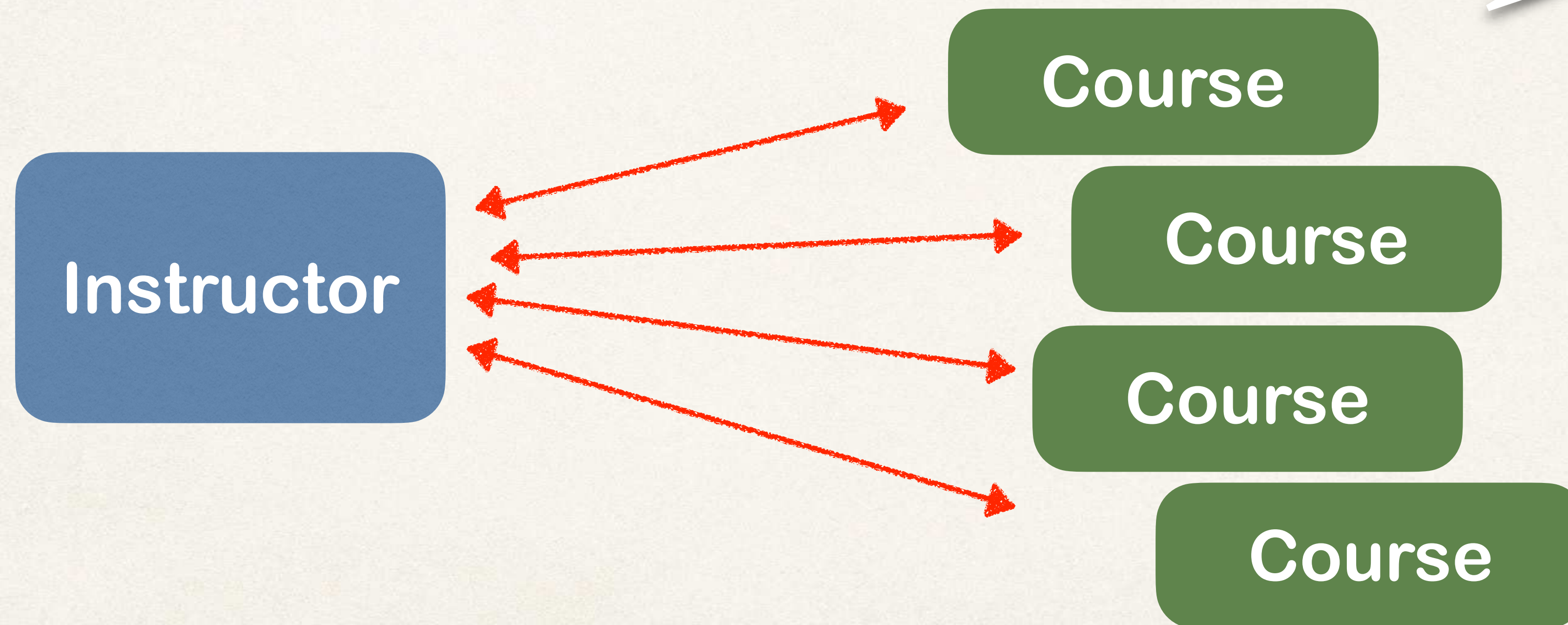
- Many courses can have one instructor
- Inverse / opposite of One-to-Many



Real-World Project Requirement

- If you delete an instructor, DO NOT delete the courses
- If you delete a course, DO NOT delete the instructor

**Do not apply
cascading deletes!**



Development Process: One-to-Many

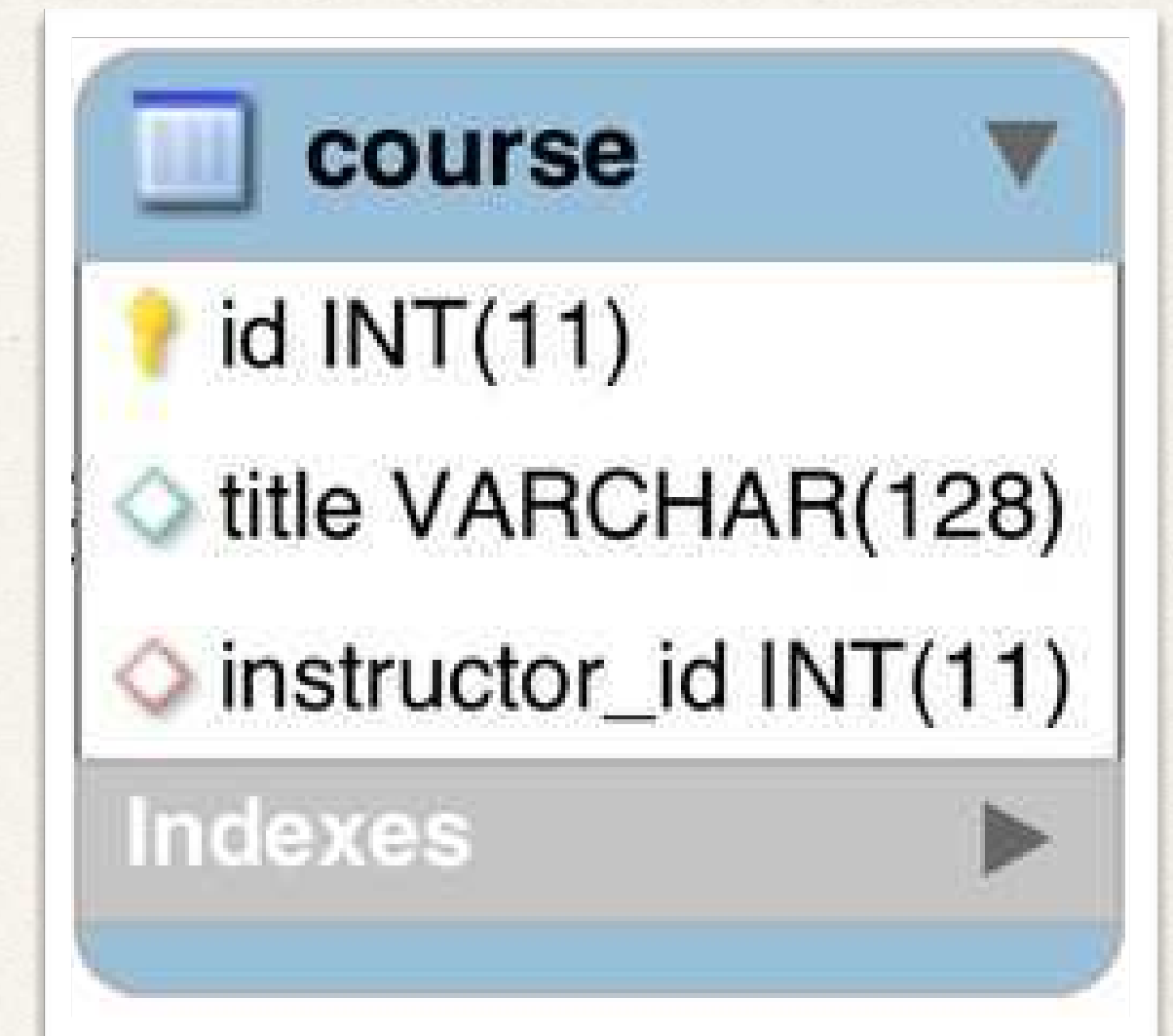
1. Prep Work - Define database tables
2. Create Course class
3. Update Instructor class
4. Create Main App

Step-By-Step

table: course

File: create-db.sql

```
CREATE TABLE `course` (  
  
  `id` int(11) NOT NULL AUTO_INCREMENT,  
  `title` varchar(128) DEFAULT NULL,  
  `instructor_id` int(11) DEFAULT NULL,  
  
  PRIMARY KEY (`id`),  
  
  UNIQUE KEY `TITLE_UNIQUE` (`title`),  
  ...  
);
```



Prevent duplicate course titles

table: course - foreign key

File: create-db.sql

```
CREATE TABLE `course` (  
  ...  
  KEY `FK_INSTRUCTOR_idx` (`instructor_id`),  
  CONSTRAINT `FK_INSTRUCTOR`  
  FOREIGN KEY (`instructor_id`)  
  REFERENCES `instructor` (`id`)  
  ...  
);
```

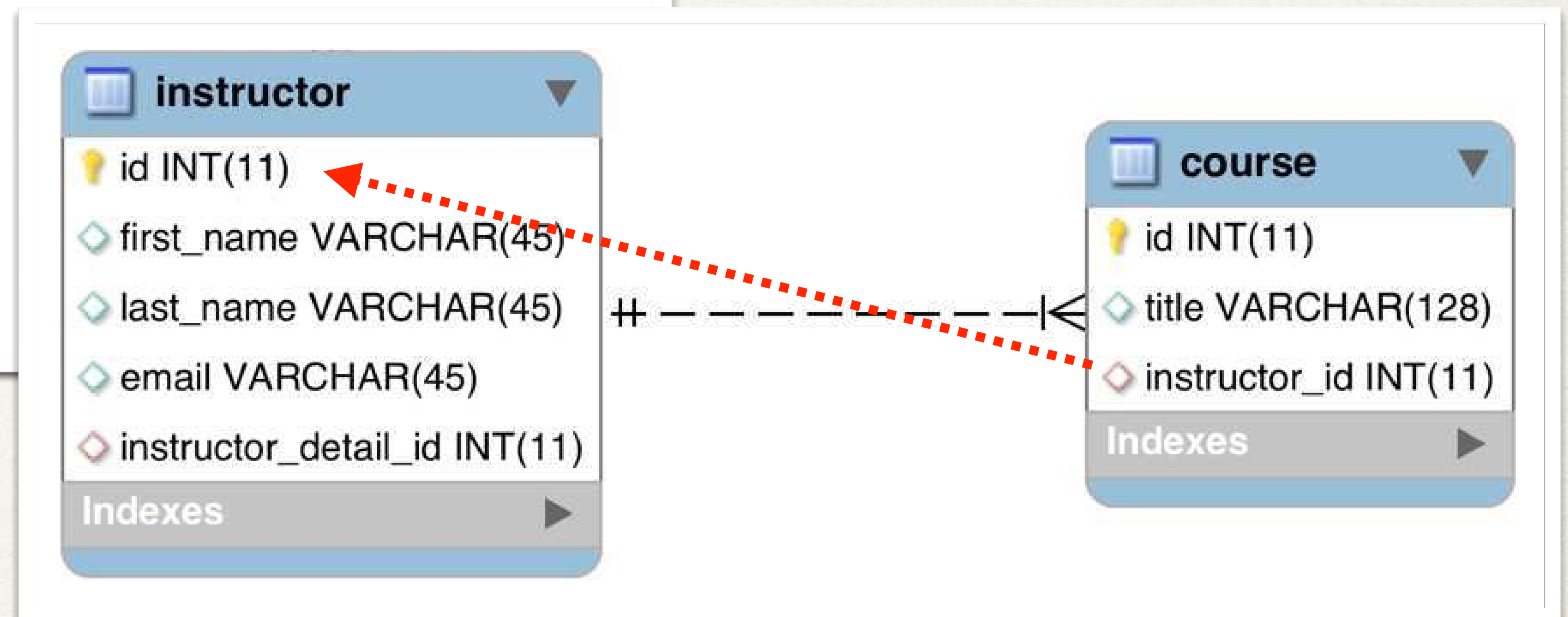
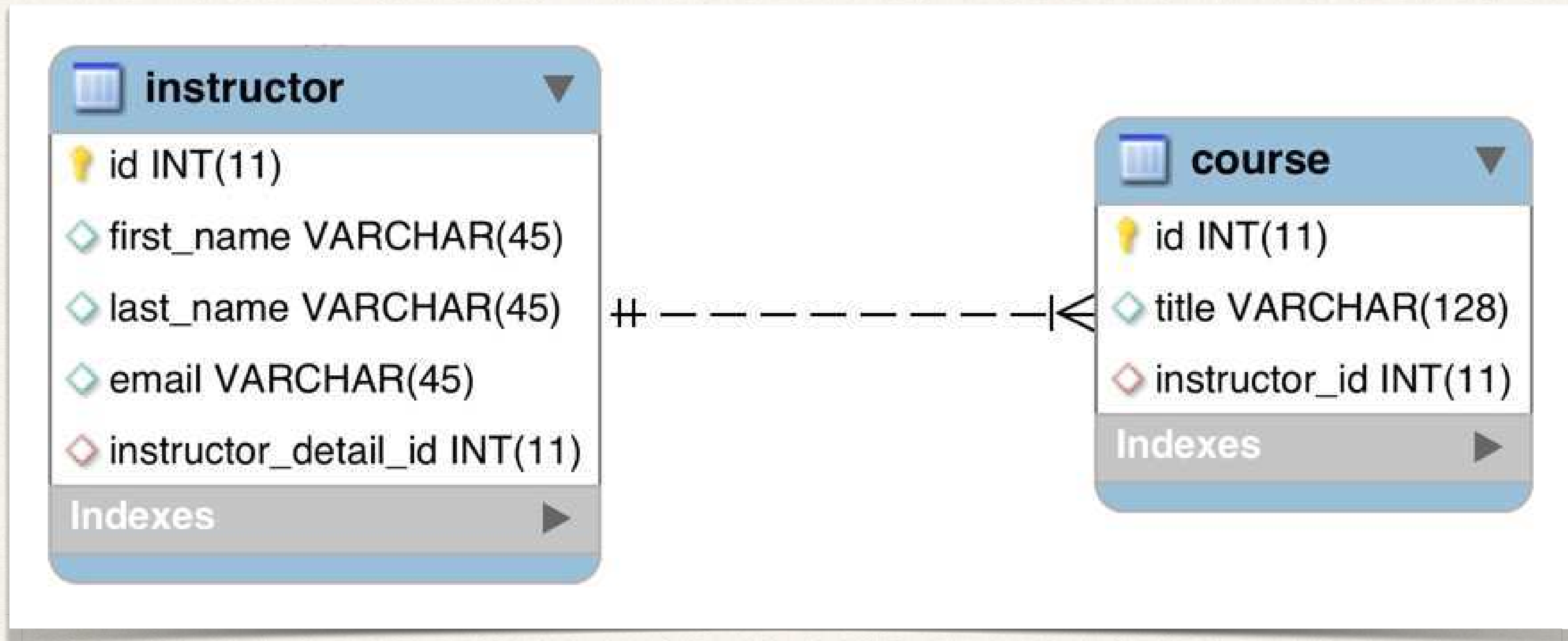


table: instructor - no changes



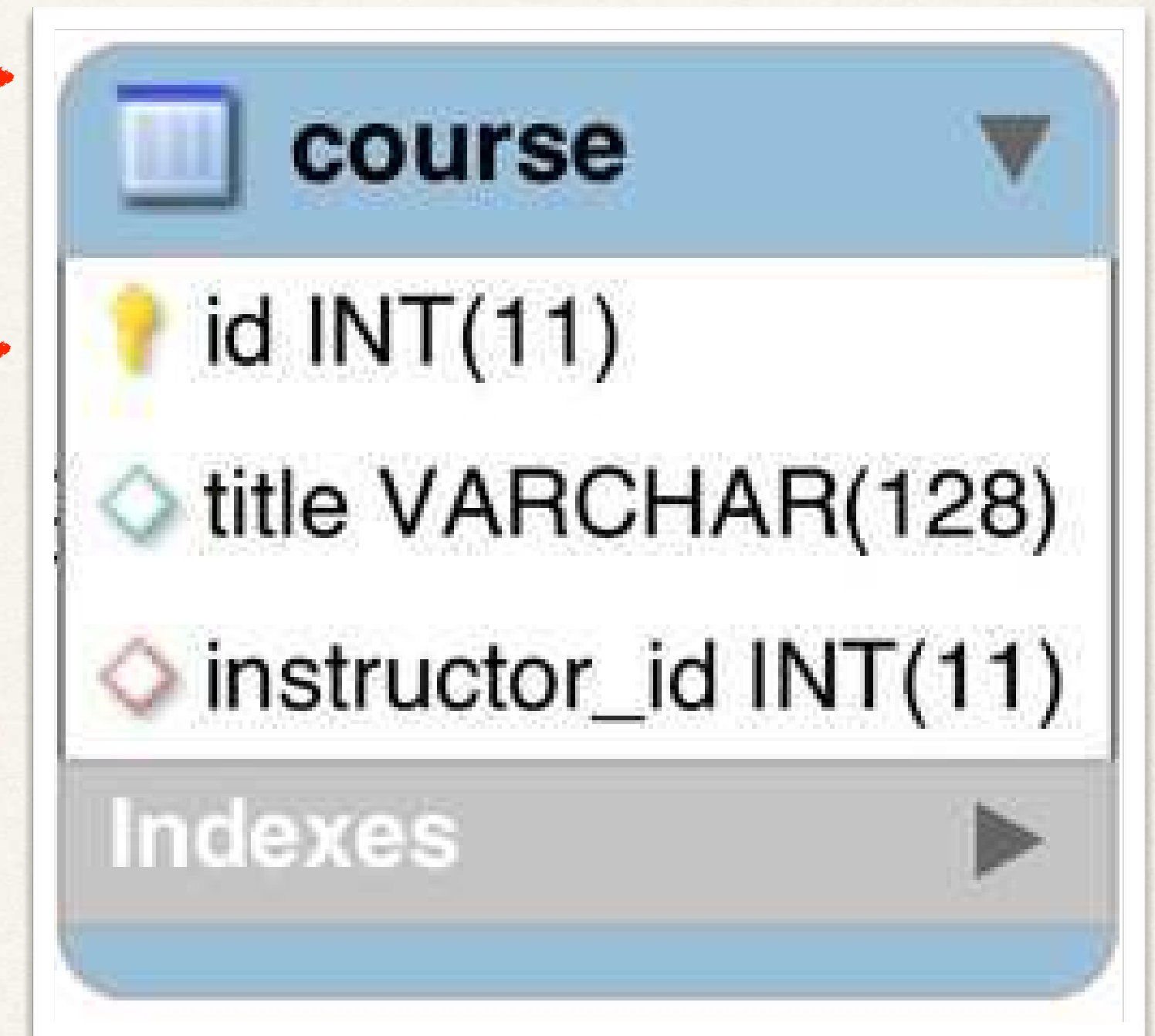
Step 2: Create Course class

```
@Entity
@Table(name="course")
public class Course {

    @Id
    @GeneratedValue(strategy=GenerationType.IDENTITY)
    @Column(name="id")
    private int id;

    @Column(name="title")
    private String title;

    ...
    // constructors, getters / setters
}
```

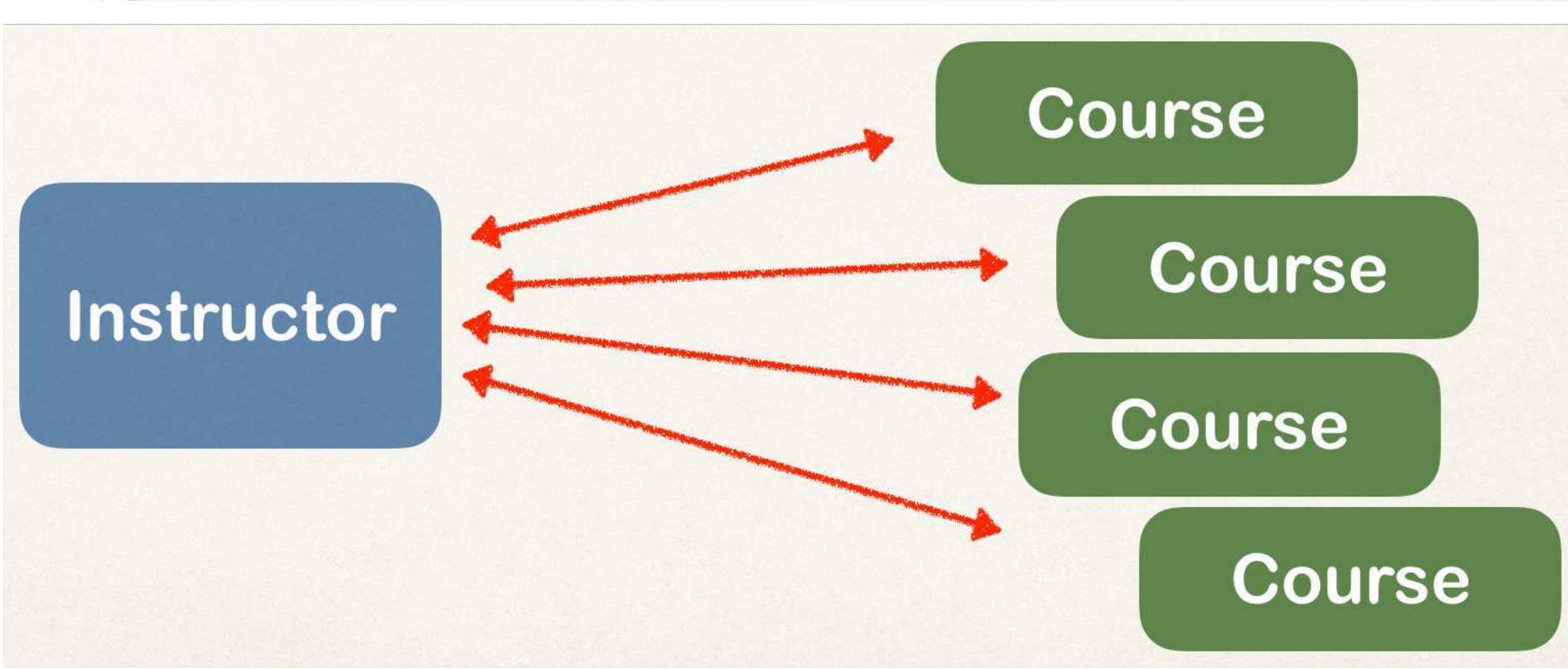
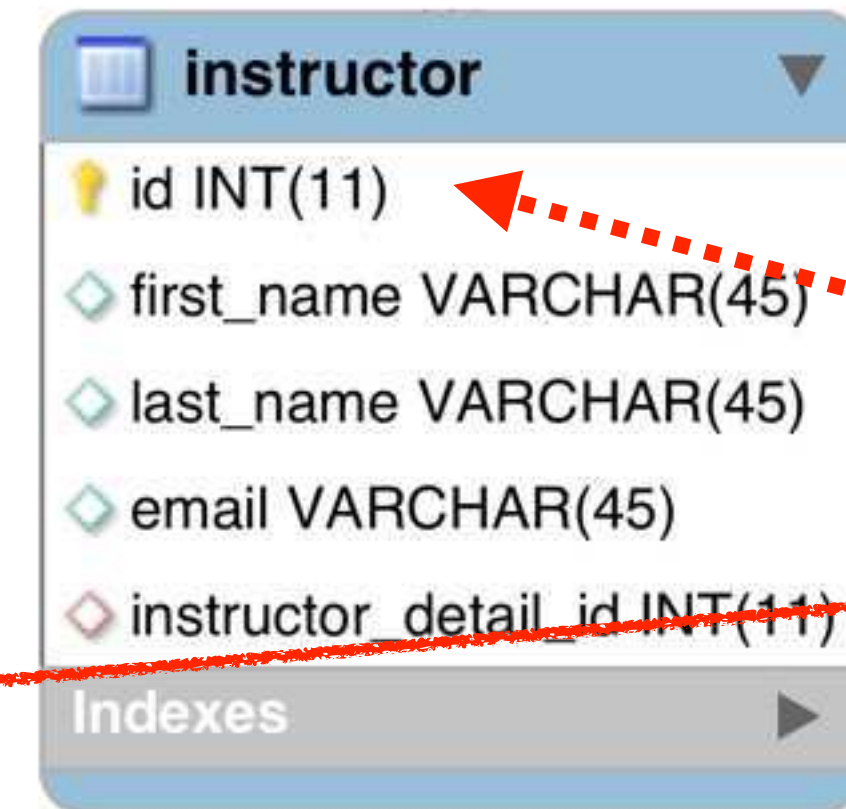


Step 2: Create Course class - @ManyToOne

```
@Entity
@Table(name="course")
public class Course {
```

```
    @ManyToOne
    @JoinColumn(name="instructor_id")
    private Instructor instructor;
```

```
    ...
    // constructors, getters / setters
}
```



Step 3: Update Instructor - reference courses

@Entity

@Table(name="instructor")

public class Instructor {

...

private List<Course> courses;

...

}

Step 3: Update Instructor - reference courses

@Entity

@Table(name="instructor")

public class Instructor {

...

private List<Course> courses;

public List<Course> getCourses() {

return courses;

}

public void setCourses(List<Course> courses) {

this.courses = courses;

}

...

}

Add @OneToMany annotation

```
@Entity
@Table(name="instructor")
public class Instructor {
    ...

    @OneToMany(mappedBy="instructor")
    private List<Course> courses;

    public List<Course> getCourses() {
        return courses;
    }

    public void setCourses(List<Course> courses) {
        this.courses = courses;
    }

    ...
}
```

Refers to “instructor” property
in “Course” class

More: mappedBy

```
public class Instructor {  
    ...  
  
    @OneToMany(mappedBy="instructor")  
    private List<Course> courses;
```

- **mappedBy** tells Hibernate
 - Look at the instructor property in the Course class
 - Use information from the Course class @JoinColumn
 - To help find associated courses for instructor



```
public class Course {  
    ...  
  
    @ManyToOne  
    @JoinColumn(name="instructor_id")  
    private Instructor instructor;
```


Add support for Cascading

```
@Entity
@Table(name="instructor")
public class Instructor {
    ...

    @OneToMany(mappedBy="instructor",
                cascade={CascadeType.PERSIST, CascadeType.MERGE
                        CascadeType.DETACH, CascadeType.REFRESH})
    private List<Course> courses;

    ...
}
```

**Do not apply
cascading deletes!**

Add support for Cascading

```
@Entity
@Table(name="course")
public class Course {
    ...

    @ManyToOne(cascade={CascadeType.PERSIST, CascadeType.MERGE
                       CascadeType.DETACH, CascadeType.REFRESH})
    @JoinColumn(name="instructor_id")
    private Instructor instructor;

    ...
    // constructors, getters / setters
}
```

**Do not apply
cascading deletes!**

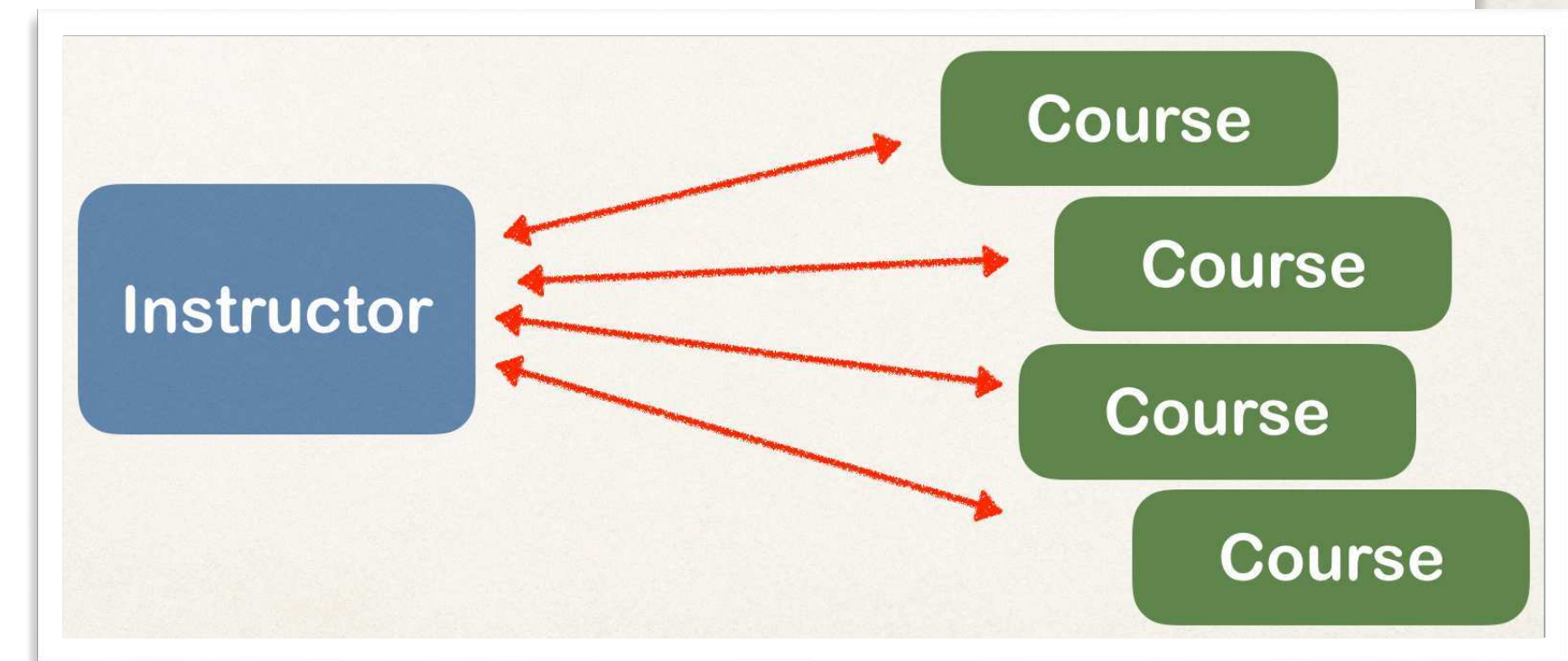
Add convenience methods for bi-directional

```
@Entity
@Table(name="instructor")
public class Instructor {
    ...
    // add convenience methods for bi-directional relationship

    public void add(Course tempCourse) {

        if (courses == null) {
            courses = new ArrayList<>();
        }

        courses.add(tempCourse);
        tempCourse.setInstructor(this);
    }
    ...
}
```



Mappings: @OneToMany

Defines physical mapping of the relationship

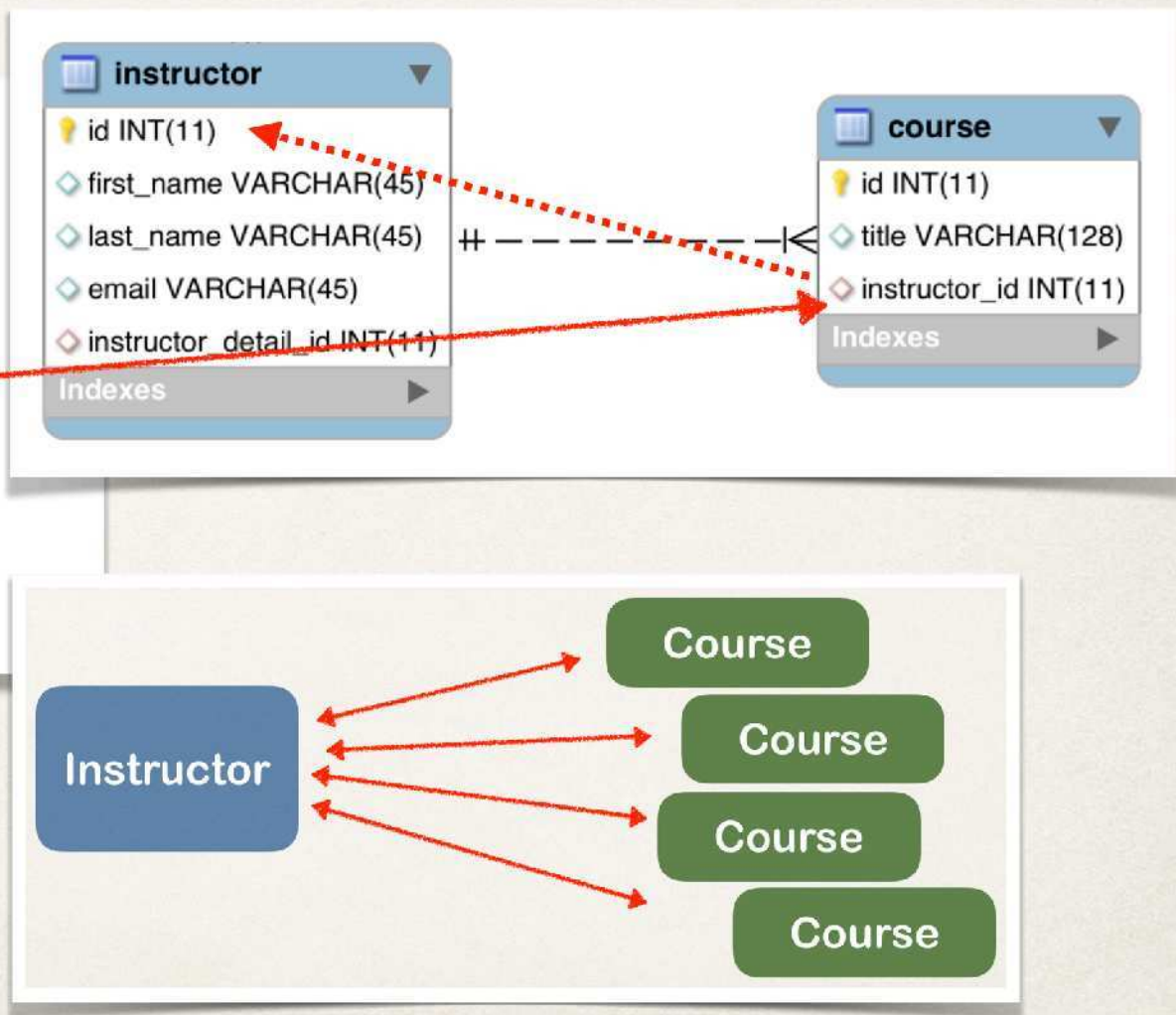
Mirrors the owning side

Mapping	Foreign Key location in database table	Owning Side	Inverse side
@OneToMany, @ManyToOne Bidirectional Instructor <-> Course	course table	Course @ManyToOne + @JoinColumn	Instructor @OneToMany(mappedBy="instructor")

```
@Entity
@Table(name="course")
public class Course {
    ...

    @ManyToOne
    @JoinColumn(name="instructor_id")
    private Instructor instructor;

    ...
    // constructors, getters / setters
}
```



```
@Entity
@Table(name="instructor")
public class Instructor {
    ...

    @OneToMany(mappedBy="instructor")
    private List<Course> courses;
}
```

Use this course's examples as a general guide.

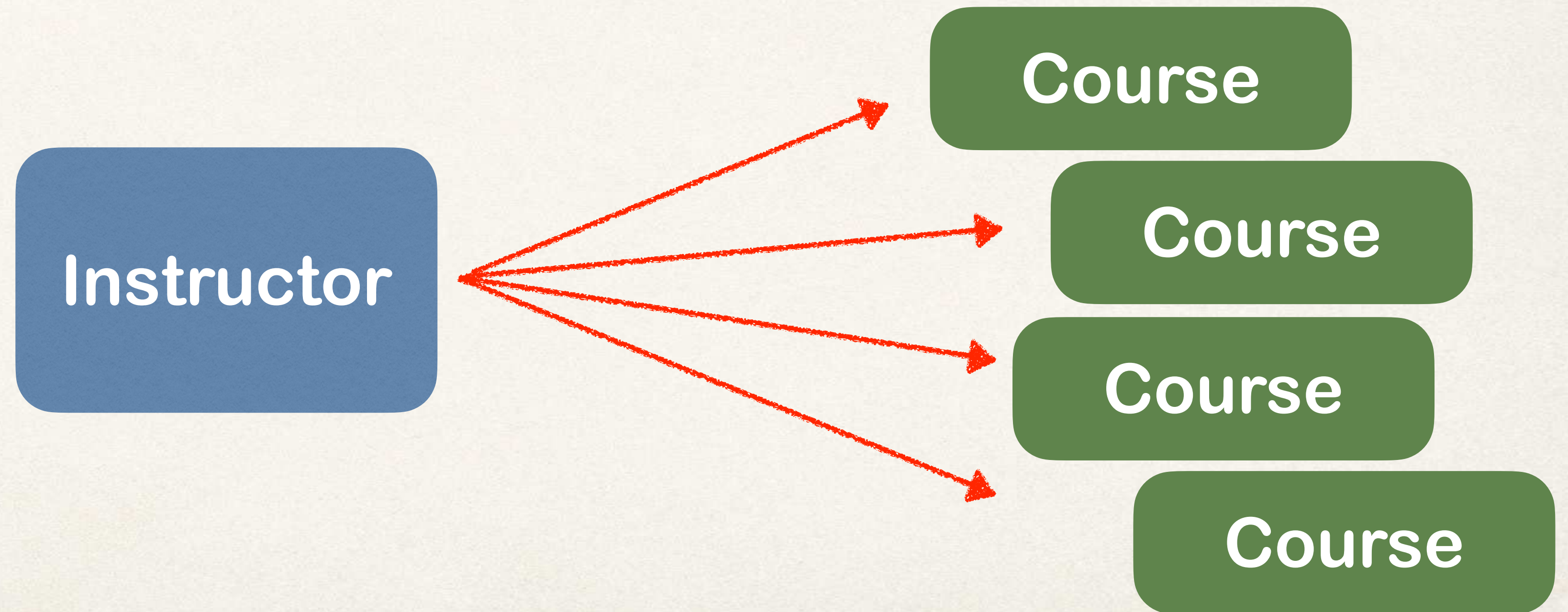
Adapt them to your application's requirements and domain needs.

Fetch Types: Eager vs Lazy



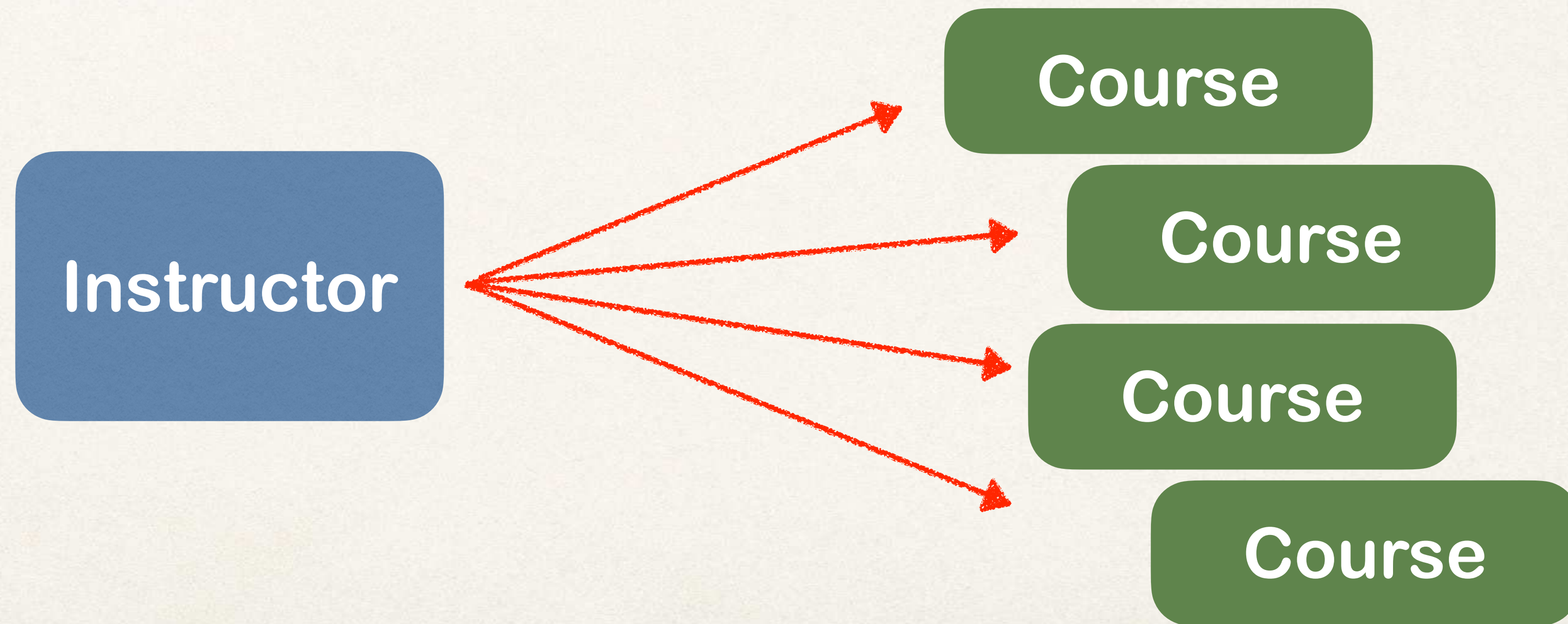
Fetch Types: Eager vs Lazy Loading

- When we fetch / retrieve data, should we retrieve EVERYTHING?
- **Eager** will retrieve everything
- **Lazy** will retrieve on request



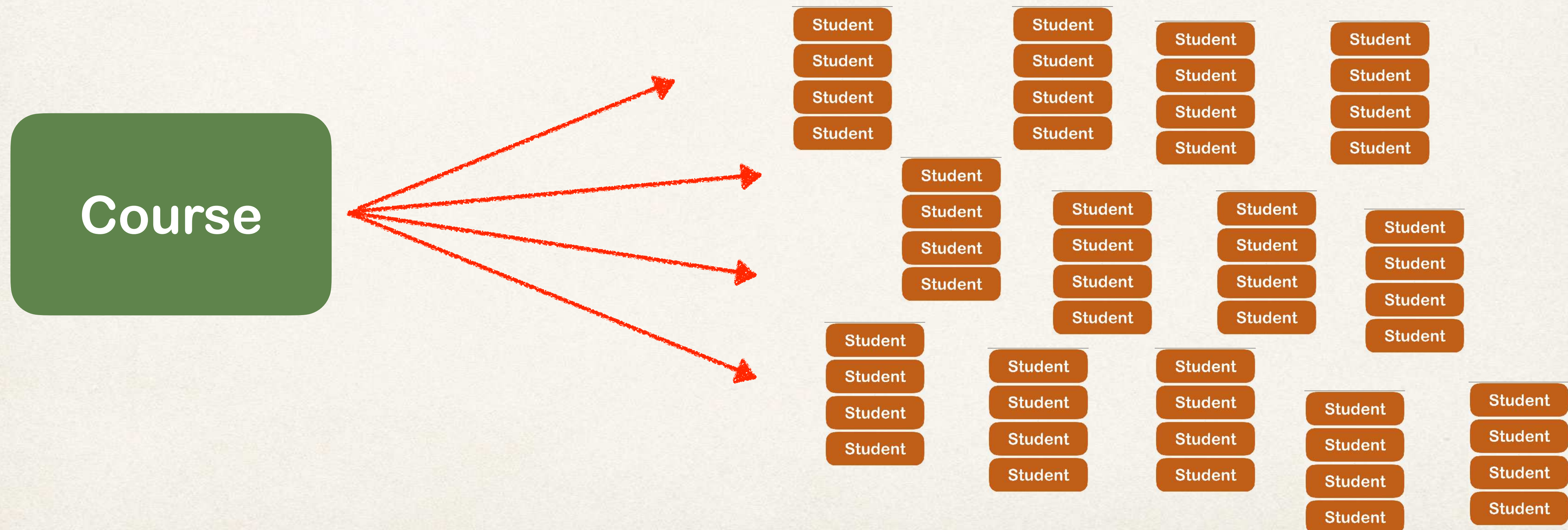
Eager Loading

- Eager loading will load all dependent entities
- Load instructor and all of their courses at once



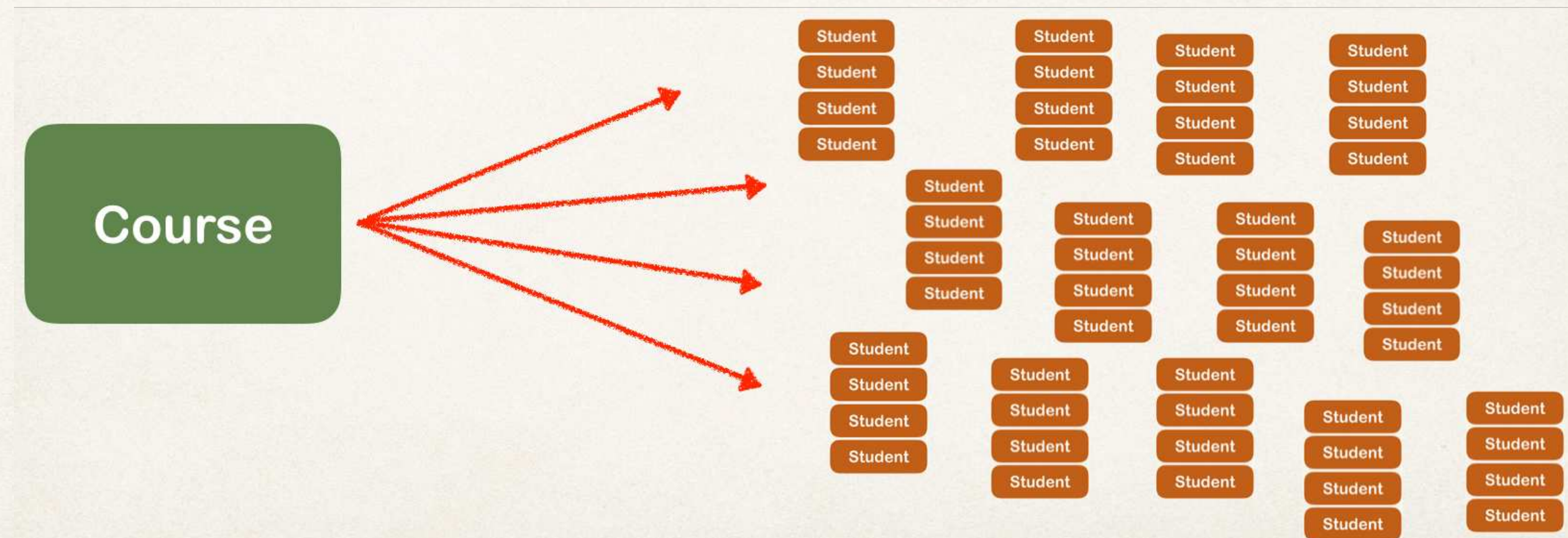
Eager Loading

- What about course and students?
- Could easily turn into a performance nightmare ...



Eager Loading

- In our app, if we are searching for a course by keyword
- Only want a list of matching courses
- Eager loading would still **load all students for each course not good!**



Best Practice

Only load data when absolutely needed

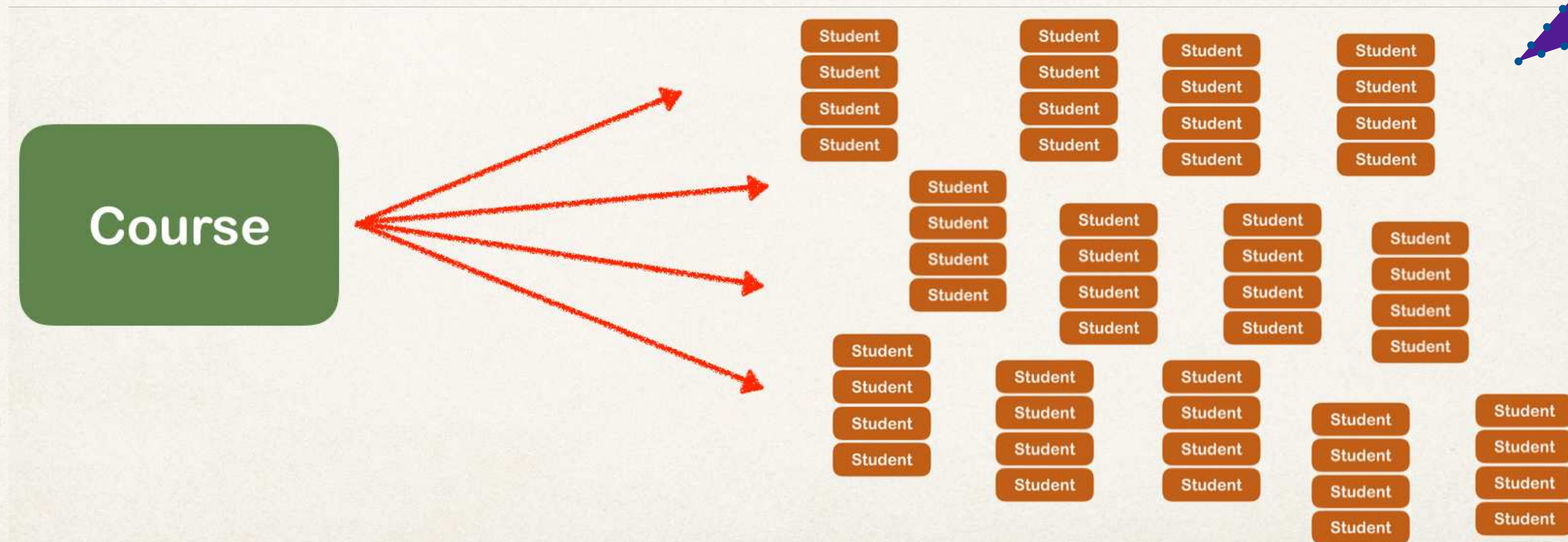
Prefer Lazy loading
instead of
Eager loading

Lazy Loading

- Lazy loading will load the main entity first
- Load dependent entities on demand (lazy)

Lazy Loading is preferred

Load course first



Load on demand (lazy)

Real-World Use Case

- Search for instructors

luv2code academy

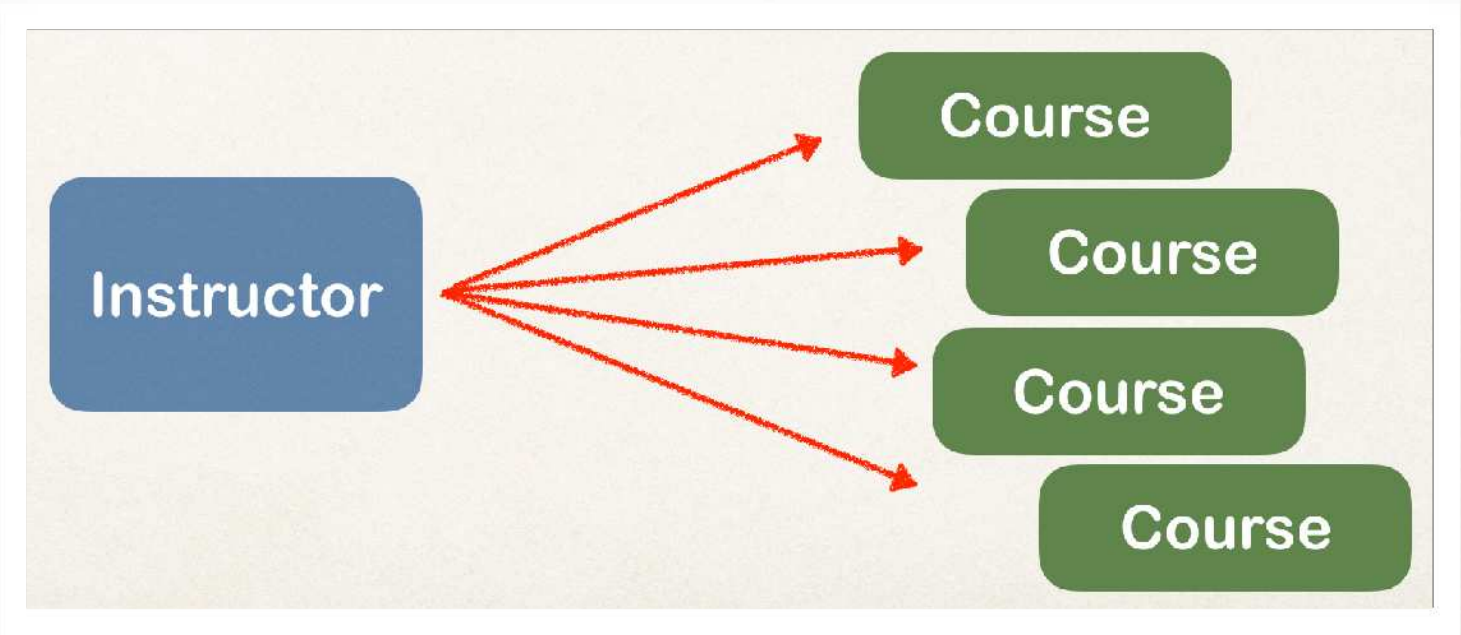
Last Name	First Name	Email	Action
Public	Mary	mary@luv2code.com	View Details
Doe	John	john@luv2code.com	View Details
Rao	Ajay	ajay@luv2code.com	View Details
Kublanov	Victor	victor@luv2code.com	View Details
O'Keefe	Paul	paul@luv2code.com	View Details
Peterson	Anthony	anthony@luv2code.com	View Details
Patel	Trupti	trupti@luv2code.com	View Details

Real-World Use Case

- In Master view, use lazy loading
- In Detail view, retrieve the entity and necessary dependent entities

Real-World Use Case - Master View

- In Master view, use lazy loading for search results
- Only load instructors ... not their courses



luv2code academy

Last Name	First Name	Email	Action
Public	Mary	mary@luv2code.com	View Details
Doe	John	john@luv2code.com	View Details
Rao	Ajay	ajay@luv2code.com	View Details
Kublanov	Victor	victor@luv2code.com	View Details
O'Keefe	Paul	paul@luv2code.com	View Details
Peterson	Anthony	anthony@luv2code.com	View Details
Patel	Trupti	trupti@luv2code.com	View Details

Real-World Use Case - Detail View

- In Detail view, retrieve the entity and necessary dependent entities
- Load instructor AND their courses

luv2code academy

Instructor Details

First name

John

Last name

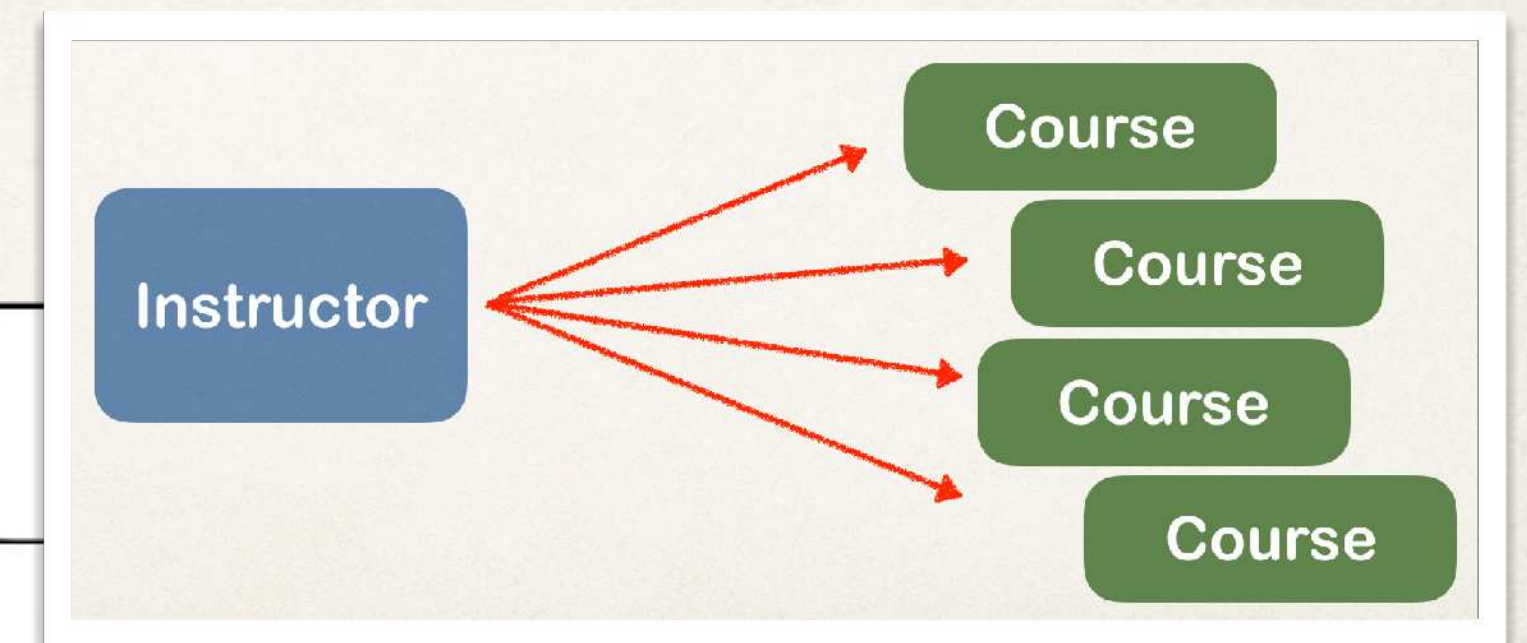
Doe

Email

john@luv2code.com

Course Title	# Students	Action
JSP & Servlets for Beginners	10147	View Students
JSF for Beginners	4562	View Students
Eclipse Tips and Tricks	47380	View Students
JDBC and MySQL	48498	View Students

[Home](#)



List of
courses

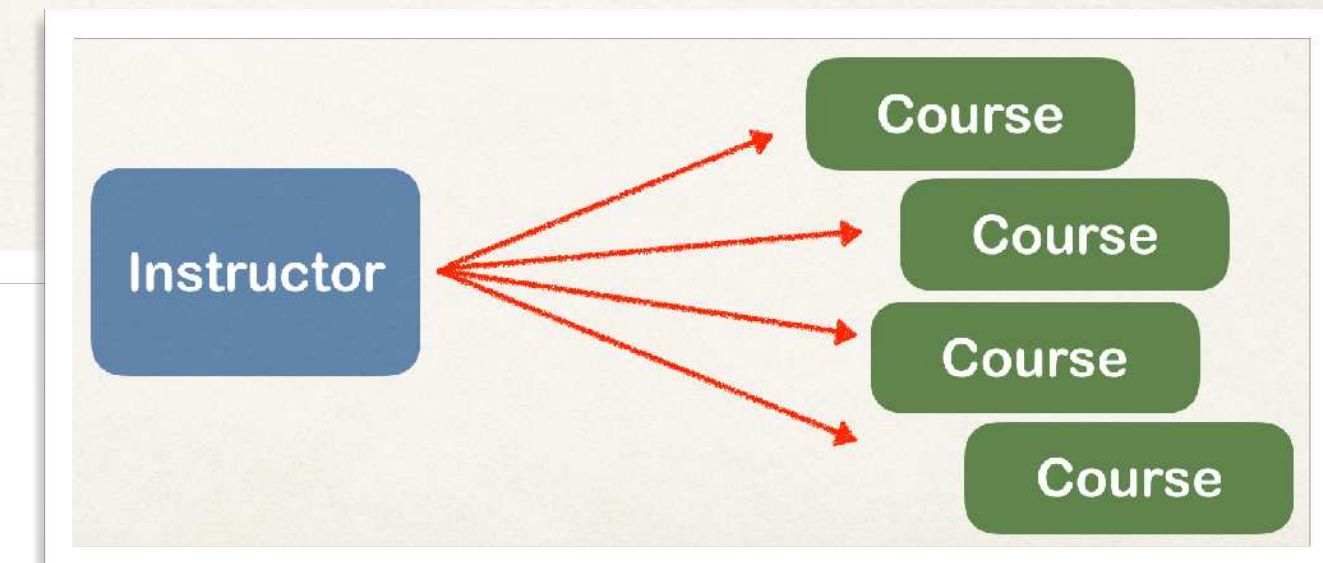
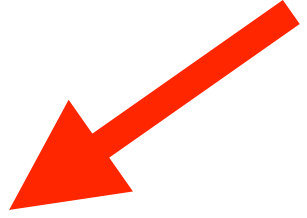
Fetch Type

- When you define the mapping relationship
- You can specify the fetch type: EAGER or LAZY

```
@Entity
@Table(name="instructor")
public class Instructor {
    ...

    @OneToMany(fetch=FetchType.LAZY, mappedBy="instructor")
    private List<Course> courses;

    ...
}
```



Default Fetch Types

Mapping	Default Fetch Type
@OneToOne	FetchType.EAGER
@OneToMany	FetchType.LAZY
@ManyToOne	FetchType.EAGER
@ManyToMany	FetchType.LAZY

Overriding Default Fetch Type

- Specifying the fetch type, overrides the defaults

Mapping	Default Fetch Type
@ManyToOne	FetchType.EAGER



```
@ManyToOne(fetch=FetchType.LAZY)  
@JoinColumn(name="instructor_id")  
private Instructor instructor;
```


More about Lazy Loading

- When you lazy load, the data is only retrieved on demand
- However, this requires an open Hibernate session
 - need an connection to database to retrieve data

More about Lazy Loading

- If the Hibernate session is closed
 - And you attempt to retrieve lazy data
 - Hibernate will throw an exception



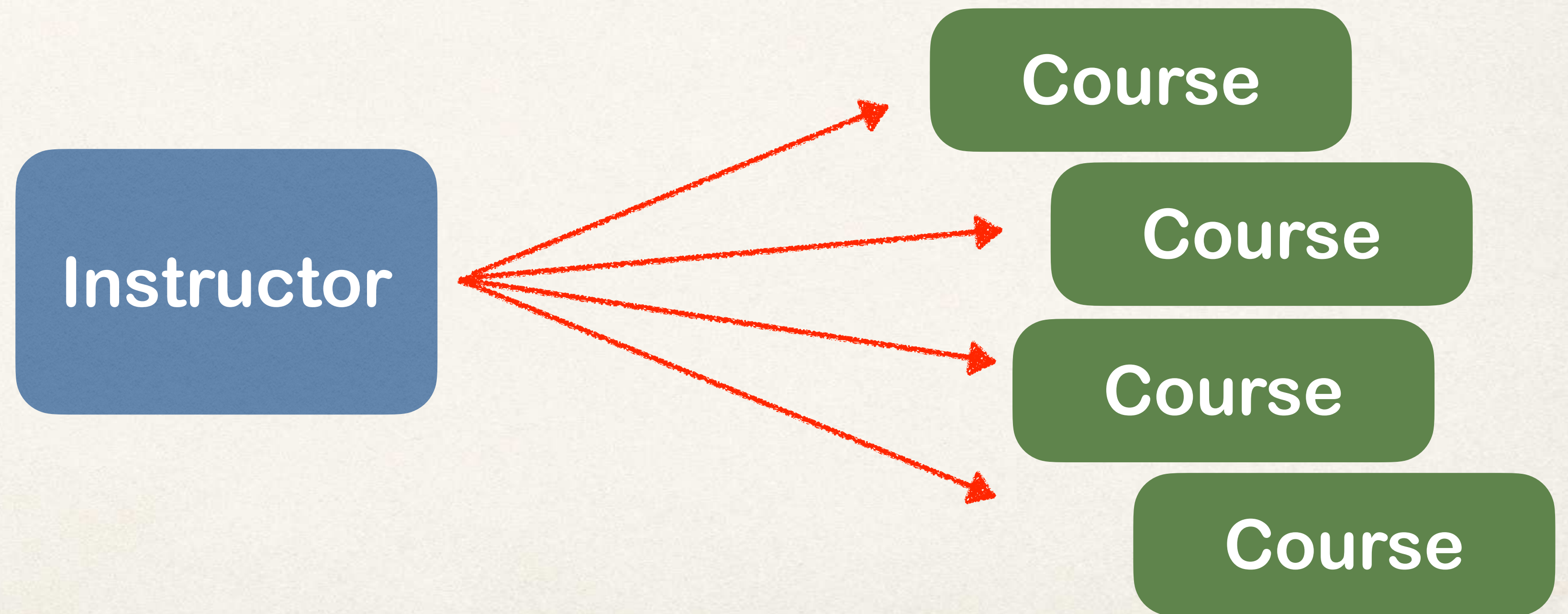
Watch out for this!

Lazy: Find Courses



Previous Solution: Eager

- **Eager** will retrieve everything ... all of the courses for an instructor
- But we may not want this *ALL* of the time
- We'd like the option to load courses as needed ...



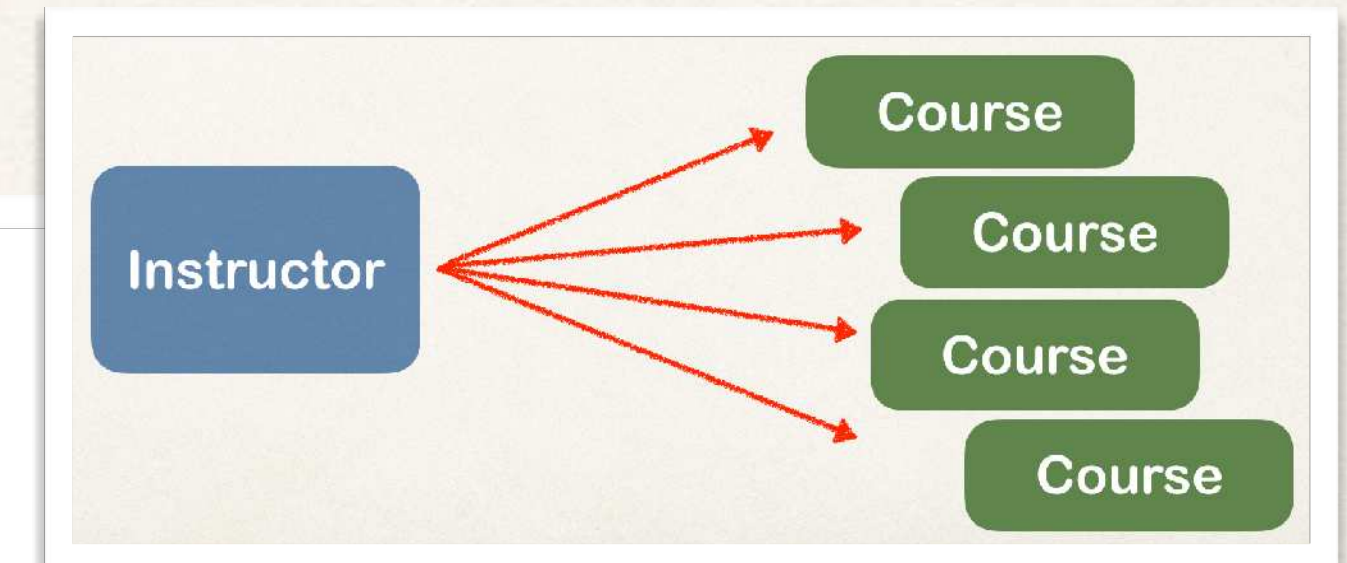
Fetch Type

- Change the fetch type back to LAZY

```
@Entity
@Table(name="instructor")
public class Instructor {
    ...

    @OneToMany(fetch=FetchType.LAZY, mappedBy="instructor")
    private List<Course> courses;

    ...
}
```



FetchType for @OneToMany defaults to lazy ...
But I will explicitly list it for readability

Add new method to find courses for instructor

File: AppDAOImpl.java

```
@Override
public List<Course> findCoursesByInstructorId(int theId) {

    // create query
    TypedQuery<Course> query = entityManager.createQuery("from Course where instructor.id = :data", Course.class);
    query.setParameter("data", theId);

    // execute query
    List<Course> courses = query.getResultList();

    return courses;
}
```

File: CruddemoApplication.java

```
private void findCoursesForInstructor(AppDAO appDAO) {

    int theId = 1;

    // find the instructor
    Instructor tempInstructor = appDAO.findInstructorById(theId);
    System.out.println("tempInstructor: " + tempInstructor);

    // find courses for instructor
    List<Course> courses = appDAO.findCoursesByInstructorId(theId);

    // associate the objects
    tempInstructor.setCourses(courses);

    System.out.println("the associated courses: " + tempInstructor.getCourses());
}
```

Since fetch type for courses is lazy

This will retrieve the instructor
WITHOUT courses

Lazy: Find Instructor with Courses



Previous Solution: Find Courses for Instructor

- Previous solution was OK ... but ...
- Required an extra query
- I wish we could have a new method that would
 - Get instructor AND courses ... in a single query
 - Also keep the LAZY option available ... don't change fetch type

Add new method to find instructor with courses

File: AppDAOImpl.java

```
@Override
public Instructor findInstructorByIdJoinFetch(int theId) {

    // create query
    TypedQuery<Instructor> query = entityManager.createQuery(
        "select i from Instructor i "
        + "JOIN FETCH i.courses "
        + "where i.id = :data", Instructor.class);

    query.setParameter("data", theId);

    // execute query
    Instructor instructor = query.getSingleResult();

    return instructor;
}
```

This code will still retrieve Instructor AND Courses

Even with Instructor
@OneToMany(fetchType=LAZY)

This code will still retrieve Instructor AND Courses

The JOIN FETCH is similar to EAGER loading

File: CruddemoApplication.java

```
private void findInstructorWithCoursesJoinFetch(AppDAO appDAO) {

    int theId = 1;

    // find the instructor
    System.out.println("Finding instructor id: " + theId);
    Instructor tempInstructor = appDAO.findInstructorByIdJoinFetch(theId);

    System.out.println("tempInstructor: " + tempInstructor);
    System.out.println("the associated courses: " + tempInstructor.getCourses());

    System.out.println("Done!");
}
```


We have options now

- If you only need Instructor ... and no courses, then call
 - `appDAO.findInstructorById(...)`
- If you need Instructor AND courses, then call
 - `appDAO.findInstructorByIdJoinFetch(...)`
- This gives us flexibility instead of having EAGER fetch hard-coded

@OneToMany: Update Instructor



Update Instructor

- Find an instructor by ID
- Change the instructor's data by calling setter method(s)
- Update the instructor using the DAO

Add new DAO method to update instructor

File: AppDAOImpl.java

```
@Override
@Transactional
public void update(Instructor tempInstructor) {
    entityManager.merge(tempInstructor);
}
```

merge(...) will update an existing entity

Main app

File: CruddemoApplication.java

```
private void updateInstructor(AppDAO appDAO) {  
  
    int theId = 1;  
  
    System.out.println("Finding instructor id: " + theId);  
    Instructor tempInstructor = appDAO.findInstructorById(theId);  
  
    System.out.println("Updating instructor id: " + theId);  
    tempInstructor.setLastName("TESTER");  
  
    appDAO.update(tempInstructor);  
  
    System.out.println("Done");  
}
```

Change instructor's data

Call DAO method
to update database

@OneToMany: Update Course



Update Course

- Find a course by ID
- Change the course's data by calling setter method(s)
- Update the course using the DAO

Add new DAO method to update course

File: AppDAOImpl.java

```
@Override
@Transactional
public void update(Course tempCourse) {
    entityManager.merge(tempCourse);
}
```

merge(...) will update an existing entity

Main app

File: CruddemoApplication.java

```
private void updateCourse(AppDAO appDAO) {  
  
    int theId = 10;  
  
    System.out.println("Finding course id: " + theId);  
    Course tempCourse = appDAO.findCourseById(theId);  
  
    System.out.println("Updating course id: " + theId);  
    tempCourse.setTitle("Enjoy the Simple Things");  
  
    appDAO.update(tempCourse);  
  
    System.out.println("Done");  
}
```

Change course's data

Call DAO method
to update database

@OneToMany: Delete Instructor



Delete instructor

- Find an instructor by ID
- Break association of all instructor's courses
- Delete the instructor

Add new DAO method to delete instructor

File: AppDAOImpl.java

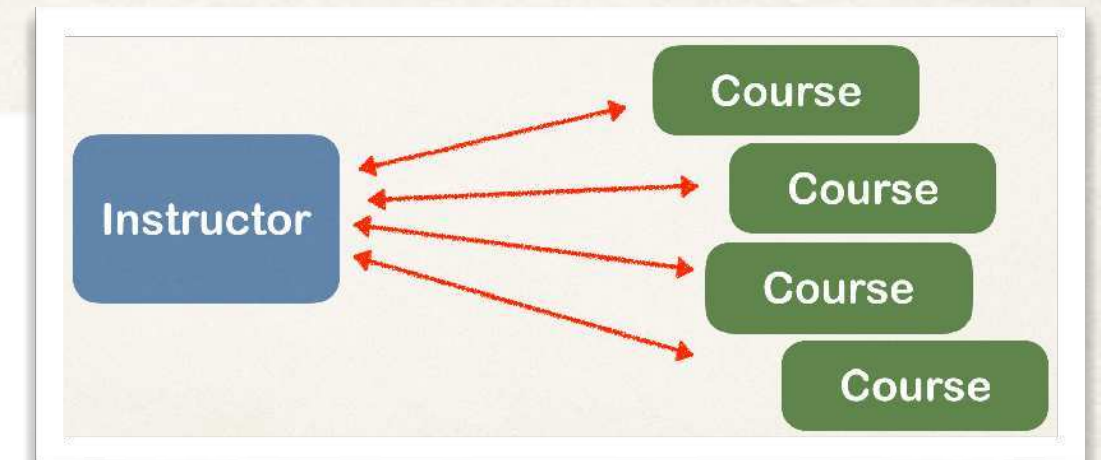
```
@Override
@Transactional
public void deleteInstructorById(int theId) {

    // retrieve the instructor
    Instructor tempInstructor = entityManager.find(Instructor.class, theId);

    List<Course> courses = tempInstructor.getCourses();

    // break associations of all courses for instructor
    for (Course tempCourse : courses) {
        tempCourse.setInstructor(null);
    }

    // delete the instructor
    entityManager.remove(tempInstructor);
}
```



Remove the instructor from the courses

We only delete the instructor ...
not the associated course
based on our cascade types

Main app

File: CruddemoApplication.java

```
private void deleteInstructor(AppDAO appDAO) {  
  
    int theId = 1;  
    System.out.println("Deleting instructor id: " + theId);  
  
    appDAO.deleteInstructorById(theId);  
  
    System.out.println("Done!");  
}
```


@OneToMany: Delete Course



Delete course

- Delete the course by ID

Add new DAO method to delete course

File: AppDAOImpl.java

```
@Override
@Transactional
public void deleteCourseById(int theId) {

    // retrieve the course
    Course tempCourse = entityManager.find(Course.class, theId);

    // delete the course
    entityManager.remove(tempCourse);
}
```


Main app

File: CruddemoApplication.java

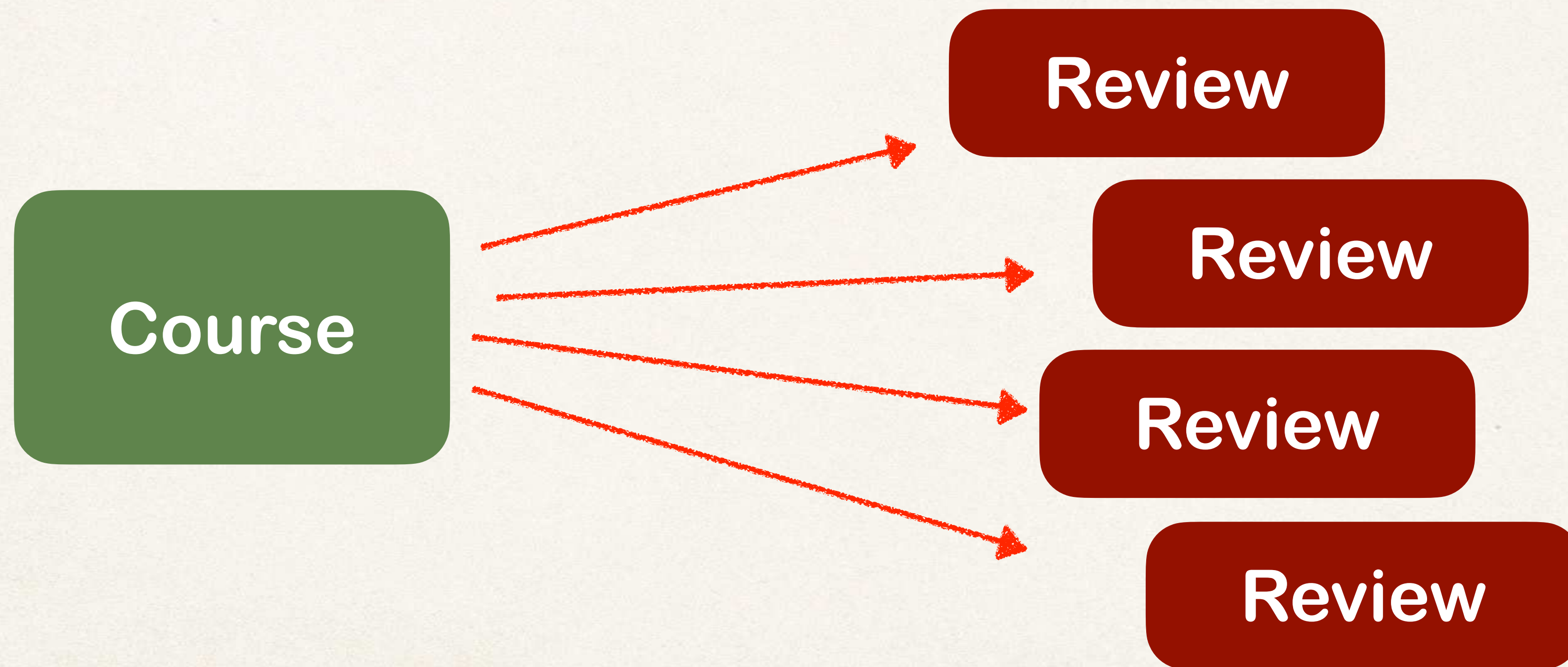
```
private void deleteCourseById(AppDAO appDAO) {  
  
    int theId = 10;  
    System.out.println("Deleting course id: " + theId);  
  
    appDAO.deleteCourseById(theId);  
  
    System.out.println("Done!");  
  
}
```


@OneToMany: Uni-Directional



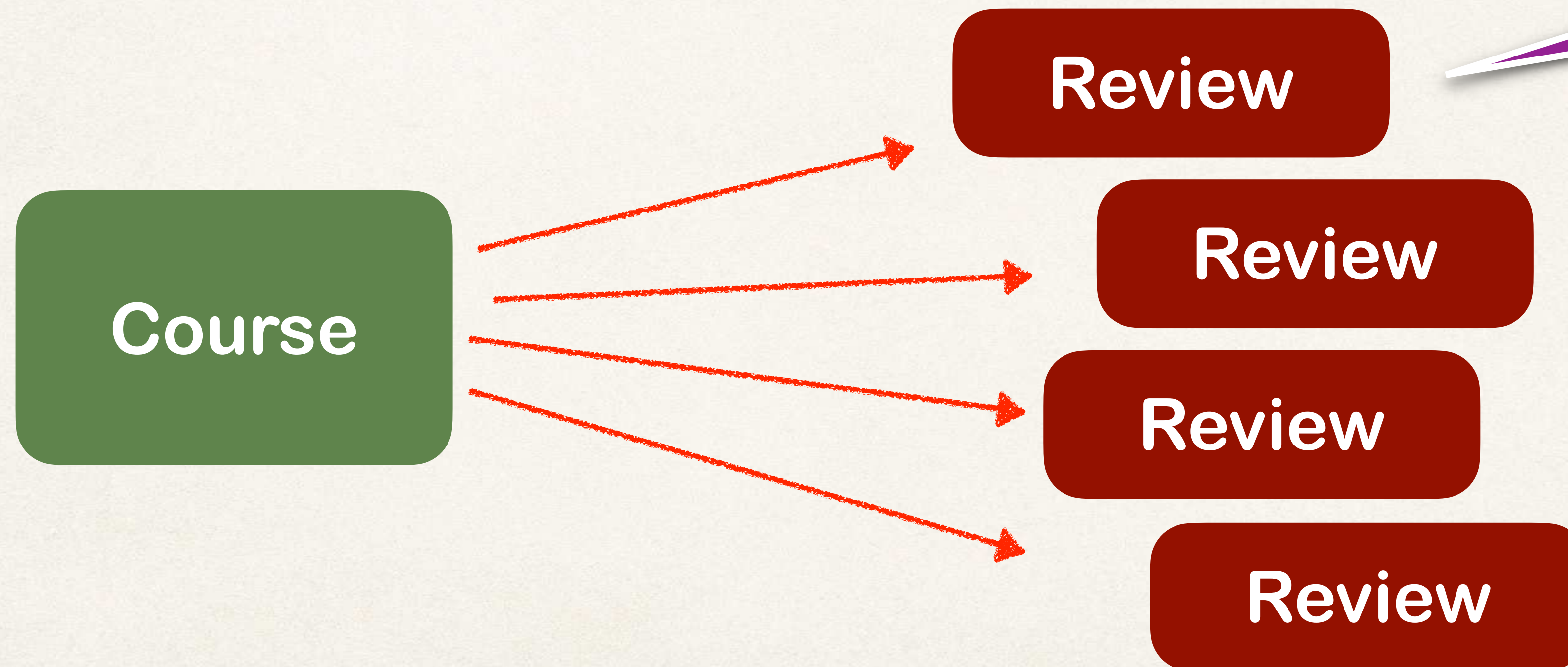
One-to-Many Mapping

- A course can have many reviews
- Uni-directional



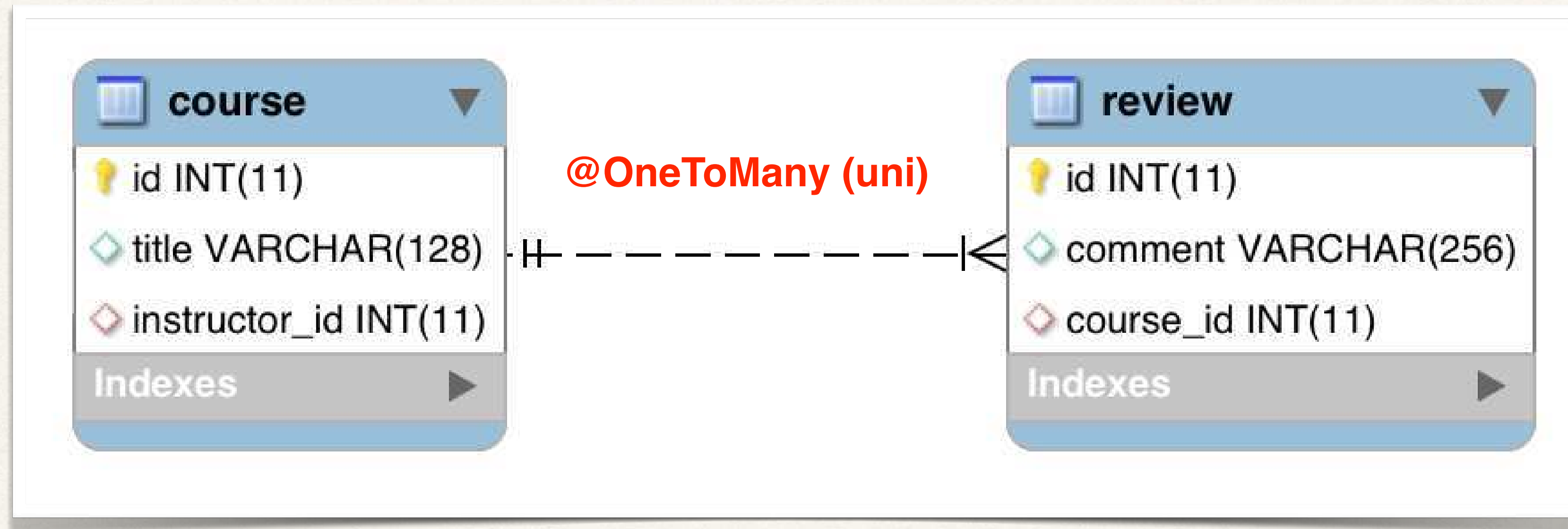
Real-World Project Requirement

- If you delete a course, also delete the reviews
- Reviews without a course ... have no meaning

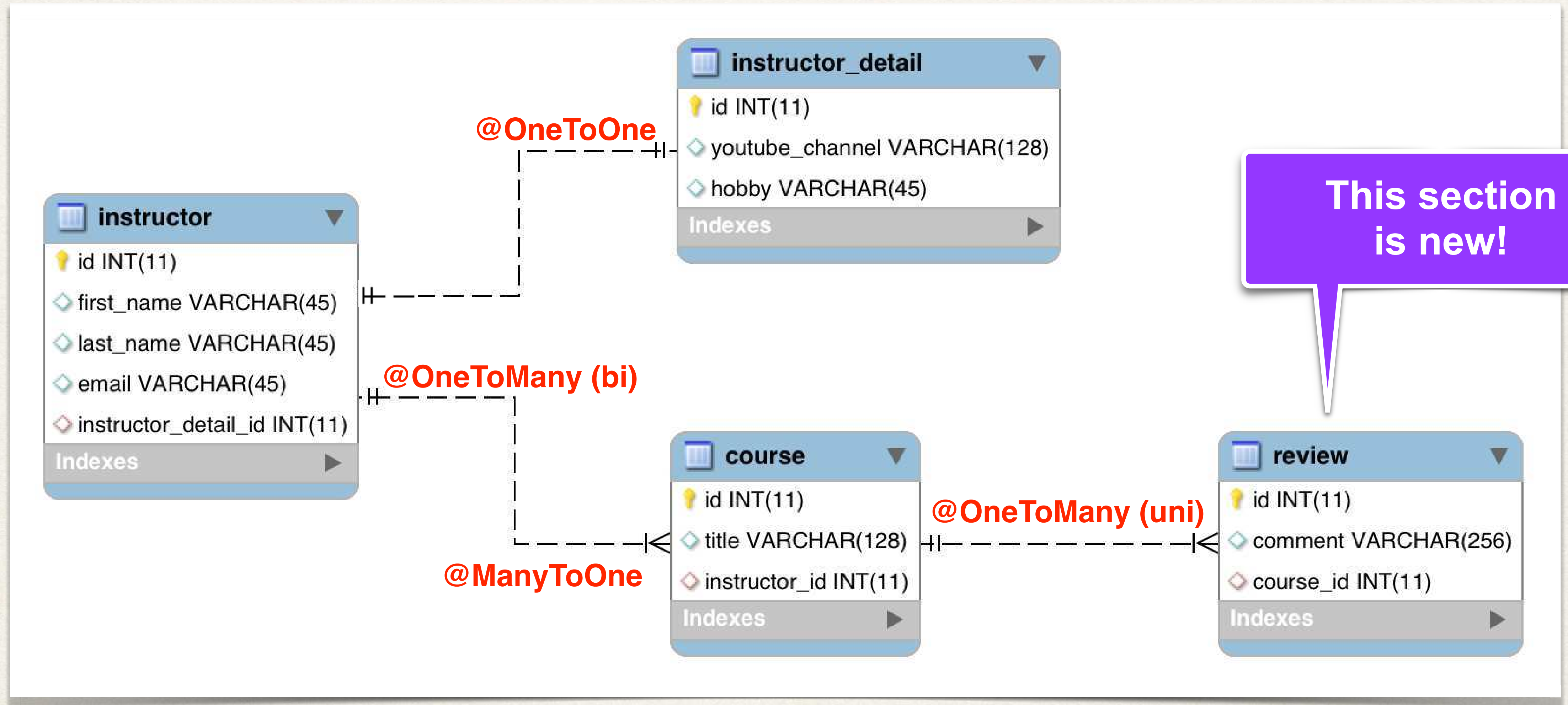


Apply cascading deletes!

@OneToMany



Look Mom ... our project is growing!



Development Process: One-to-Many

Step-By-Step

1. Prep Work - Define database tables
2. Create **Review** class
3. Update **Course** class

table: review

File: create-db.sql

```
CREATE TABLE `review` (  
  `id` int(11) NOT NULL AUTO_INCREMENT,  
  `comment` varchar(256) DEFAULT NULL,  
  `course_id` int(11) DEFAULT NULL,  
  ...  
);
```

comment:
"Wow ... this course is awesome!"

review	
💡	id INT(11)
🔹	comment VARCHAR(256)
🔹	course_id INT(11)
Indexes ▶	

table: review - foreign key

File: create-db.sql

```
CREATE TABLE `review` (  
...  
  KEY `FK_COURSE_ID_idx` (`course_id`),  
  CONSTRAINT `FK_COURSE`  
  FOREIGN KEY (`course_id`)  
  REFERENCES `course` (`id`)  
...  
);
```

Table

Column

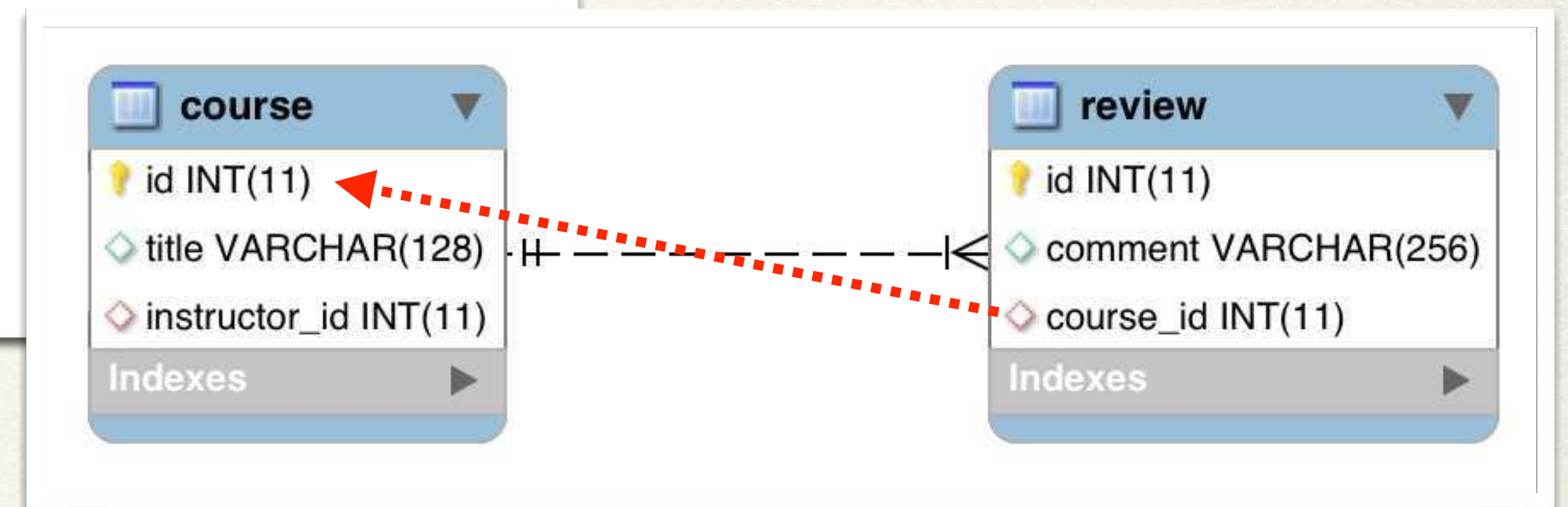
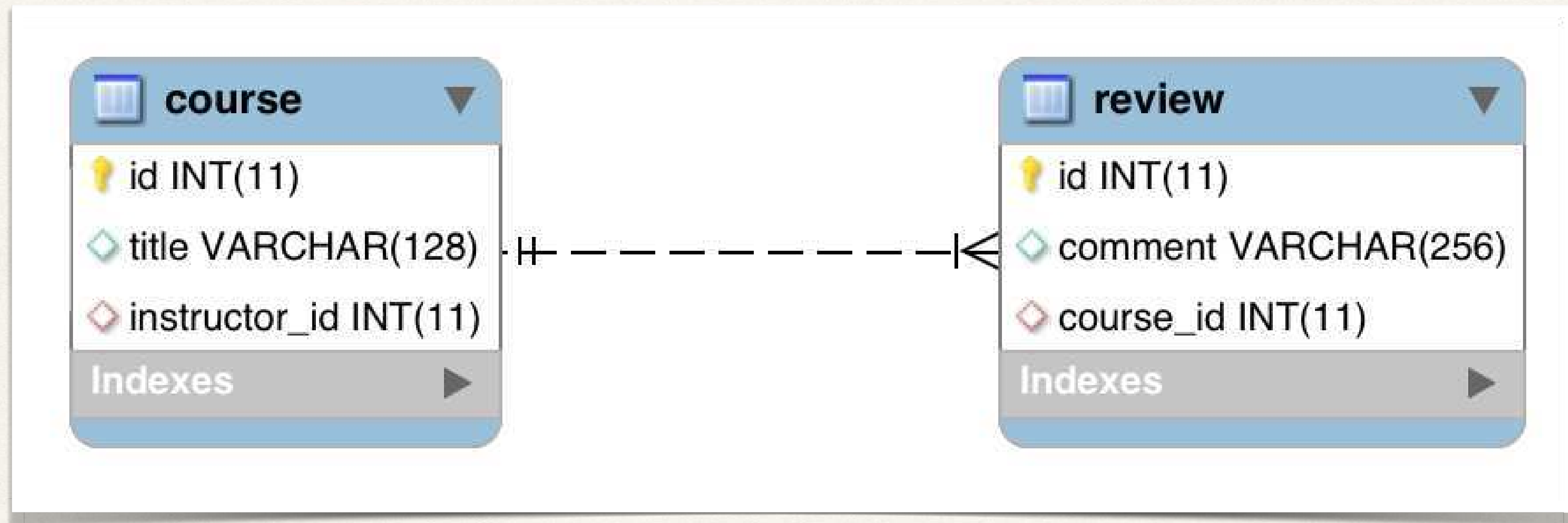


table: course - no changes



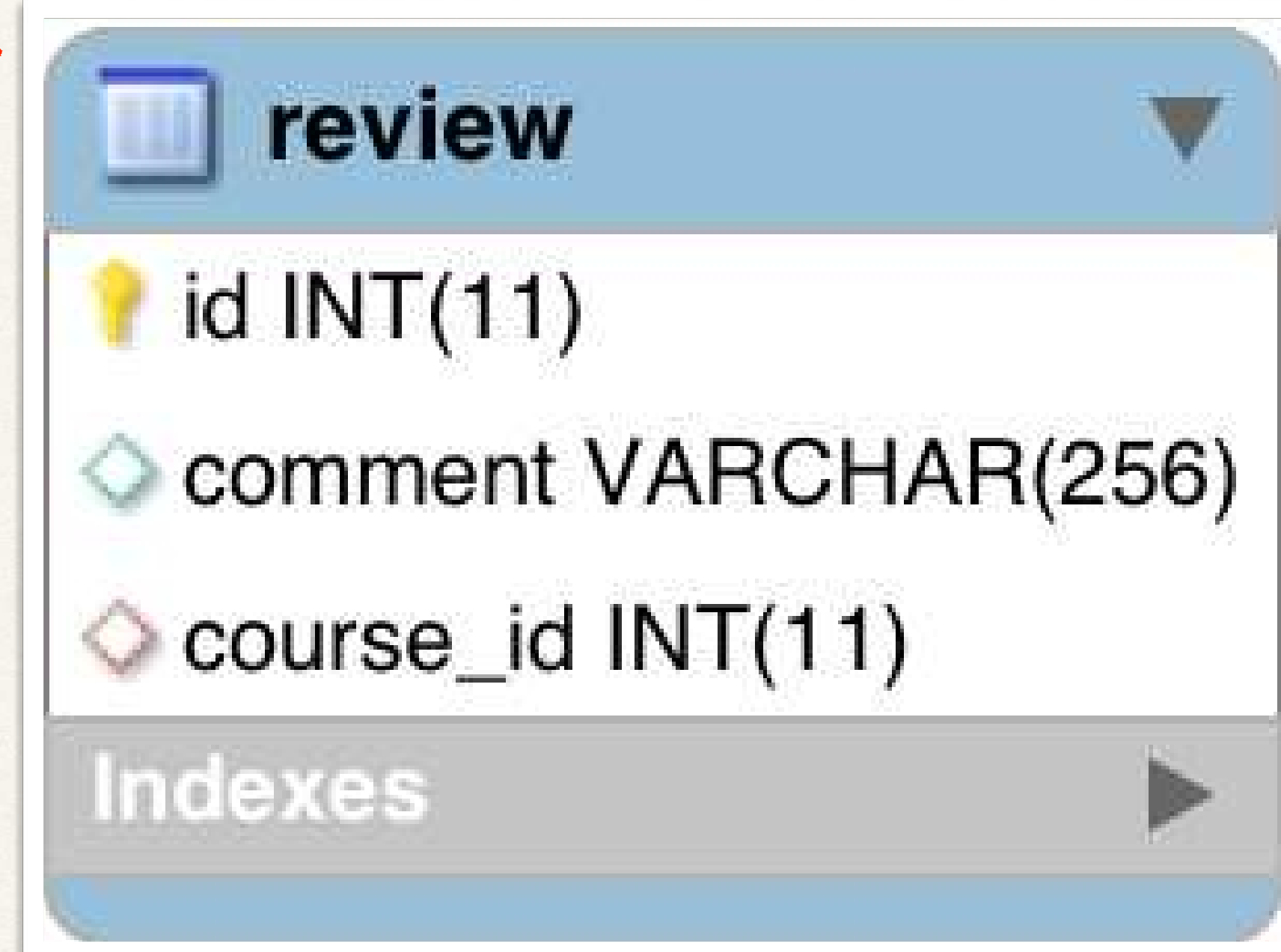
Step 2: Create Review class

```
@Entity
@Table(name="review")
public class Review {

    @Id
    @GeneratedValue(strategy=GenerationType.IDENTITY)
    @Column(name="id")
    private int id;

    @Column(name="comment")
    private String comment;

    ...
    // constructors, getters / setters
}
```



Step 3: Update Course - reference reviews

```
@Entity
@Table(name="course")
public class Course {
```

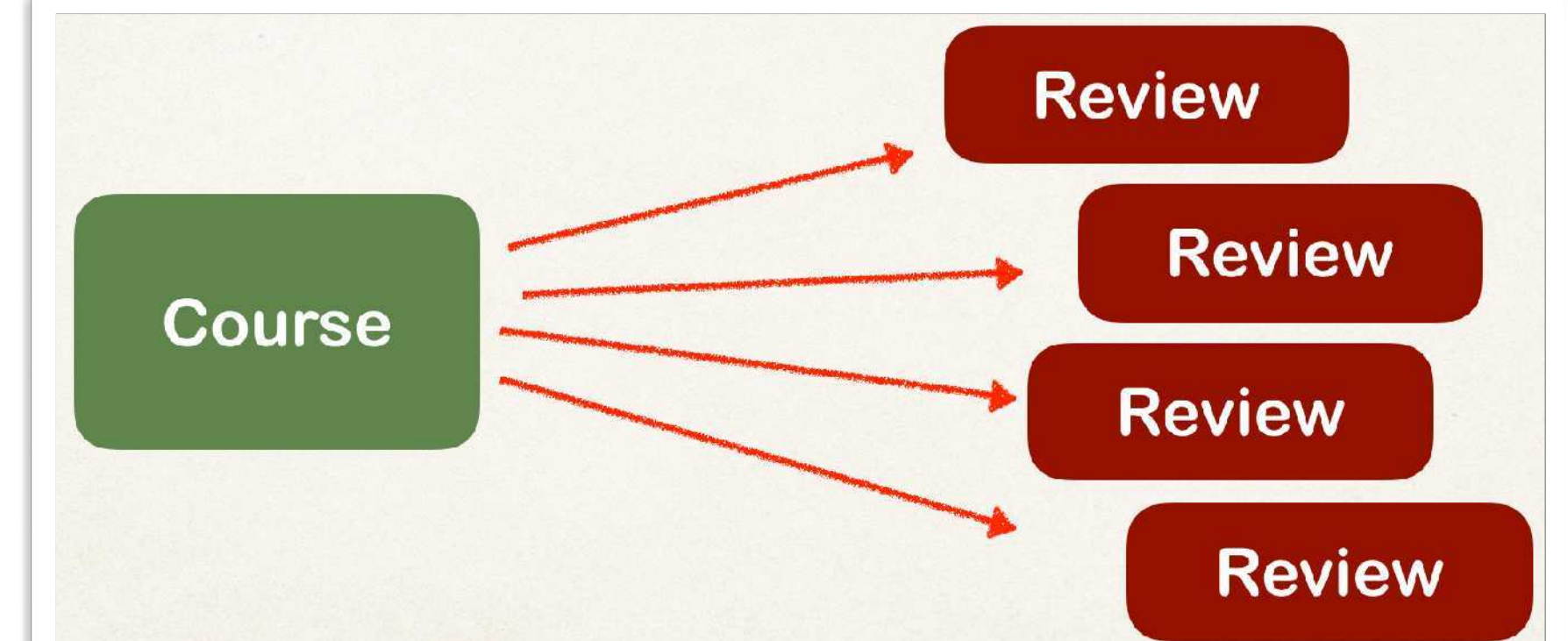
```
...
```

```
private List<Review> reviews;
```

```
// getter / setters
```

```
...
```

```
}
```



Add @OneToMany annotation

```
@Entity
@Table(name="course")
public class Course {
    ...
```

Refers to "course_id" column
in "review" table

```
@OneToMany
@JoinColumn(name="course_id")
private List<Review> reviews;
```

```
// getter / setters
```

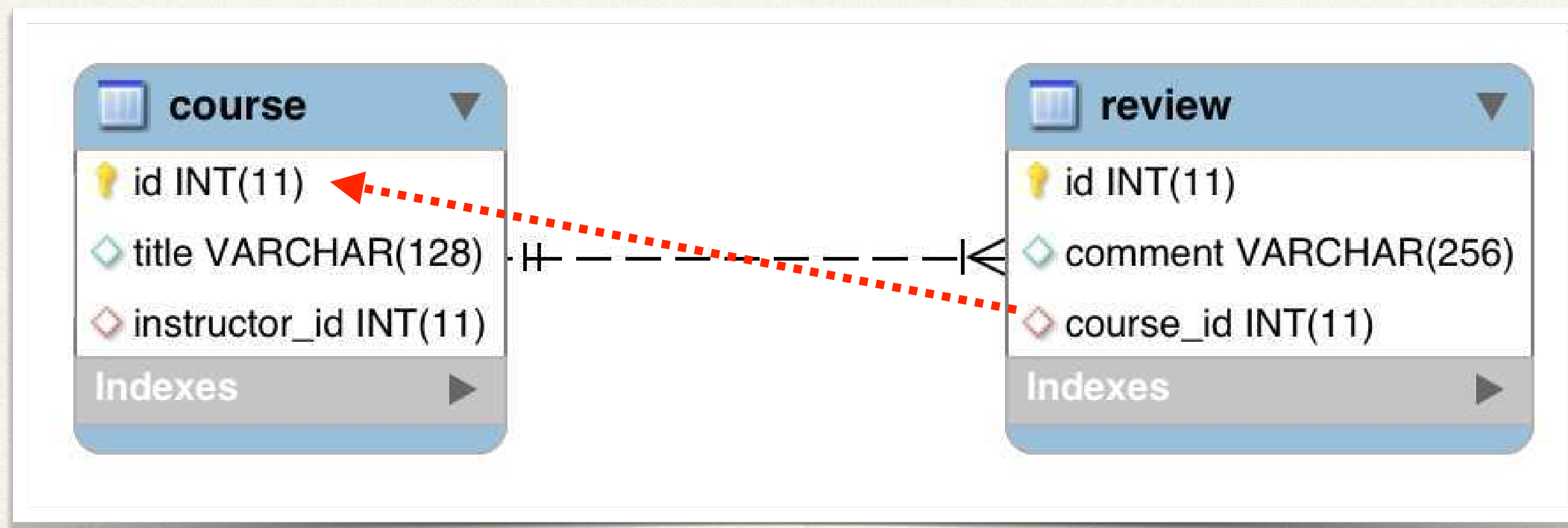
```
...
```

```
}
```


More: @JoinColumn

- In this scenario, **@JoinColumn** tells Hibernate
- Look at the **course_id** column in the **review** table
- Use this information to help find associated reviews for a course

```
public class Course {  
    ...  
  
    @OneToMany  
    @JoinColumn(name="course_id")  
    private List<Review> reviews;  
}
```



Add support for Cascading

```
@Entity
@Table(name="course")
public class Course {
    ...

    @OneToMany(cascade=CascadeType.ALL)
    @JoinColumn(name="course_id")
    private List<Review> reviews;

    ...
}
```

Cascade all operations
including deletes!

Add support for Lazy loading

```
@Entity
@Table(name="course")
public class Course {
    ...

    @OneToMany(fetch=FetchType.LAZY, cascade=CascadeType.ALL)
    @JoinColumn(name="course_id")
    private List<Review> reviews;

    ...
}
```

Lazy load the
reviews

Add convenience method for adding review

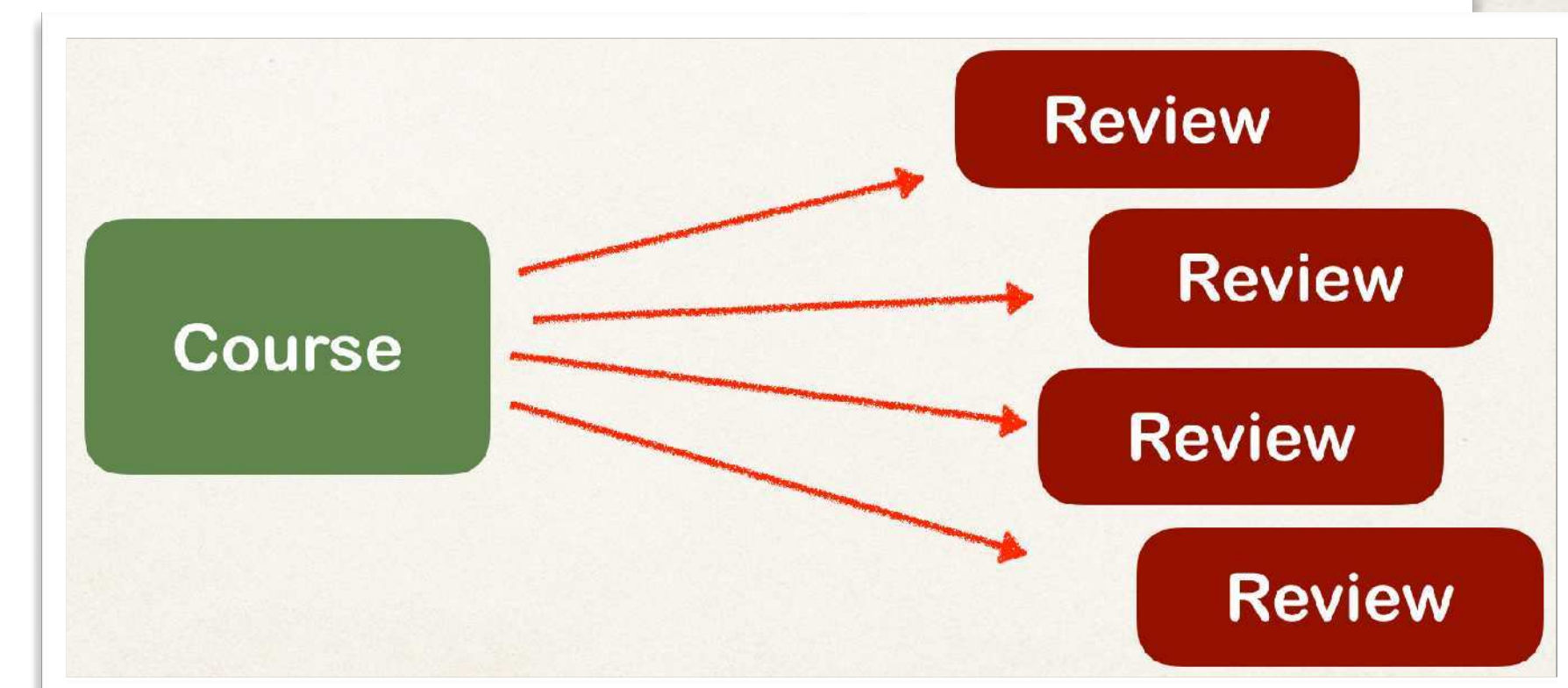
```
@Entity
@Table(name="course")
public class Course {
    ...
    // add convenience methods for adding reviews

    public void add(Review tempReview) {

        if (reviews == null) {
            reviews = new ArrayList<>();
        }

        reviews.add(tempReview);

    }
    ...
}
```

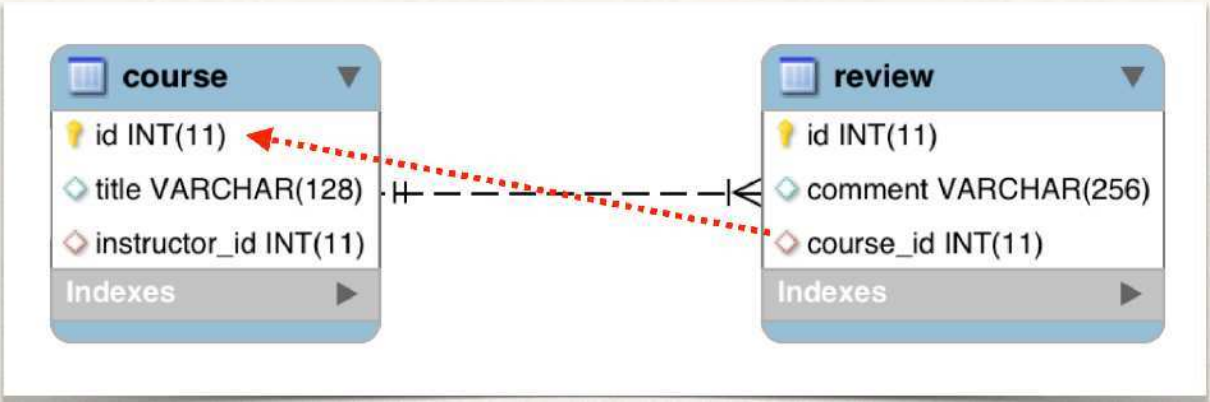
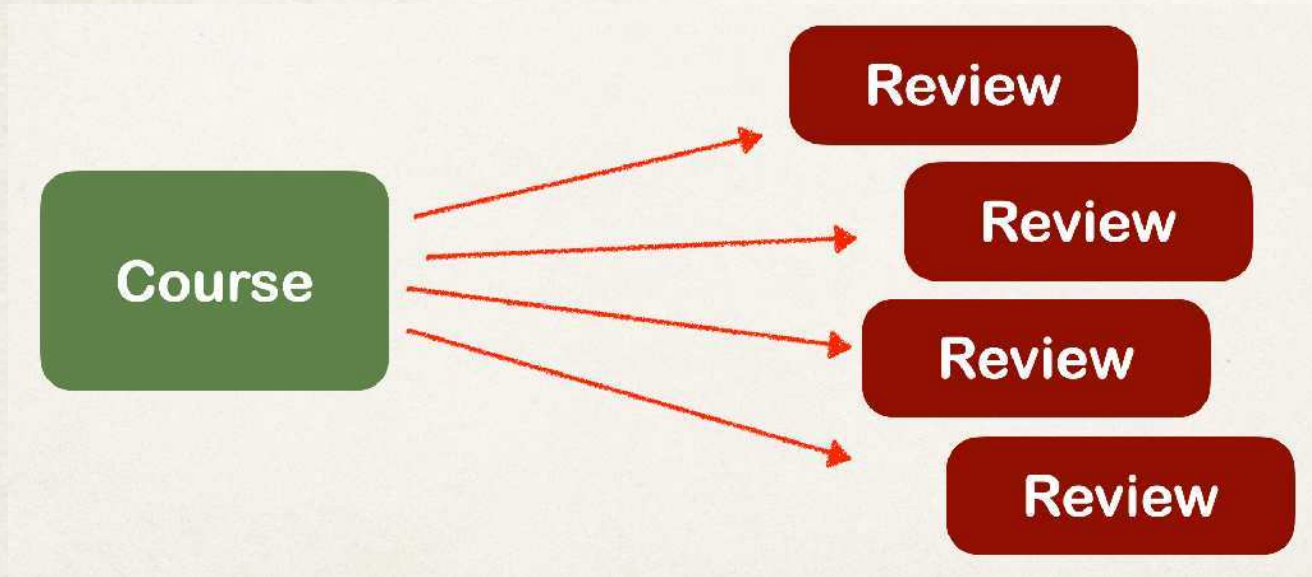


Mappings: @OneToMany

Defines physical mapping of the relationship

only applies for bi-directional relationships

Mapping	Foreign Key location in database table	Owning Side	Inverse side
@OneToMany – Unidirectional Course -> Review	review table	Course @OneToMany + @JoinColumn	-



```
@Entity
@Table(name="course")
public class Course {
    ...

    @OneToMany
    @JoinColumn(name="course_id")
    private List<Review> reviews;

    // getter / setters
}
```

Use this course's examples as a general guide.

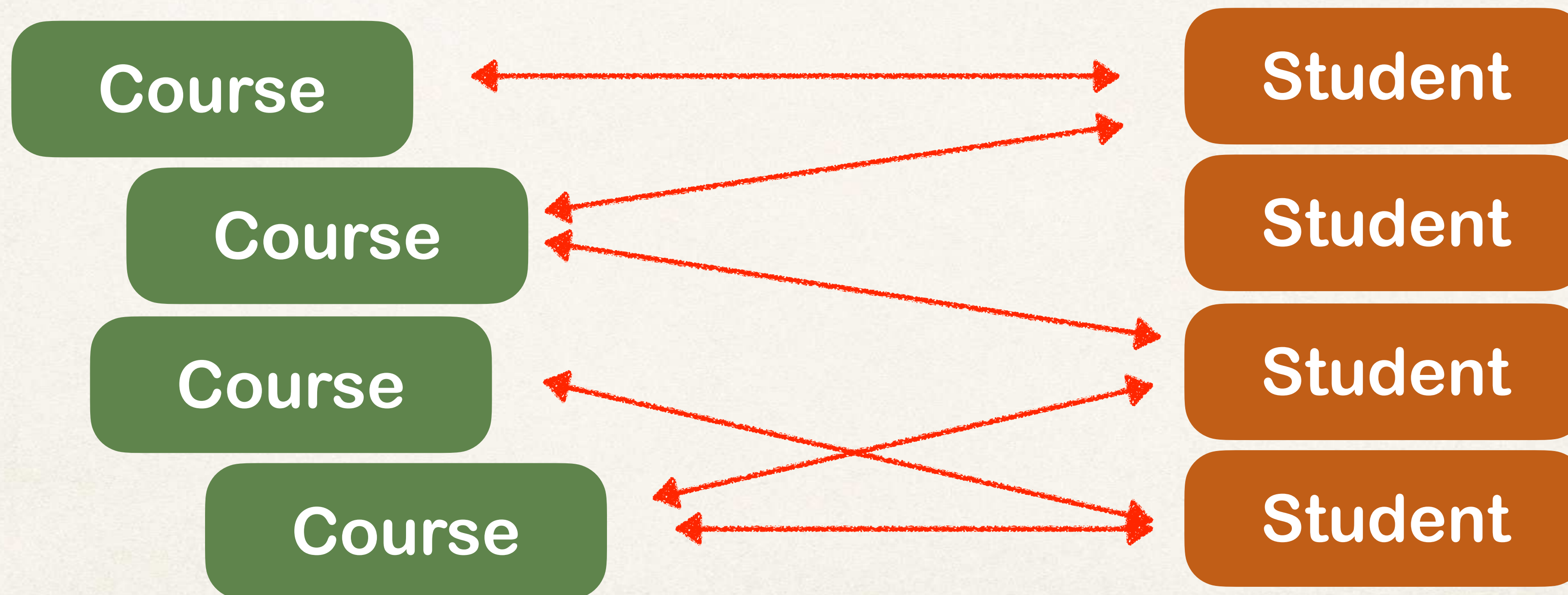
Adapt them to your application's requirements and domain needs.

@ManyToMany



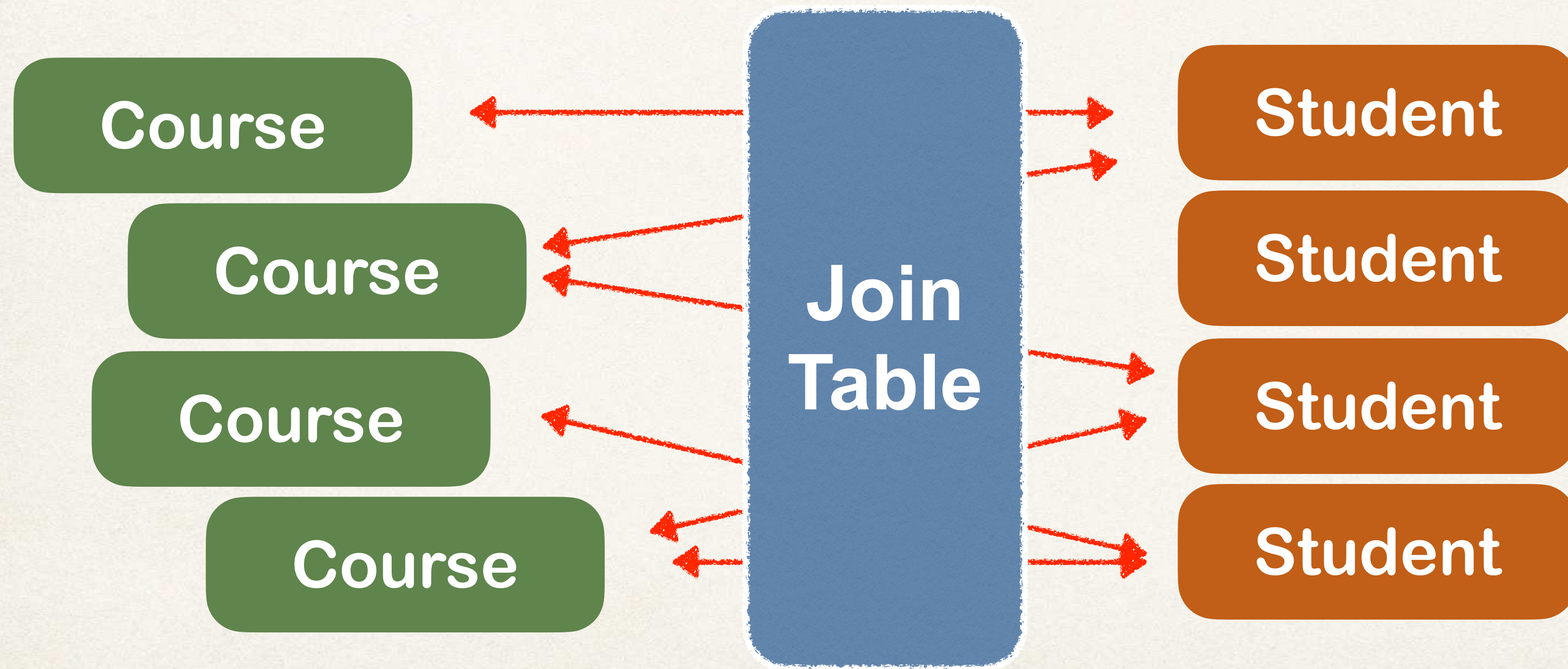
Many-to-Many Mapping

- A course can have many students
- A student can have many courses



Keep track of relationships

- Need to track which student is in which course and vice-versa



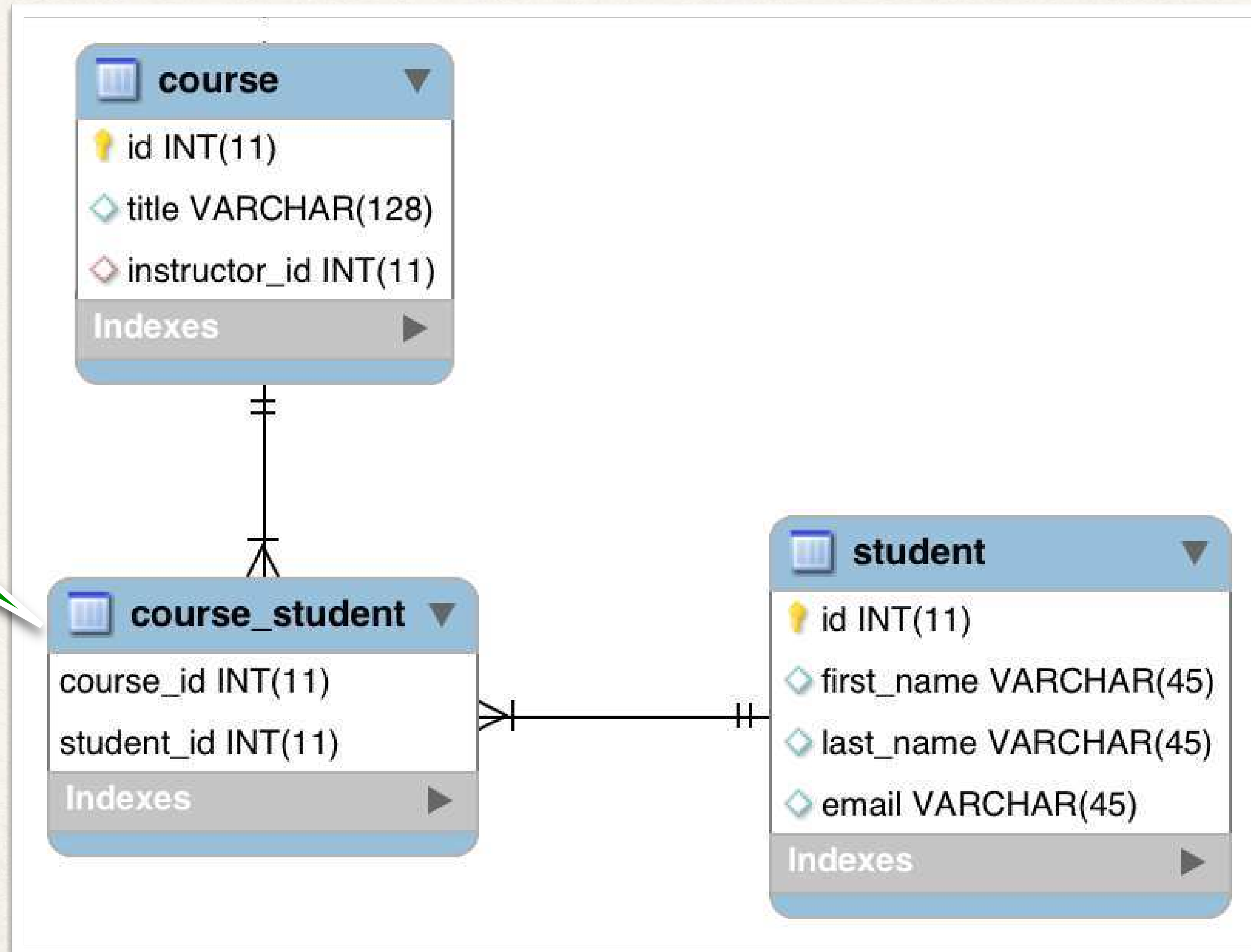
Join Table

A table that provides a mapping between two tables.

**It has foreign keys for each table
to define the mapping relationship.**

@ManyToMany

Join Table

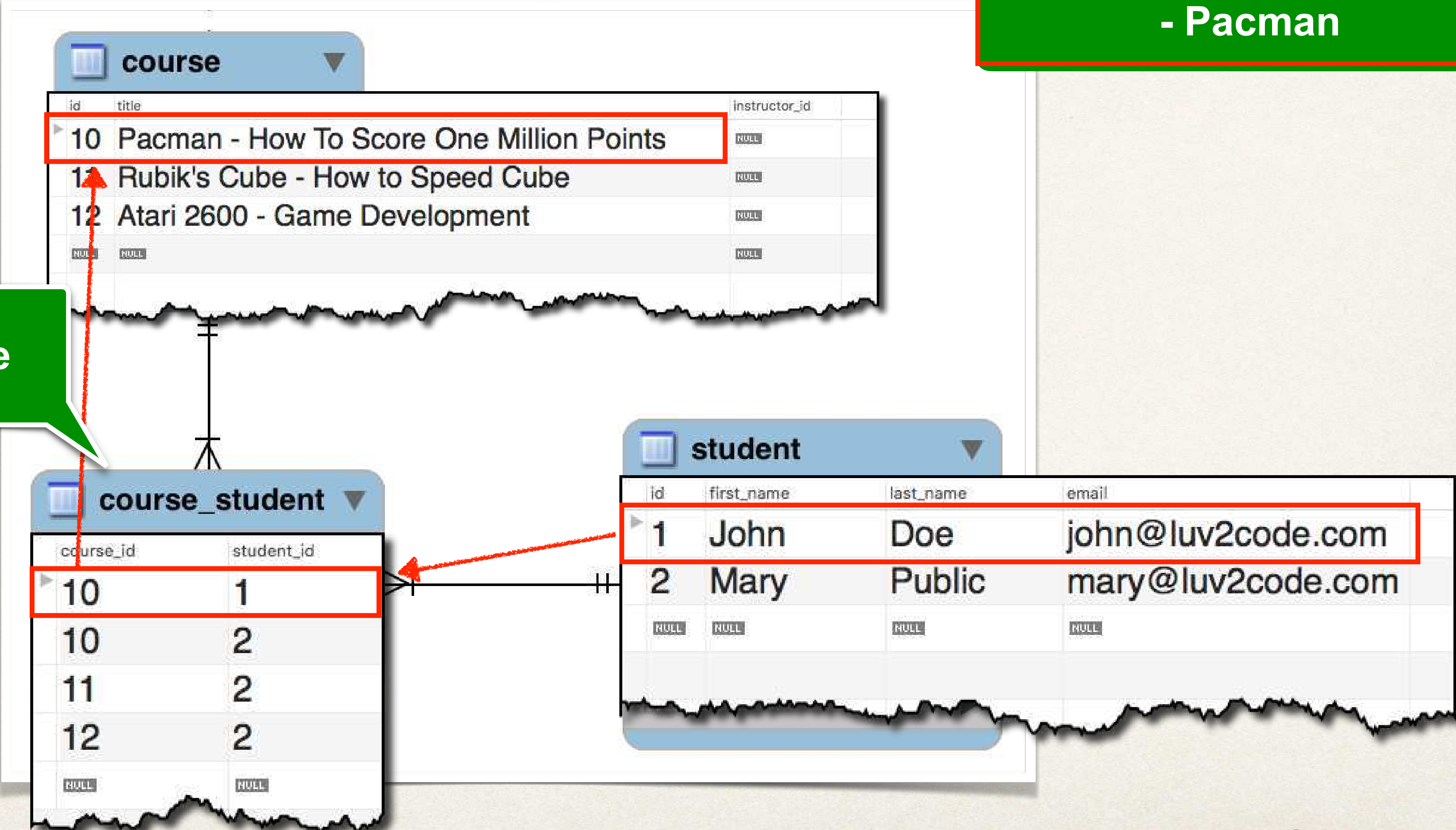


Join Table Example

Find John's Courses

- Pacman

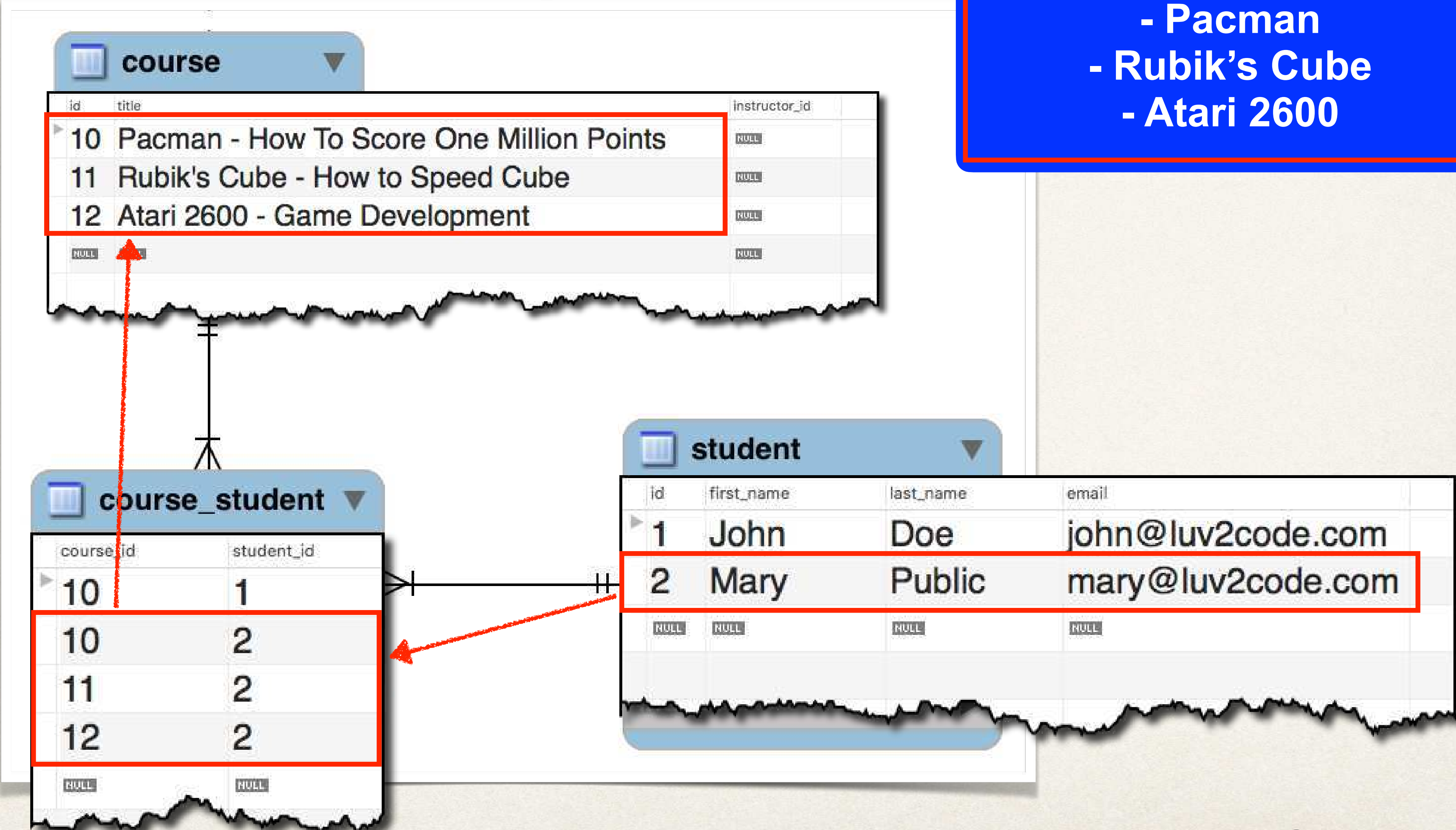
Join Table



Join Table Example

Find Mary's Courses

- Pacman
- Rubik's Cube
- Atari 2600



Development Process: Many-to-Many

1. Prep Work - Define database tables
2. Update **Course** class
3. Update **Student** class

Step-By-Step

join table: course_student

File: create-db.sql

```
CREATE TABLE `course_student` (  
  
  `course_id` int(11) NOT NULL,  
  `student_id` int(11) NOT NULL,  
  
  PRIMARY KEY (`course_id`, `student_id`),  
  
  ...  
);
```

course_student ▼
course_id INT(11)
student_id INT(11)
Indexes ▶

join table: course_student - foreign keys

```
CREATE TABLE `course_student` (
```

```
...
```

```
  CONSTRAINT `FK_COURSE_05`  
  FOREIGN KEY (`course_id`)  
  REFERENCES `course` (`id`),
```

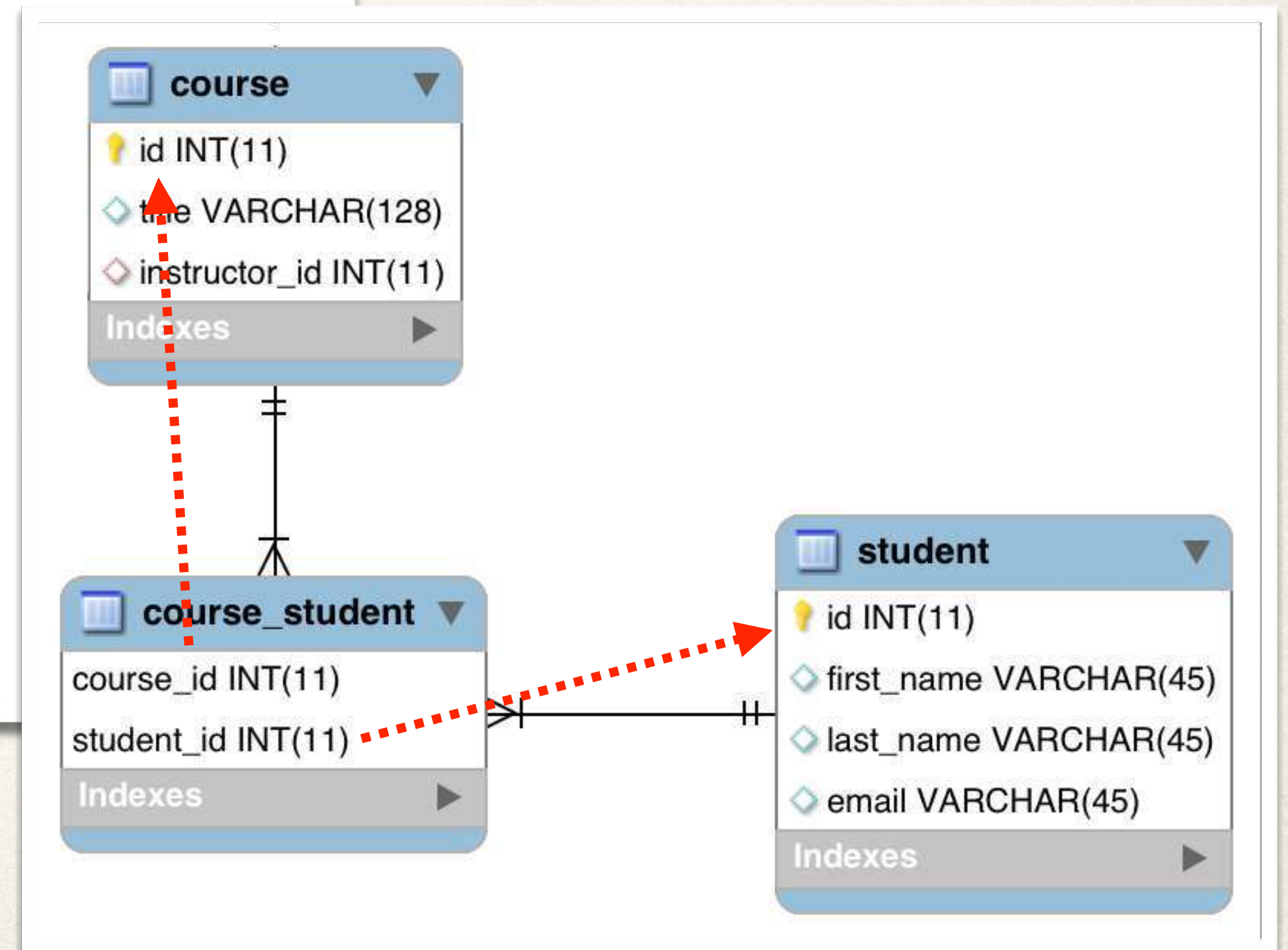
```
  CONSTRAINT `FK_STUDENT`  
  FOREIGN KEY (`student_id`)  
  REFERENCES `student` (`id`)
```

```
...
```

```
);
```

Table

Column



join table: course_student - foreign keys

```
CREATE TABLE `course_student` (
```

```
...
```

```
  CONSTRAINT `FK_COURSE_05`  
  FOREIGN KEY (`course_id`)  
  REFERENCES `course` (`id`),
```

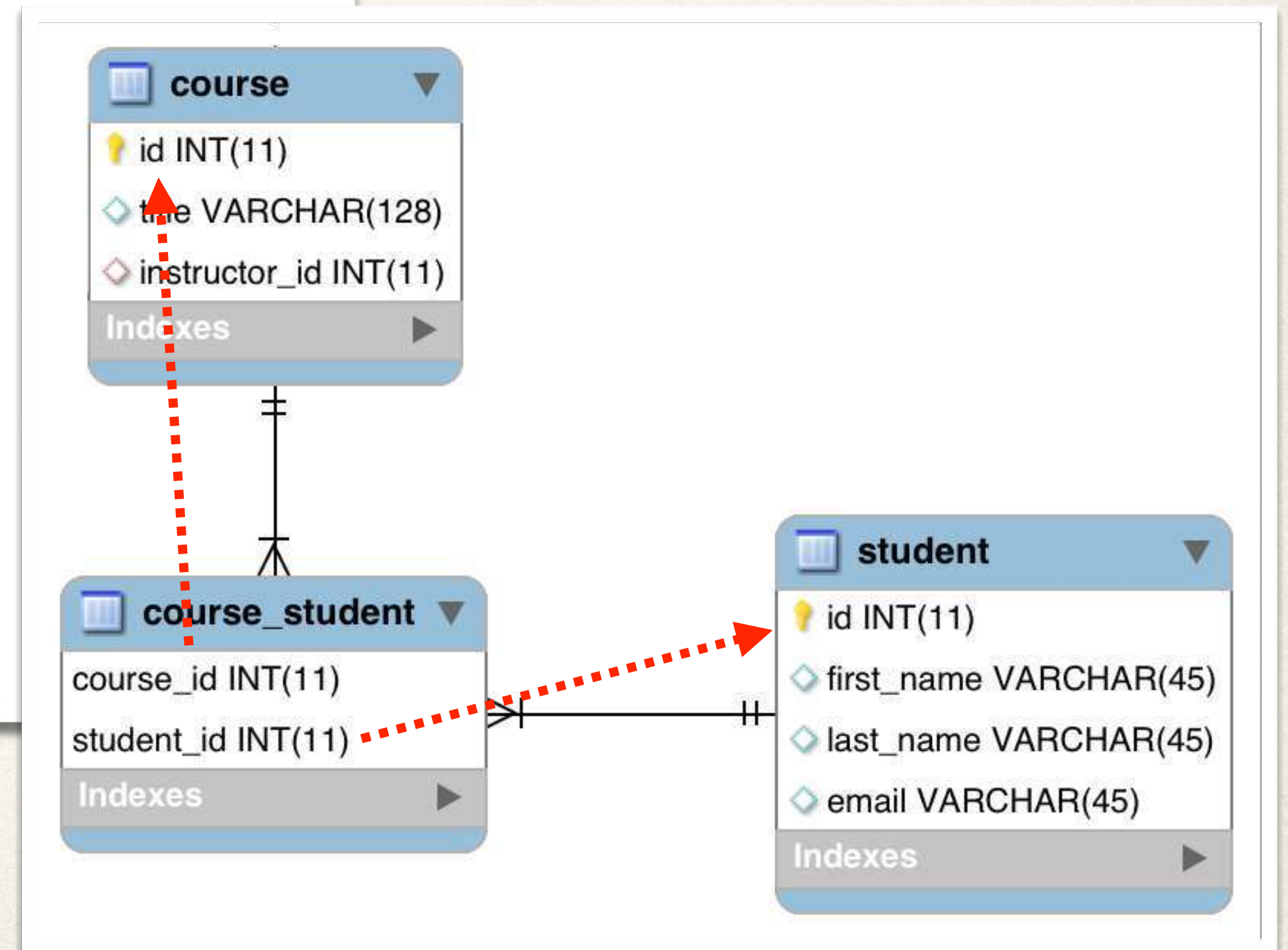
```
  CONSTRAINT `FK_STUDENT`  
  FOREIGN KEY (`student_id`)  
  REFERENCES `student` (`id`),
```

```
...
```

```
);
```

Table

Column



Step 2: Update Course - reference students

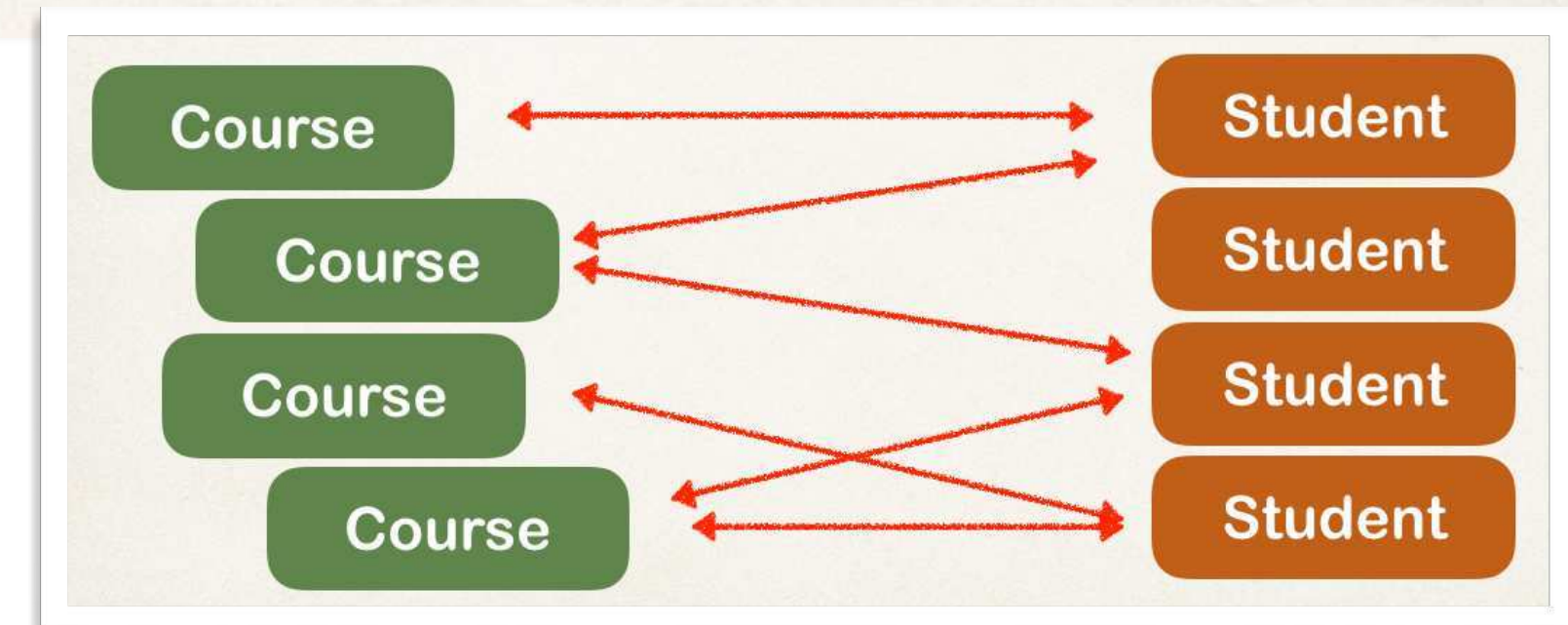
```
@Entity
@Table(name="course")
public class Course {
    ...
```

```
    private List<Student> students;
```

```
    // getter / setters
```

```
    ...
```

```
}
```



Add @ManyToMany annotation

```
@Entity
@Table(name="course")
public class Course {
    ...
}
```

Refers to "course_id" column
in "course_student" join table

```
@ManyToMany
@JoinTable(
    name="course_student",
    joinColumns=@JoinColumn(name="course_id"),
    inverseJoinColumns=@JoinColumn(name="student_id")
)
```

```
private List<Student> students;
```

```
// getter / setters
```

```
...
```

```
}
```

course_student
course_id INT(11)
student_id INT(11)
Indexes

Add @ManyToMany annotation

```
@Entity
@Table(name="course")
public class Course {
    ...
}
```

Refers to "student_id" column
in "course_student" join table

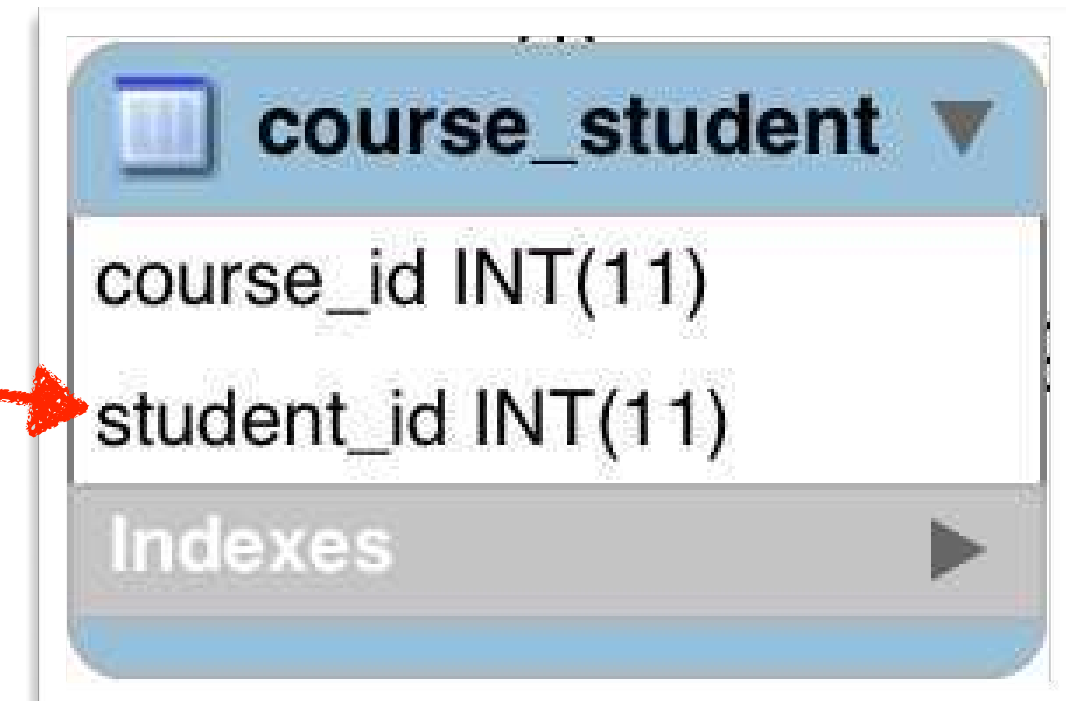
```
@ManyToMany
@JoinTable(
    name="course_student",
    joinColumns=@JoinColumn(name="course_id"),
    inverseJoinColumns=@JoinColumn(name="student_id")
)
```

```
private List<Student> students;
```

```
// getter / setters
```

```
...
```

```
}
```



The diagram shows a table named 'course_student' with two columns: 'course_id INT(11)' and 'student_id INT(11)'. Below the columns is a section labeled 'Indexes' with a right-pointing arrow. A red arrow points from the 'student_id' column in the table to the '@JoinColumn(name="student_id")' annotation in the code block above.

course_student ▼	
course_id	INT(11)
student_id	INT(11)
Indexes ▶	

Add @ManyToMany annotation

@Entity

@Table(name="course")

public class Course {

...

@ManyToMany

@JoinTable(

name="course_student",

joinColumns=@JoinColumn(name="course_id"),

inverseJoinColumns=@JoinColumn(name="student_id")

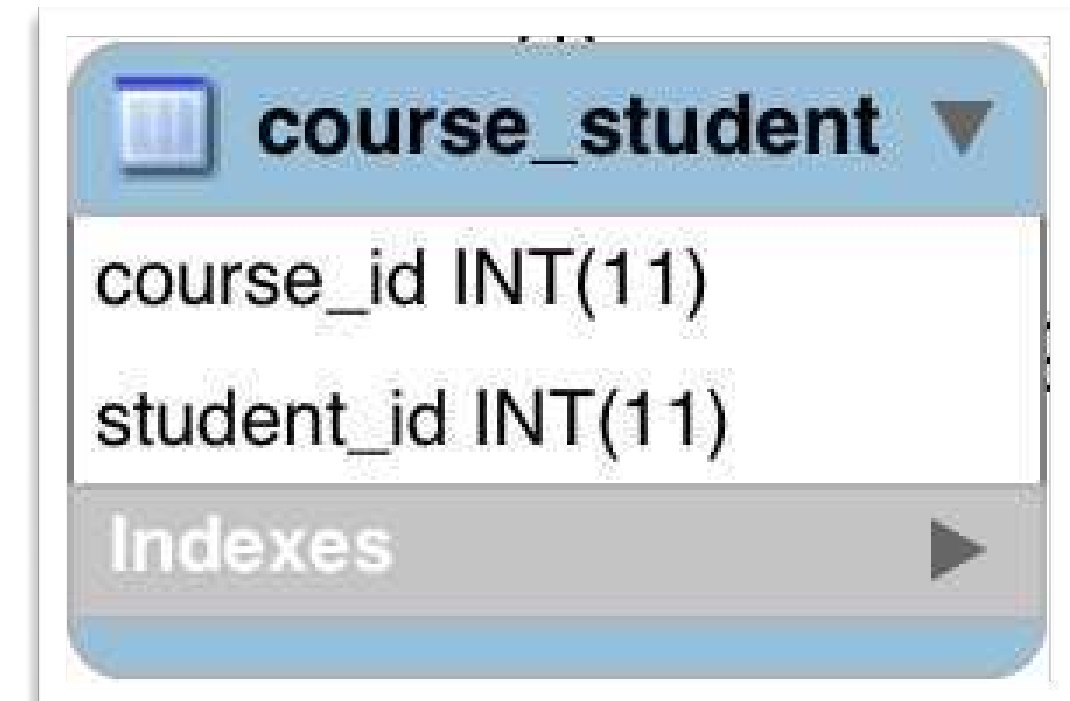
)

private List<Student> students;

// getter / setters

...

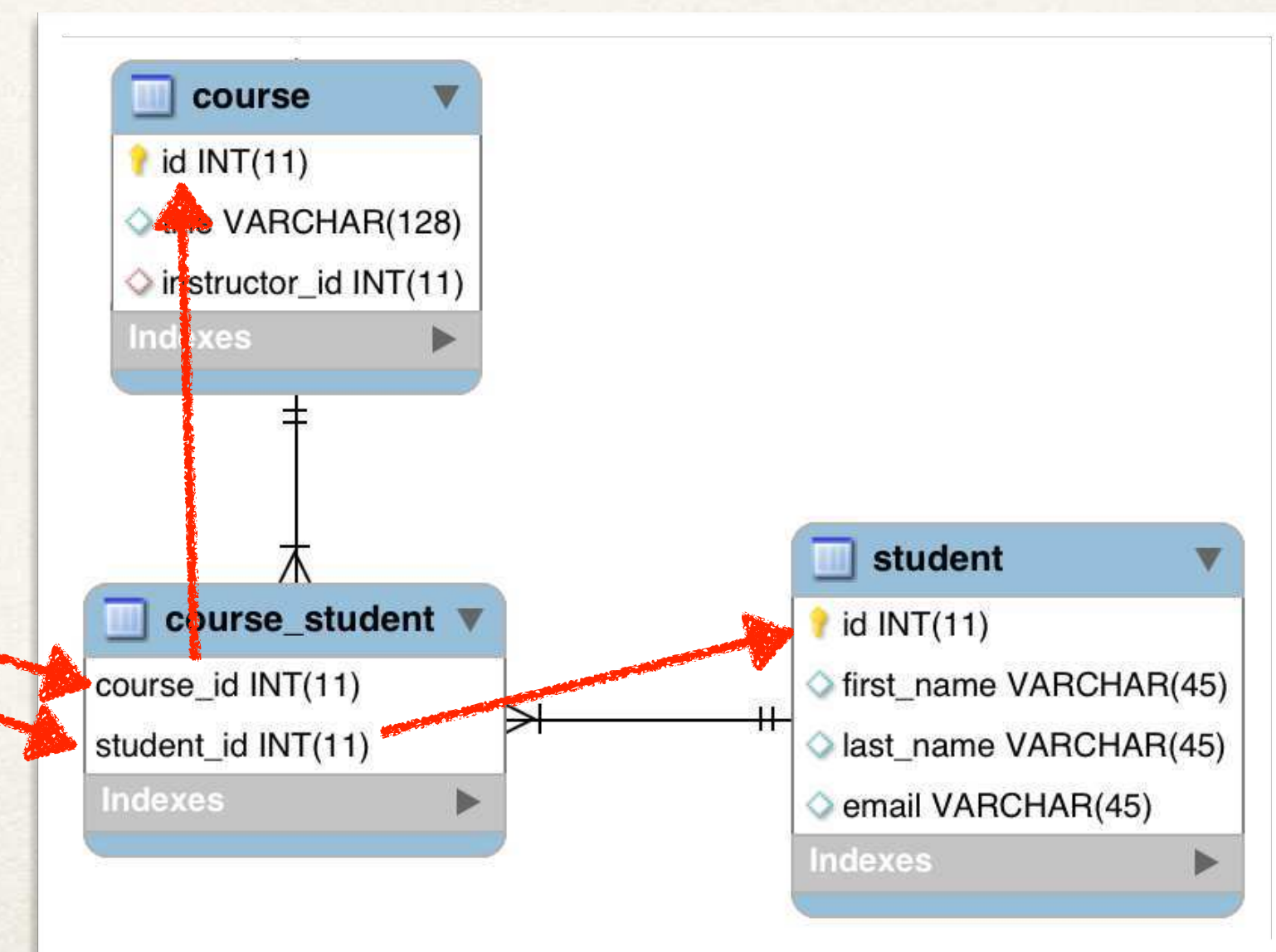
}



More: @JoinTable

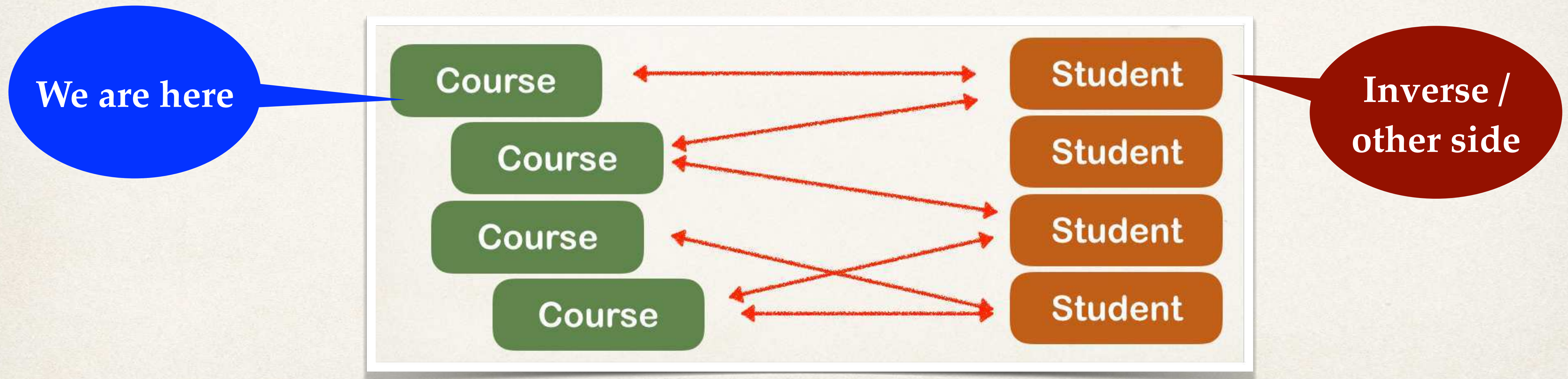
- **@JoinTable** tells Hibernate
 - Look at the **course_id** column in the **course_student** table
 - For other side (inverse), look at the **student_id** column in the **course_student** table
 - Use this information to find relationship between **course** and **students**

```
public class Course {  
  
    @ManyToMany  
    @JoinTable(  
        name="course_student",  
        joinColumns=@JoinColumn(name="course_id"),  
        inverseJoinColumns=@JoinColumn(name="student_id")  
    )  
    private List<Student> students;  
  
}
```



More on “inverse”

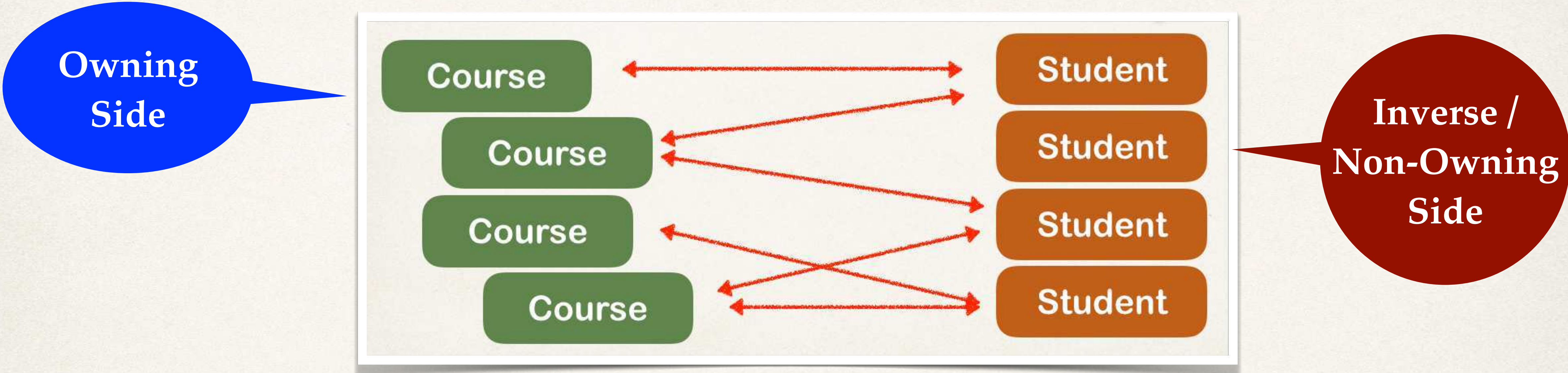
- In this context, we are defining the relationship in the **Course** class
- The **Student** class is on the “other side” ... so it is considered the “inverse”
- “Inverse” refers to the “other side” of the relationship



Owning Side and Inverse / Non-Owning Side

- Every many-to-many association has two sides:
 - The owning side and the inverse / non-owning side
- If the association is bidirectional
 - Either side may be designated as the owning side
 - The non-owning side must use the **mappedBy** element of the **ManyToMany** annotation
 - To specify the relationship field or property of the owning side

Owning Side and Inverse / Non-Owning Side



Now, let's update the mapping for the Student

Step 3: Update Student - reference courses

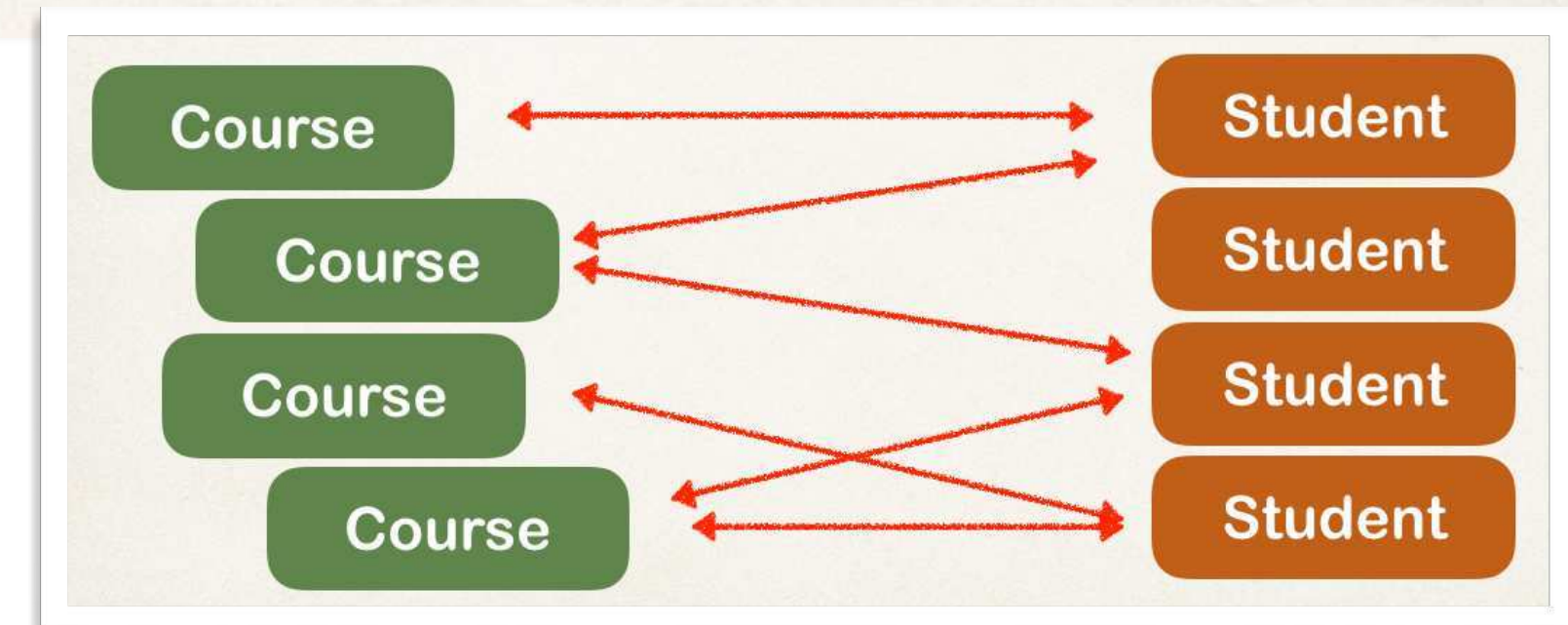
```
@Entity
@Table(name="student")
public class Student {
    ...
```

```
    private List<Course> courses;
```

```
    // getter / setters
```

```
    ...
```

```
}
```



Add @ManyToMany annotation

```
@Entity
@Table(name="student")
public class Student {
    ...
```

Refers to “students” property
in “Course” class

```
@ManyToMany(mappedBy="students")
private List<Course> courses;

// getter / setters
...
}
```


Add @ManyToMany annotation

```
@Entity
@Table(name="student")
public class Student {
    ...

    @ManyToMany(mappedBy="students")
    private List<Course> courses;

    // getter / setters
    ...
}
```


More: mappedBy

- **mappedBy** tells Hibernate
 - Look at the students property in the Course class
 - Use information from the Course class @JoinTable
 - To help find associated courses for student

```
public class Student {  
    ...  
  
    @ManyToMany(mappedBy="students")  
    private List<Course> courses;  
    .....  
}
```

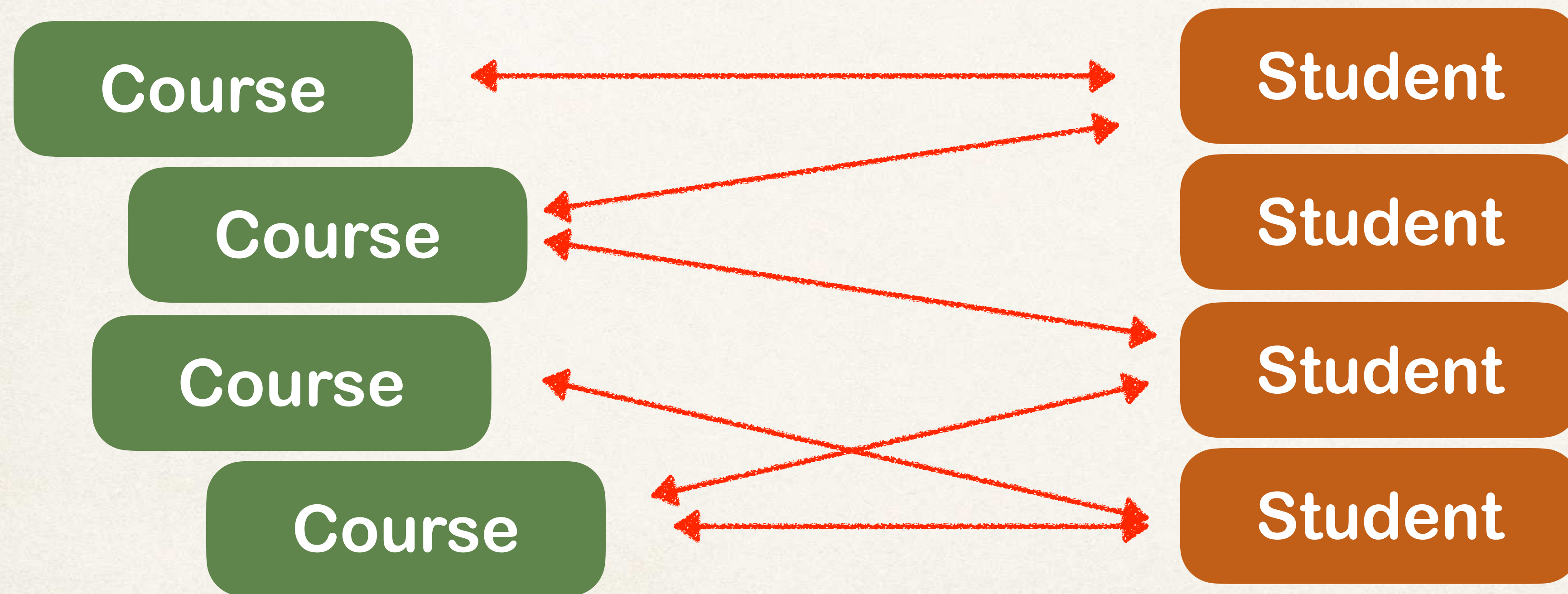


```
public class Course {  
  
    @ManyToMany  
    @JoinTable(  
        name="course_student",  
        joinColumns=@JoinColumn(name="course_id"),  
        inverseJoinColumns=@JoinColumn(name="student_id")  
    )  
    private List<Student> students;  
    .....  
}
```


Real-World Project Requirement

- If you delete a course, DO NOT delete the students

**DO NOT
apply cascading
deletes!**

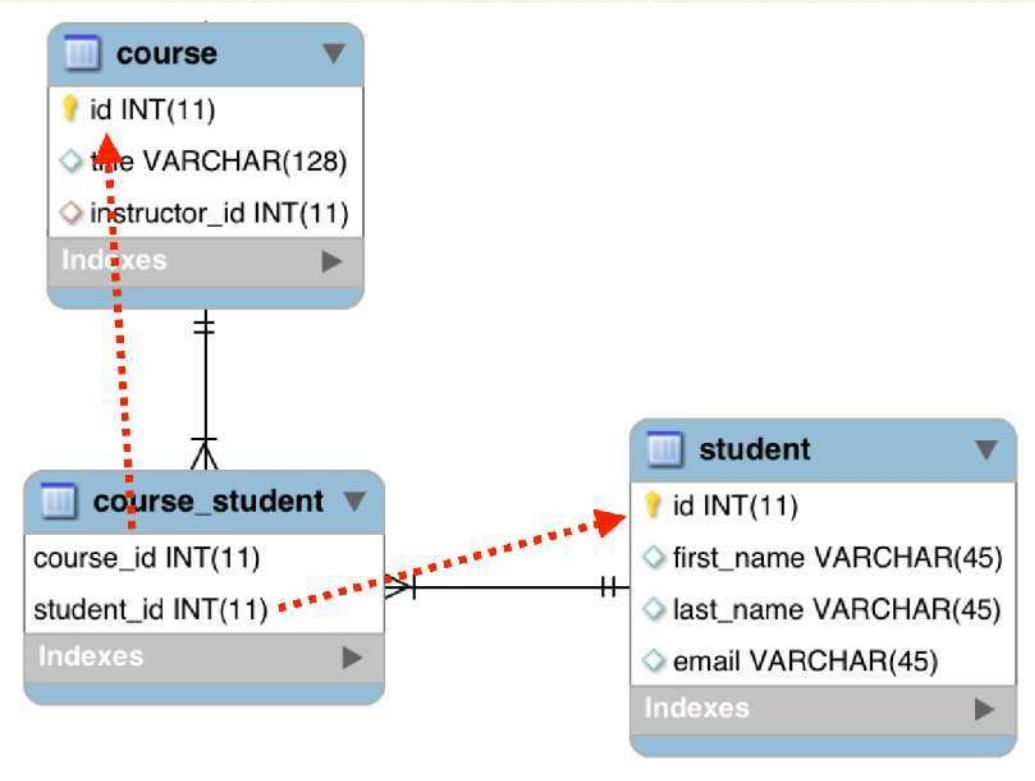
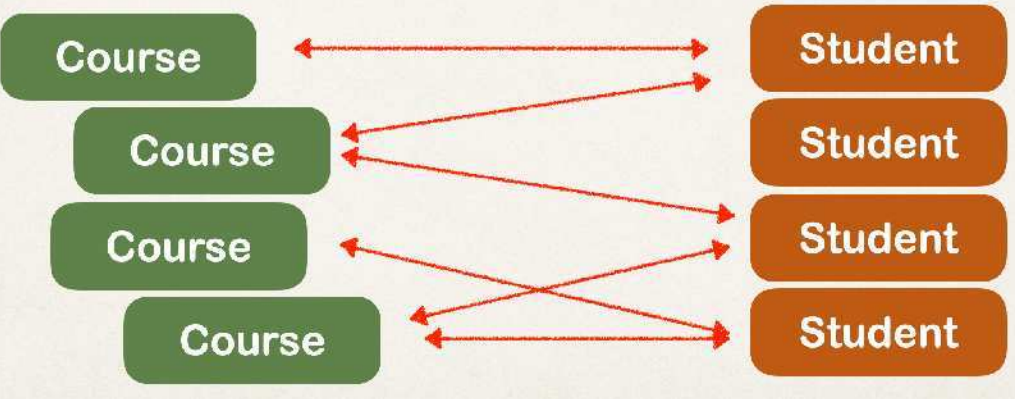


Mappings: @ManyToMany

Defines physical mapping of the relationship

Mirrors the owning side

Mapping	Foreign Key location in database table	Owning Side	Inverse side
@ManyToMany - Bidirectional Course <-> Student	join table: course_student	Course @ManyToMany + @JoinTable	Student @ManyToMany (mappedBy="students")



```
@Entity
@Table(name="course")
public class Course {
    ...

    @ManyToMany
    @JoinTable(
        name="course_student",
        joinColumns=@JoinColumn(name="course_id"),
        inverseJoinColumns=@JoinColumn(name="student_id")
    )
    private List<Student> students;
```

```
@Entity
@Table(name="student")
public class Student {
    ...

    @ManyToMany(mappedBy="students")
    private List<Course> courses;
    // getter / setters
```

Use this course's examples as a general guide.

Adapt them to your application's requirements and domain needs.

Other features

- In the next set of videos, we'll add support for other features
- **Lazy Loading** of students and courses
- **Cascading** to handle cascading saves ... but NOT deletes
 - If we delete a course, DO NOT delete students
 - If we delete a student, DO NOT delete courses

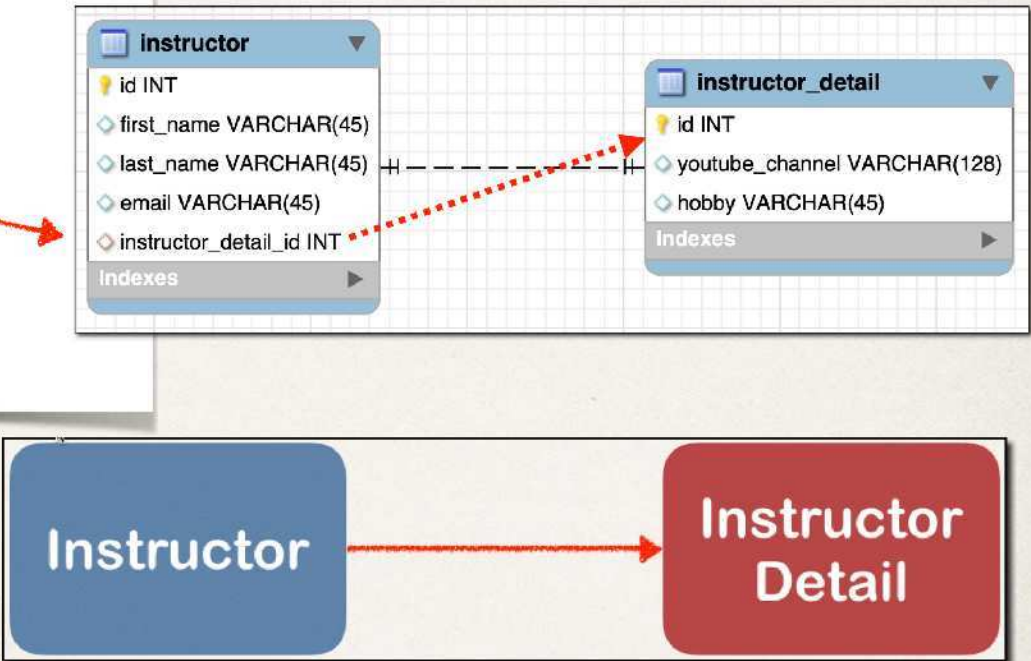
Mappings Review



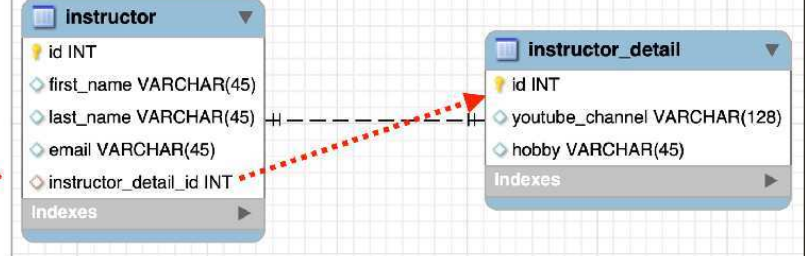
Mappings: @OneToOne

Mapping	Foreign Key location in database table	Owning Side	Inverse side
@OneToOne - Unidirectional Instructor -> InstructorDetails	instructor table	Instructor @OneToOne + @JoinColumn	-
@OneToOne - Bidirectional Instructor <-> InstructorDetails	instructor table	Instructor @OneToOne + @JoinColumn	InstructorDetail @OneToOne(mappedBy="instructorDetail")

```
@Entity
@Table(name="instructor")
public class Instructor {
    ...
    @OneToOne
    @JoinColumn(name="instructor_detail_id")
    private InstructorDetail instructorDetail;
    ...
    // constructors, getters / setters
}
```



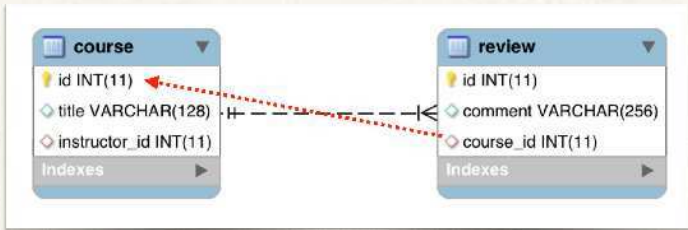
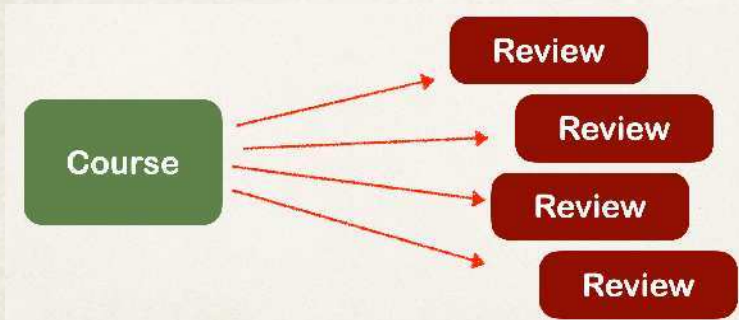
```
@Entity
@Table(name="instructor")
public class Instructor {
    ...
    @OneToOne
    @JoinColumn(name="instructor_detail_id")
    private InstructorDetail instructorDetail;
    ...
    // constructors, getters / setters
}
```



```
@Entity
@Table(name="instructor_detail")
public class InstructorDetail {
    ...
    @OneToOne(mappedBy="instructorDetail")
    private Instructor instructor;
    ...
}
```


Mappings: @OneToMany

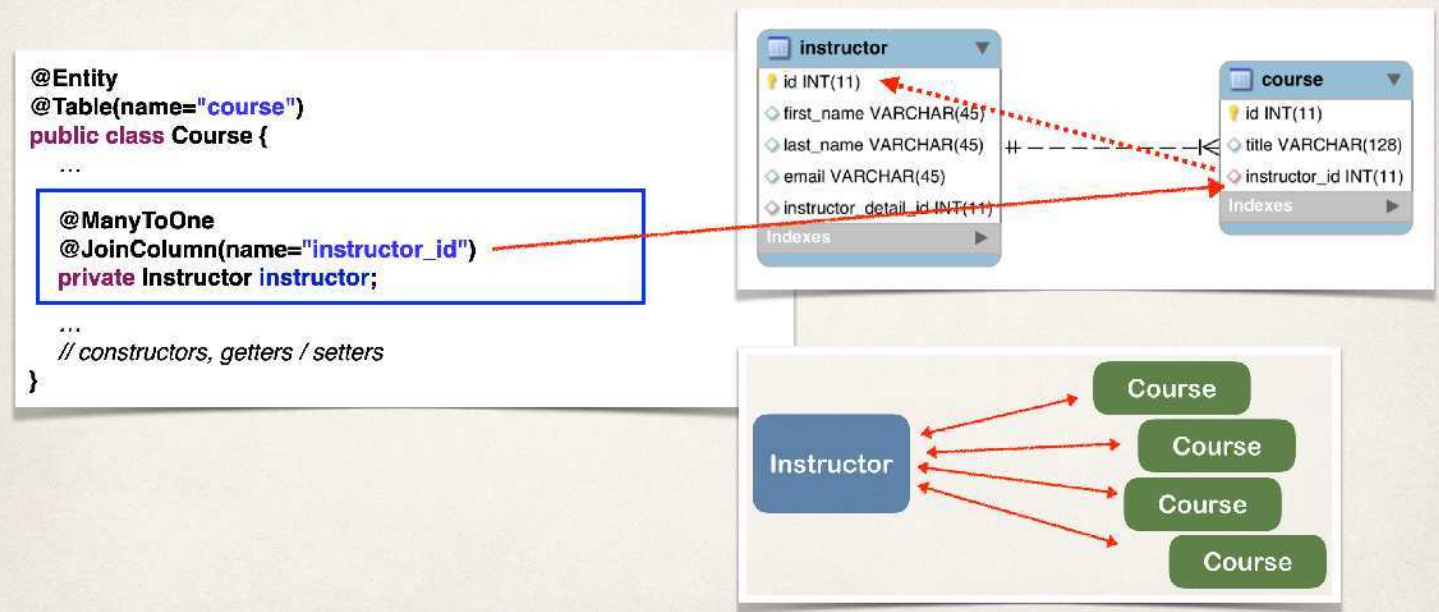
Mapping	Foreign Key location in database table	Owning Side	Inverse side
@OneToMany - Unidirectional Course -> Review	review table	Course @OneToMany + @JoinColumn	-
@OneToMany, @ManyToOne Bidirectional Instructor <-> Course	course table	Course @ManyToOne + @JoinColumn	Instructor @OneToMany(mappedBy="instructor")



```
@Entity
@Table(name="course")
public class Course {
    ...

    @OneToMany
    @JoinColumn(name="course_id")
    private List<Review> reviews;

    // getter / setters
}
```



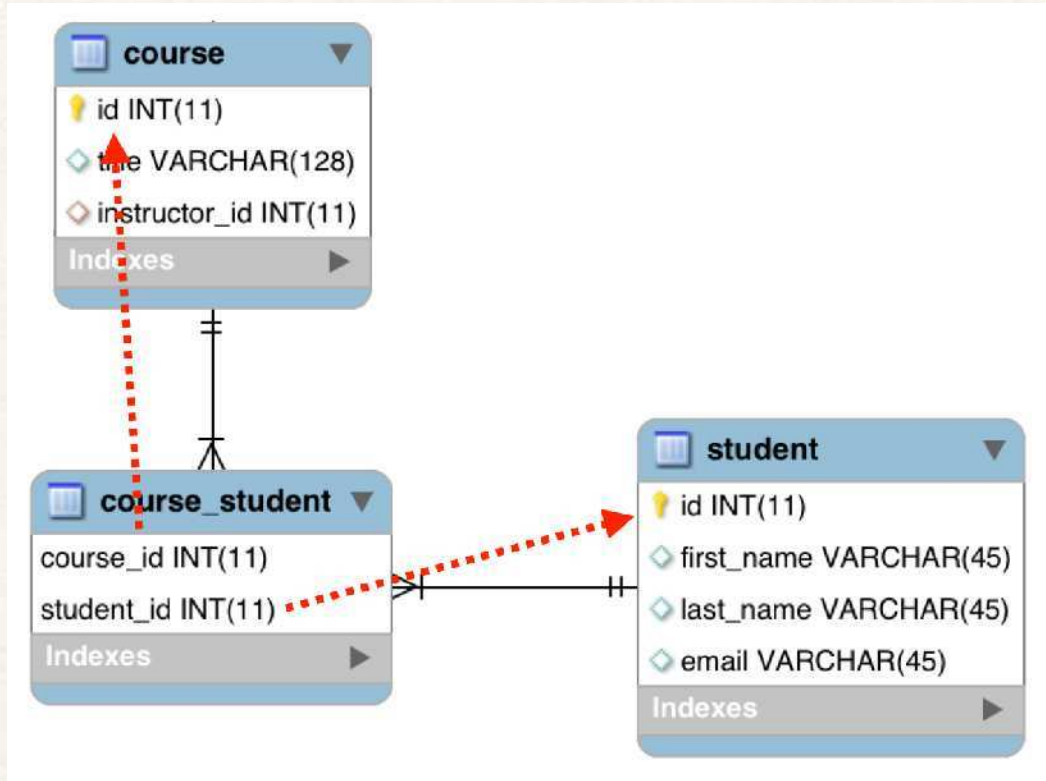
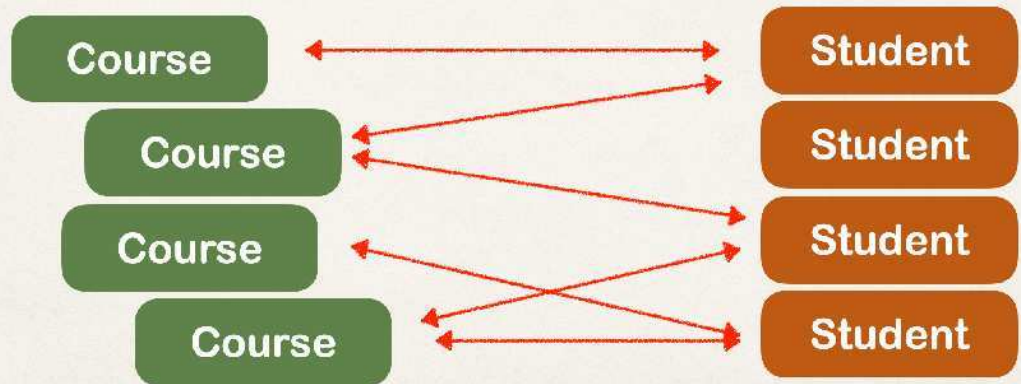
```
@Entity
@Table(name="instructor")
public class Instructor {
    ...

    @OneToMany(mappedBy="instructor")
    private List<Course> courses;

    // constructors, getters / setters
}
```


Mappings: @ManyToMany

Mapping	Foreign Key location in database table	Owning Side	Inverse side
@ManyToMany - Bidirectional Course <-> Student	join table: course_student	Course @ManyToMany + @JoinTable	Student @ManyToMany (mappedBy="students")



```
@Entity
@Table(name="course")
public class Course {
    ...

    @ManyToMany
    @JoinTable(
        name="course_student",
        joinColumns=@JoinColumn(name="course_id"),
        inverseJoinColumns=@JoinColumn(name="student_id")
    )
    private List<Student> students;
```

```
@Entity
@Table(name="student")
public class Student {
    ...

    @ManyToMany(mappedBy="students")
    private List<Course> courses;
    // getter / setters
```


Multiple Valid Designs

- For JPA / Hibernate relationships, this isn't one "right" mapping
- JPA / Hibernate supports several ways to model
 - @OneToOne, @OneToMany / @ManyToOne and @ManyToMany
- You may find other solutions online with different approaches
- In this course,
 - Treat the examples as a general guide
 - Adapt when your application requirements and domain needs differ